Book XI of Euclid's Elements

Formal Reconstruction — Technical Documentation

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Proposition XI.1

1.1 Introduction

Proposition XI.1 opens Euclid's solid geometry with a statement whose mathematical content is fundamentally incidence-theoretic. No angle, congruence, order, parallelism, or continuity is involved. The intended content is that once two distinct points of a straight line lie in a plane, the whole straight line lies in that plane.

This proposition is therefore an especially useful comparison point between Euclid's organization of Book XI and Hilbert's axiomatic organization of space. In the present formal reconstruction the proposition is not proved by reproducing Euclid's historical circle argument. Its precise incidence content is already one of the basic spatial incidence principles: Hilbert I.6.

The production file is

```
CGJteamLab/Proposition11_1.lean
```

and it imports only

```
CGJteamLab.Hilbert3DInterface
```

from the project-level spatial infrastructure.

1.2 Euclid's Statement

Euclid states Proposition XI.1 as follows:

> A part of a straight line cannot be in the plane of reference and a part in plane more elevated.

The natural precise incidence reading is:

> If two distinct points of a straight line lie in a plane, then the whole straight line lies in that plane.

This is the interpretation used by the formal reconstruction.

1.3 Classical Configuration

Let l be a straight line and let π be the reference plane. Choose two distinct points A, B on l such that

$$A \in \pi, \qquad B \in \pi.$$

The formal target is

$$l \subset \pi.$$

Euclid describes the same situation diagrammatically by regarding a segment AB as lying in the reference plane and supposing, for contradiction, that a continuation of the same straight line leaves that plane.

There is no essential metric configuration. The proposition concerns only the relation between a line, two of its points, and a plane.

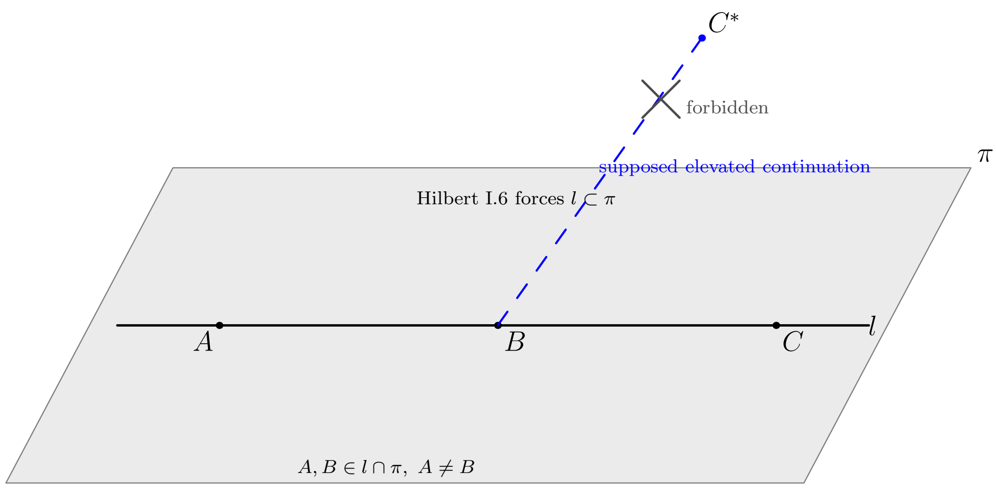


Figure 1.1: The incidence content of Proposition XI.1. The black carrier l contains the distinct points A, B of the reference plane π, hence Hilbert I.6 forces the whole line to remain in π. The blue dashed segment toward C^* does not represent a realizable straight continuation; it records only Euclid's contradictory supposition that the same straight line could leave the plane.

1.4 Euclid's Classical Proof

The received proof proceeds by contradiction. Euclid supposes that a part AB of a straight line ABC lies in the reference plane while the continuation BC is elevated out of that plane.

He then chooses in the reference plane another straight continuation BD of AB. The lines ABC and ABD would therefore share the segment AB. Euclid declares this impossible and appeals to a circle with center B and radius AB.

The logical difficulty is that this circle argument is not an adequate spatial incidence foundation. In three-dimensional space the center and radius do not determine a unique circle unless its plane is also specified. More importantly, the desired conclusion is itself an elementary closure property of line–plane incidence and should not require metric circle geometry.

For the formal reconstruction we therefore retain the mathematical incidence content of XI.1 but not this historical argument.

1.5 Classical Dependency Pattern

At the level of the printed proof, the structure is roughly

$$\begin{gathered}
AB \text{ lies in the reference plane} \\
\Downarrow \\
\text{suppose the continuation } BC \text{ leaves the plane} \\
\Downarrow \\
\text{construct another continuation } BD \text{ in the plane} \\
\Downarrow \\
ABC \text{ and } ABD \text{ share } AB \\
\Downarrow \\
\text{circle-based contradiction} \\
\Downarrow \\
\text{the straight line cannot leave the plane.}
\end{gathered}$$

This is not the dependency graph used by the formal proof.

1.6 What Is Genuinely Three-Dimensional Here?

XI.1 is spatial only because the primitive object π is an ambient plane and the conclusion concerns containment of an ambient line in that plane. There is no change of plane and no interaction between two distinct planes.

The genuine three-dimensional datum is simply

$$A, B \in l \cap \pi, \qquad A \neq B.$$

From these data the conclusion is the closure statement

$$l \subset \pi.$$

Thus XI.1 belongs entirely to the spatial incidence layer. It does not require an induced planar geometry $\mathrm{PlaneGeo}(\pi)$, because there is no planar theorem to prove inside π.

1.7 Formal Statement

The production theorem is:

```
theorem euclid_proposition_11_1
    [H : HilbertIncidence Geo]
    [S : HilbertSpacePrimitive Geo]
    [HSI : HilbertSpaceIncidence Geo]
    (pi : S.Plane)
    (l : Geo.Line)
    (A B : Geo.Point)
    (hAB : Ne A B)
    (hAl : H.OnLine A l)
    (hBl : H.OnLine B l)
    (hApi : S.OnPlane A pi)
    (hBpi : S.OnPlane B pi) :
    HilbertLineInPlane Geo l pi := by
  exact
    HilbertSpaceIncidence.line_in_plane
      (Geo := Geo)
      A B hAB
      l hAl hBl
      pi hApi hBpi
```

The conclusion

```
HilbertLineInPlane Geo l pi
```

means that every point incident with l is incident with π.

1.8 Hilbert Incidence Architecture

The decisive declaration is the spatial incidence field

```
HilbertSpaceIncidence.line_in_plane :
  forall A B : Geo.Point,
    Ne A B ->
    forall l : Geo.Line,
      H.OnLine A l ->
      H.OnLine B l ->
      forall pi : S.Plane,
        S.OnPlane A pi ->
        S.OnPlane B pi ->
        HilbertLineInPlane Geo l pi
```

This is Hilbert incidence axiom I.6 in the current 3D organization:

> If two distinct points of a line lie in a plane, then every point of the line lies in that plane.

Consequently the formal Euclid theorem is intentionally a thin proposition wrapper over the general spatial incidence interface.

This is structurally analogous to several Book I reconstruction points where a Euclid proposition becomes short after its reusable mathematical content has already been isolated in a general Hilbert theorem or axiom interface.

1.9 Ambient Space and PlaneGeo

No `PlaneGeo` slice is created in XI.1.

The objects remain ambient throughout:

```
A B : Geo.Point
l   : Geo.Line
pi  : S.Plane
```

and the incidence relations are

```
H.OnLine A l
H.OnLine B l
S.OnPlane A pi
S.OnPlane B pi
```

The proof does not require

```
HilbertPlaneIncidence Geo
HilbertSpaceOrder Geo
HilbertSpaceCongruence Geo
HilbertSpaceEuclidean Geo
```

as theorem parameters.

This is the minimal spatial layer appropriate to the proposition.

1.10 Lean Proof Architecture

The Lean proof has exactly one mathematical step:

$$\begin{gathered} A \neq B \\ A \in l, \quad B \in l \\ A \in \pi, \quad B \in \pi \\ \Downarrow \\ \texttt{HilbertSpaceIncidence.line\_in\_plane} \\ \Downarrow \\ l \subset \pi. \end{gathered}$$

There are no proposition-local helper lemmas and no case distinctions.

The formal proof does not construct a second line, a circle, an auxiliary plane, or any metric object. The historical contradiction is replaced by the exact spatial incidence principle that expresses the intended content.

1.11 Wyler-Style Flat Reconstruction

The Wyler-inspired reading of XI.1 replaces the informal idea that a line cannot “leave” a plane by a closure statement about generated flats. This is a structural reinterpretation of the proof, not a historical claim about Wyler’s own treatment of Euclid.

Let A, B be the two distinct points of l already known to lie in π. The relevant rank-one chain is

$$\mathrm{carrier}(l) \subseteq \mathrm{Span}\{A, B\} \subseteq \mathrm{carrier}(\pi).$$

The first inclusion expresses that the line carrier is generated by any two of its distinct points; in the flat API the corresponding span is in fact the same rank-one carrier. The second inclusion is the least-flat property of the span: since the plane carrier is a flat containing both generators, it must contain their generated flat.

Thus the public conclusion $l \subseteq \pi$ is recovered by composition:

```
A,B in l,  A != B
        |
        v
carrier(l) subset Span{A,B}
        |
A,B in pi and carrier(pi) is flat
        |
        v
Span{A,B} subset carrier(pi)
        |
        v
l subset pi
```

XI.1 is therefore the smallest nontrivial test of the flat vocabulary. A classical incidence statement becomes a rank-one closure statement, while the public theorem remains exactly Euclid XI.1 rather than being replaced by a span identity.

1.12 Hidden Formal Data

Although the theorem is short, two pieces of data that are implicit in Euclid’s prose are explicit in Lean.

1.12.1 The two plane points must be distinct

The hypothesis

```
hAB : Ne A B
```

is essential. A single point shared by a line and a plane does not force the whole line to lie in that plane.

1.12.2 The line carrier is explicit

Euclid speaks of the straight line ABC. Lean receives an explicit carrier

```
l : Geo.Line
```

together with the incidence witnesses

```
hAl : H.OnLine A l
hBl : H.OnLine B l
```

The conclusion is not merely that a third named point lies in the plane; it is the universal carrier statement `HilbertLineInPlane Geo l pi`.

1.13 Relationship with Earlier and Later Book XI Propositions

XI.1 has no earlier Book XI proposition dependency.

Its significance is forward-looking. The closure principle expressed by XI.1 is repeatedly needed whenever a later spatial argument knows two points of a line to lie in a plane and must promote that pointwise information to whole-line containment.

In the present Hilbert architecture later propositions normally use the general declaration

```
HilbertSpaceIncidence.line_in_plane
```

directly rather than importing `euclid_proposition_11_1`. Thus the mathematical content of XI.1 is foundational, while the Euclid-numbered theorem records the correspondence between the *Elements* and the general 3D incidence layer.

XI.2 is the first immediate classical consumer of XI.1: Euclid uses the principle that a line determined inside a plane cannot subsequently leave that plane. In the formal development XI.2 can instead call the stronger general spatial theorem for two intersecting lines.

1.14 Relationship with Hilbert's Group I

XI.1 is a particularly direct meeting point between Euclid and Hilbert.

In the current project, Hilbert's spatial incidence layer contains:

- existence of points on lines;
- existence and uniqueness of planes through noncollinear points;
- line closure under two plane points;
- the common-point principle for two planes;
- existence of noncoplanar points.

XI.1 corresponds exactly to the line-closure clause, Hilbert I.6.

No order axiom from Group II is used. No congruence axiom from Group III is used. No Euclidean parallel axiom from Group IV is used. No continuity assumption is used.

1.15 Classical DAG versus Formal DAG

The central structural difference can be summarized as follows.

Euclid.

$$\begin{array}{c}\text{supposed line leaving the reference plane}\\ \Downarrow \\ \text{second continuation in the reference plane}\\ \Downarrow \\ \text{shared segment}\\ \Downarrow \\ \text{circle-based contradiction}\\ \Downarrow \\ \text{XI.1.}\end{array}$$

Lean / Hilbert.

$$\begin{array}{c}A \neq B, \quad A, B \in l, \quad A, B \in \pi\\ \Downarrow \\ \text{Hilbert I.6}\\ \Downarrow \\ \texttt{HilbertSpaceIncidence.line\_in\_plane}\\ \Downarrow \\ \texttt{euclid\_proposition\_11\_1}.\end{array}$$

The second DAG is shorter because the incidence principle needed to make solid geometry possible is stated explicitly at the axiomatic level.

1.16 Structural Lesson

XI.1 exposes a foundational issue that is easy to miss when reading Book XI diagrammatically. To begin solid geometry one needs rules governing how lines and planes are incident. A planar circle construction cannot replace those rules.

The Hilbert reconstruction makes this boundary explicit. The proposition contains no metric content to reconstruct: its correct mathematical home is the spatial incidence system itself.

This also explains why the production theorem is deliberately thin. The formalization is not avoiding Euclid's mathematical content; it is locating that content at the level where it functions as a foundational rule for all later spatial arguments.

1.17 Audit Status

This documentation was checked against the current production declaration `Geometry.euclid_proposition_11_` and the spatial incidence interface used by its proof. The formal statement, the Hilbert incidence interpretation, and the classical/formal dependency comparison in this note reflect the current production architecture.

1.18 Conclusion

Proposition XI.1 is the incidence threshold of Book XI. Its intended mathematical statement is simple:

$$A \neq B, \quad A, B \in l, \quad A, B \in \pi \quad \Longrightarrow \quad l \subset \pi.$$

In the current formal architecture this is exactly Hilbert I.6. The Lean theorem therefore contains no auxiliary construction and no proposition-local proof machinery: it exposes the already explicit spatial incidence rule under Euclid's proposition number.

The contrast with the historical proof is instructive. Euclid attempts to exclude a line leaving the plane by a circle-based contradiction. The Hilbert reconstruction instead identifies the missing foundational content directly as line–plane incidence closure. That is both logically cleaner and exactly the form required by the subsequent development of solid geometry.

Chapter 2

Proposition XI.2

2.1 Introduction

Proposition XI.2 contains two closely related incidence statements:

1. two distinct straight lines which intersect lie in one plane;
2. every nondegenerate triangle lies in one plane.

The proposition is still entirely incidence-theoretic. No order, congruence, perpendicularity, parallelism, or continuity is needed.

Its importance is structural. Euclid wants to begin reasoning with configurations of intersecting lines and triangles in space, but such reasoning first requires an existence principle for planes. The present Hilbert reconstruction makes that existence explicit.

The production file is

```
CGJteamLab/Proposition11_2.lean
```

and the decisive reusable theorem is

```
hilbert_plane_through_two_intersecting_lines
```

from

```
CGJteamLab/Hilbert3DInterface.lean
```

2.2 Euclid's Statement

Euclid states:

> If two straight lines cut one another, then they lie in one plane; and every triangle lies in one plane.

Thus XI.2 has two conclusions, not merely one. In the formal development they are exposed as two public theorems:

```
euclid_proposition_11_2
euclid_proposition_11_2_triangle
```

The first handles intersecting lines. The second isolates the triangle clause.

2.3 Classical Configuration

Let the two straight lines AB and CD intersect at E.

Euclid then selects points on the rays EC and EB and considers the triangle ECB. The intended plane containing this triangle becomes the plane in which the two intersecting carriers AB and CD must lie.

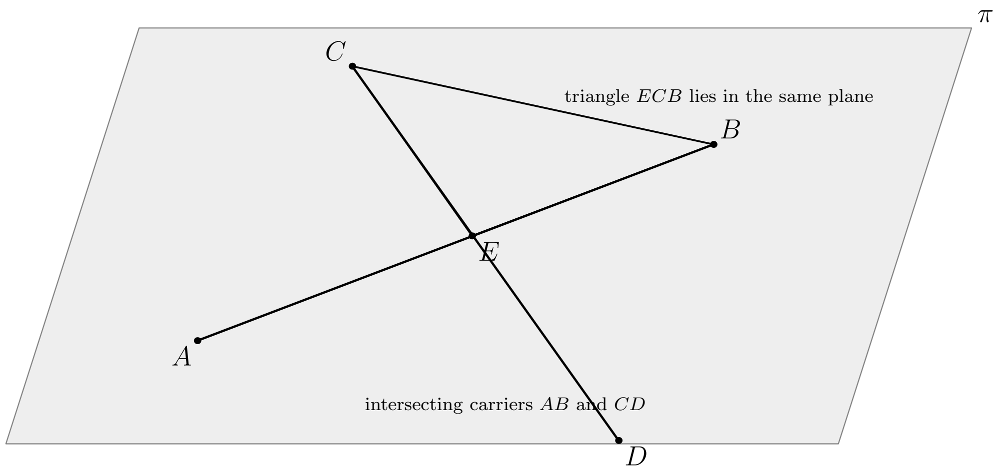


Figure 2.1: The incidence configuration of Proposition XI.2. The two distinct carriers AB and CD meet at E. The triangle ECB determines the plane π; once EC and EB lie in that plane, the full carriers CD and AB lie there as well.

The essential data are therefore

$$E \in AB, \qquad E \in CD, \qquad AB \neq CD.$$

For the triangle clause, the corresponding data are simply three noncollinear points.

2.4 Euclid's Classical Proof

Euclid first considers the triangle ECB. He argues that no part of this triangle can lie in one plane while another part lies in another, because that would force part of one of its straight sides to lie in one plane and part in another, contradicting XI.1.

Once the triangle ECB is taken to lie in one plane, its two sides EC and EB lie in that plane. Since EC is part of the carrier CD and EB is part of the carrier AB, XI.1 is used again to promote the side segments to the whole straight lines. Hence both intersecting lines lie in one plane.

The second conclusion, that every triangle lies in one plane, is treated by the same idea.

The classical local dependency is therefore

$$\begin{gathered} AB \cap CD = \{E\} \\ \Downarrow \\ \text{consider triangle } ECB \\ \Downarrow \\ \text{XI.1} \\ \Downarrow \\ ECB \text{ lies in one plane} \\ \Downarrow \\ EC, EB \text{ lie in that plane} \\ \Downarrow \\ \text{XI.1} \\ \Downarrow \\ AB, CD \text{ lie in that plane.} \end{gathered}$$

2.5 The Missing Plane-Existence Step

The classical argument contains a foundational difficulty. It speaks of "the plane of reference" and reasons about parts of the triangle lying in that plane, but no plane containing the relevant spatial configuration has yet been constructed.

This is exactly the sort of obligation that a modern incidence axiomatization must make explicit.

A natural repair is the Hilbert incidence principle that three noncollinear points determine a plane. Once such a plane exists, XI.1-like line closure can be applied rigorously.

The formal reconstruction therefore does not imitate the circular "reference plane" language. It explicitly produces the required plane from incidence data.

2.6 What Is Genuinely Three-Dimensional Here?

XI.2 is the first Book XI proposition whose conclusion is an existential statement about a plane determined by spatial data.

For the intersecting-line clause the input consists of two ambient lines

$$l, m$$

and a common point

$$P \in l \cap m, \qquad l \neq m.$$

The conclusion is the existence of an ambient plane

$$\pi$$

such that

$$l \subset \pi, \qquad m \subset \pi.$$

For the triangle clause the input is a noncollinear triple

$$A, B, C,$$

and the conclusion is a plane containing all three points.

No induced planar theorem is needed after the plane has been obtained. Thus XI.2 is genuinely spatial, but purely at the incidence level.

2.7 Formal Statement: Intersecting Lines

The first production theorem is:

```
theorem euclid_proposition_11_2
    [H : HilbertIncidence Geo]
    [HilbertPlaneIncidence Geo]
    [S : HilbertSpacePrimitive Geo]
    [HSI : HilbertSpaceIncidence Geo]
    (l m : Geo.Line)
    (P : Geo.Point)
    (hlm : Ne l m)
    (hPl : H.OnLine P l)
    (hPm : H.OnLine P m) :
    exists pi : S.Plane,
      HilbertLineInPlane Geo l pi /\
      HilbertLineInPlane Geo m pi := by

  have h :=
    hilbert_plane_through_two_intersecting_lines
      (Geo := Geo)
      l m hlm P hPl hPm

  cases h with
  | intro pi hData =>
      exact
        Exists.intro pi
          (And.intro
            hData.1
            hData.2.1)
```

The theorem intentionally exposes only the existence conclusion required by Euclid.

The reusable Hilbert theorem is stronger: it also proves uniqueness of the plane.

2.8 The Stronger Hilbert Theorem

The interface theorem has the form

```
theorem hilbert_plane_through_two_intersecting_lines
    [H : HilbertIncidence Geo]
    [HilbertPlaneIncidence Geo]
    [S : HilbertSpacePrimitive Geo]
    [HilbertSpaceIncidence Geo]
    (l m : Geo.Line)
    (hlm : Ne l m)
    (P : Geo.Point)
    (hPl : H.OnLine P l)
    (hPm : H.OnLine P m) :
    exists pi : S.Plane,
      HilbertLineInPlane Geo l pi /\
      HilbertLineInPlane Geo m pi /\
      forall rho : S.Plane,
        HilbertLineInPlane Geo l rho ->
        HilbertLineInPlane Geo m rho ->
        rho = pi
```

Thus the Hilbert layer proves not only

$$\exists \pi, \quad l, m \subset \pi,$$

but also

$$\forall \rho, \quad l, m \subset \rho \Longrightarrow \rho = \pi.$$

The Euclid theorem discards this uniqueness component because it is not part of XI.2's stated conclusion.

2.9 How the Stronger Theorem Is Proved

The reusable theorem is a genuine derived incidence result.

Starting from the common point P, it chooses a second point A on l and a second point B on m, with

$$A \neq P, \qquad B \neq P.$$

Because $l \neq m$, the three points A, P, B are noncollinear. Hilbert spatial incidence then supplies a plane through them.

The two line-containment conclusions follow from the XI.1-type incidence principle:

$$A, P \in l \cap \pi \Longrightarrow l \subset \pi,$$

$$B, P \in m \cap \pi \Longrightarrow m \subset \pi.$$

Finally, if another plane ρ contained both l and m, then it would contain the same noncollinear triple A, P, B. Plane uniqueness therefore gives

$$\rho = \pi.$$

Schematically,

$$\begin{array}{c}
l \neq m, \quad P \in l \cap m \\
\Downarrow \\
A \in l,\ A \neq P, \qquad B \in m,\ B \neq P \\
\Downarrow \\
A, P, B \text{ noncollinear} \\
\Downarrow \\
\text{Hilbert I.4} \\
\Downarrow \\
\exists \pi :\ A, P, B \in \pi \\
\Downarrow \\
\text{Hilbert I.6} \\
\Downarrow \\
l, m \subset \pi \\
\Downarrow \\
\text{Hilbert I.5} \\
\Downarrow \\
\pi \text{ is unique.}
\end{array}$$

This is the real formal incidence content behind the short public wrapper.

2.10 Formal Statement: Triangle Clause

The second public theorem is:

```
theorem euclid_proposition_11_2_triangle
    [H : HilbertIncidence Geo]
    [S : HilbertSpacePrimitive Geo]
    [HSI : HilbertSpaceIncidence Geo]
    (A B C : Geo.Point)
    (hABC : Not (PrimCollinear Geo A B C)) :
    exists pi : S.Plane,
      S.OnPlane A pi /\
      S.OnPlane B pi /\
      S.OnPlane C pi := by

  exact
    HilbertSpaceIncidence.plane_through
      (Geo := Geo)
      A B C hABC
```

This is precisely the existence part needed to say that a nondegenerate triangle lies in a plane.

The theorem returns the three vertex-incidence facts. If explicit side carriers are later introduced, Hilbert I.6 promotes each pair of vertex incidences to whole-line containment.

2.11 Why the Triangle Theorem Is Minimal

It would be possible to formulate a larger theorem returning explicit side lines AB, BC, and CA together with proofs that all three lie in the same plane.

The current theorem deliberately avoids that extra representation layer. At the incidence level a nondegenerate triangle is already determined by three noncollinear vertices. The existence of their plane is the exact content of the second clause of XI.2.

Whenever side carriers are required, they can be constructed separately by the ordinary two-point line theorem and placed in the same plane using `line_in_plane`.

2.12 Spatial Architecture

The intersecting-line theorem uses only:

```
HilbertIncidence Geo
HilbertPlaneIncidence Geo
HilbertSpacePrimitive Geo
HilbertSpaceIncidence Geo
```

The triangle theorem uses an even smaller signature:

```
HilbertIncidence Geo
HilbertSpacePrimitive Geo
HilbertSpaceIncidence Geo
```

There is no use of

```
HilbertSpaceOrder
HilbertSpaceCongruence
HilbertSpaceEuclidean
PlaneGeo
```

Thus XI.2 belongs completely to Hilbert's spatial Group I incidence layer.

2.13 Wyler-Style Flat Reconstruction

XI.2 exhibits the first genuine rank jump. The two clauses of the proposition have two parallel normal forms in the flat calculus.

For two distinct intersecting lines l, m, the common point guarantees that they generate a plane and the exact carrier identity is

$$\mathrm{Join}(\mathrm{carrier}(l), \mathrm{carrier}(m)) = \mathrm{carrier}(\pi).$$

The public incidence conclusion is then recovered because both generators are contained in their join.

For the triangle clause, three noncollinear points form a rank-two generating set:

$$\mathrm{Span}\{A, B, C\} = \mathrm{carrier}(\pi).$$

Hence the two parts of XI.2 may be read as the two standard ways to generate a plane in three-space:

```
two distinct intersecting lines     three noncollinear points
               |                                 |
               v                                 v
       Join(l,m) = plane                Span{A,B,C} = plane
               |                                 |
               v                                 v
          l,m subset pi                      A,B,C in pi
```

This is the first place where the point–line–plane hierarchy becomes a rank ladder rather than a list of unrelated incidence constructions. No stronger flat statement replaces the theorem: the exact Euclidean clauses are still the public outputs.

2.14 Hidden Formal Data

Several facts suppressed by the classical diagram must be explicit in Lean.

2.14.1 The two carriers must be distinct

The hypothesis

```
hlm : Ne l m
```

is essential. If $l = m$, infinitely many planes may contain that one line, so the unique-plane theorem is false.

2.14.2 The common point must be explicit

The intersection is represented by

```
P : Geo.Point
hPl : H.OnLine P l
hPm : H.OnLine P m
```

rather than by a primitive relation saying merely that the lines intersect.

2.14.3 Noncollinearity must be proved

The stronger helper does not assume three ready-made noncollinear points. It constructs one auxiliary point on each line and derives noncollinearity from the distinctness of the two carriers.

2.14.4 The witness plane survives

The conclusion is existential:

```
exists pi : S.Plane, ...
```

so the plane is a genuine proof object, not an implicit background surface supplied by the diagram.

2.15 Relationship with Proposition XI.1

Classically XI.1 is the immediate dependency of XI.2.

Euclid repeatedly reasons that if part of a side lies in a plane, then the whole straight line must lie there. That is precisely the content of XI.1.

The formal DAG is organized differently. The public theorem `euclid_proposition_11_2` does not call `euclid_proposition_11_1`. Instead, the stronger general theorem `hilbert_plane_through_two_inter` uses the underlying Hilbert I.6 field directly.

Hence:

$$\text{classical dependency: XI.1} \to \text{XI.2},$$

whereas

$$\text{formal dependency: Hilbert I.2–I.6} \to \texttt{hilbert\_plane\_through\_two\_intersecting\_lines} \to \text{XI.2}.$$

This is not a loss of historical structure. It records the fact that XI.1's mathematical content has already been moved into the reusable spatial incidence interface.

2.16 Relationship with Later Book XI Propositions

XI.2 is a basic coplanarity theorem and becomes part of the classical infrastructure for later Book XI arguments.

In particular, later propositions repeatedly need to know that two intersecting carriers determine a common plane before a planar theorem can be applied to them.

In the formal development the stronger interface theorem often replaces direct calls to XI.2. This is analogous to XI.1: the Euclid-numbered theorem records the correspondence with the *Elements*, while downstream Lean proofs are free to consume the stronger reusable result.

2.17 Classical DAG versus Formal DAG

The distinction is especially clear for XI.2.

Euclid.

$$\begin{gathered} AB, CD \text{ intersect at } E \\ \Downarrow \\ \text{triangle } ECB \\ \Downarrow \\ \text{XI.1} \\ \Downarrow \\ \text{triangle lies in one plane} \\ \Downarrow \\ \text{XI.1} \\ \Downarrow \\ AB, CD \text{ lie in that plane.} \end{gathered}$$

The weak point is that the plane itself is never produced from an explicit plane-existence principle.

Lean / Hilbert.

$$\begin{gathered} l \neq m, \quad P \in l \cap m \\ \Downarrow \\ \text{two auxiliary line points} \\ \Downarrow \\ \text{noncollinear triple} \\ \Downarrow \\ \text{Hilbert I.4} \\ \Downarrow \\ \text{explicit plane witness } \pi \\ \Downarrow \\ \text{Hilbert I.6} \\ \Downarrow \\ l, m \subset \pi \\ \Downarrow \\ \text{Hilbert I.5} \\ \Downarrow \\ \pi \text{ unique} \\ \Downarrow \\ \texttt{euclid\_proposition\_11\_2}. \end{gathered}$$

For the triangle clause the formal DAG is even shorter:

$$A, B, C \text{ noncollinear} \Longrightarrow \text{Hilbert I.4} \Longrightarrow \exists \pi,\ A, B, C \in \pi.$$

2.18 Structural Lesson

XI.2 makes explicit why solid geometry cannot begin merely by reusing planar reasoning in an unspecified ambient setting.

Before one can say that two intersecting spatial lines are a planar configuration, one must prove the existence of the plane containing them. Before one can treat a triangle as planar, one must prove the existence of the plane through its vertices.

The Hilbert incidence system supplies exactly these missing existence and uniqueness principles. Consequently XI.2 becomes a short Euclid interface over a stronger and reusable spatial theorem.

This proposition is therefore an early example of a recurring pattern in Book XI:

$$\text{diagrammatic coplanarity} \quad \longrightarrow \quad \text{explicit existential plane witness.}$$

That distinction becomes increasingly important in later propositions, where several different planes coexist and their identities or intersections can no longer be left implicit.

2.19 Audit Status

This documentation was checked against the current production declarations `Geometry.euclid_proposition_11` and `Geometry.euclid_proposition_11_2_triangle`, together with the stronger spatial incidence theorem used by the reconstruction. The two formal statements and the classical/formal dependency comparison in this note reflect the current production architecture.

2.20 Conclusion

Proposition XI.2 is the first explicit plane-existence theorem of the Euclidean solid-geometry sequence.

Its two conclusions are

$$l \neq m, \quad P \in l \cap m \quad \Longrightarrow \quad \exists \pi,\ l, m \subset \pi,$$

and

$$A, B, C \text{ noncollinear} \quad \Longrightarrow \quad \exists \pi,\ A, B, C \in \pi.$$

The historical proof tries to obtain these conclusions through XI.1 while speaking of a reference plane that has not itself been produced. The Hilbert reconstruction resolves the issue at the incidence level: noncollinear points determine a plane, two plane points force their whole line into the plane, and three noncollinear common plane points determine that plane uniquely.

The public Euclid theorem is therefore short, but unlike XI.1 its first clause sits on top of a genuine derived spatial theorem. That stronger theorem is the correct reusable form for the remainder of the formal development.

Chapter 3

Proposition XI.3

3.1 Introduction

Proposition XI.3 states that if two distinct planes meet, then their common section is a straight line. In Euclid's sequence this is the natural continuation of XI.2: once planes containing spatial configurations have been introduced, one must also know how two such planes intersect.

At first sight the proposition seems elementary. In fact it is one of the first explicitly dimension-sensitive steps of Book XI. In genuine three-dimensional incidence geometry, two distinct planes with a common point intersect in a line. In higher-dimensional ambient settings this need not be true: two distinct 2-planes may meet in a single point only. Thus XI.3 is not merely a convenient diagrammatic fact. It is a structural property of the three-dimensional spatial layer.

The production file is

```
CGJteamLab/Proposition11_3.lean
```

and the decisive reusable theorem is

```
hilbert_plane_intersection_line
```

from

```
CGJteamLab/Hilbert3DInterface.lean
```

3.2 Euclid's Statement

Euclid states:

> If two planes cut one another, then their common section is a straight line.

The key phrase is "common section". In modern incidence language this means the full intersection of the two planes, not merely the existence of one line lying in both.

The present formal reconstruction therefore states XI.3 extensionally: it returns a line l contained in both planes and proves that a point belongs to both planes if and only if it lies on l.

3.3 Classical Configuration

Let π and ρ be two distinct planes having a common point A.

The intended conclusion is that there exists a line l such that

$$A \in l, \qquad l \subset \pi, \qquad l \subset \rho,$$

and indeed that the whole common section satisfies

$$\pi \cap \rho = l$$

in the extensional incidence sense.

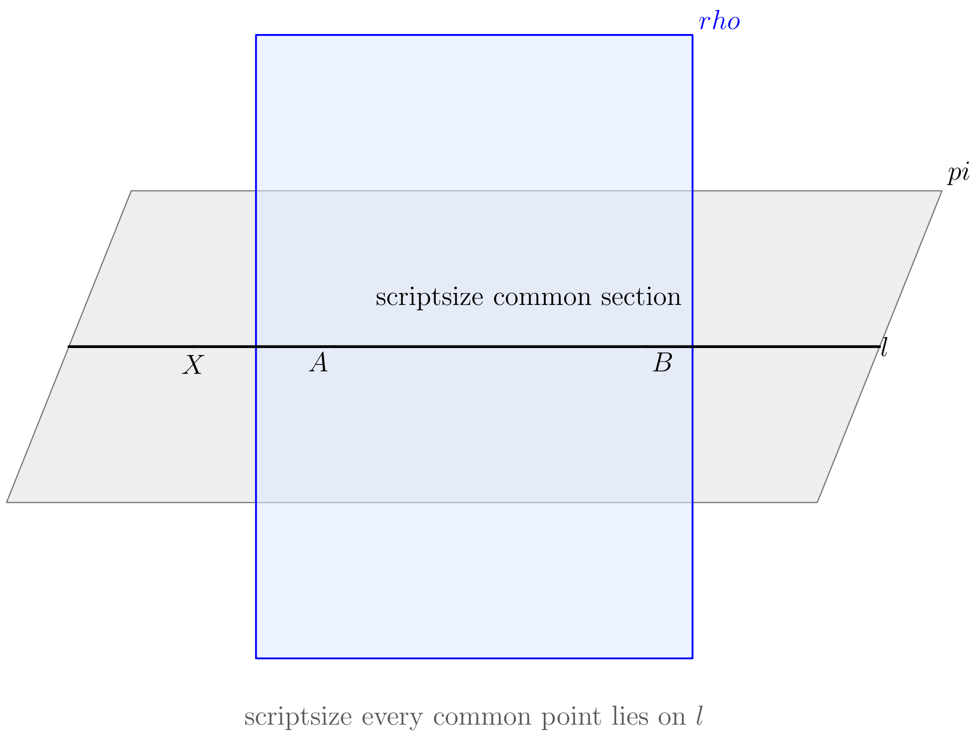


Figure 3.1: The incidence configuration of Proposition XI.3. The two distinct planes π and ρ have the common point A. Their common section is the black line l. Any point belonging simultaneously to π and ρ lies on this same line.

The important point is that the line is not background decoration. It is the entire common section to be proved.

3.4 Euclid's Classical Proof

Euclid argues by choosing another common point of the two planes in addition to the given one, and then taking the straight line through the two common points. He then reasons that any point common to both planes must lie on this same straight line, so the common section is a straight line.

At the historical level the proof is short and intuitive. But once one asks what is required for a formal derivation, two hidden obligations appear:

1. from one common point A, one must obtain a second common point B of the two distinct planes;
2. one must prove not only that some line lies in both planes, but that every common point lies on that line.

These are exactly the obligations discharged by the Hilbert 3D interface.

3.5 Why XI.3 Is Dimension-Sensitive

XI.3 is the first place in Book XI where the three-dimensional nature of the ambient setting becomes mathematically decisive.

In ordinary 3-space, two distinct planes sharing one point must share a whole line. But in a higher-dimensional ambient geometry two distinct 2-planes can intersect in a single point only. Thus the implication

$$A \in \pi \cap \rho, \quad \pi \neq \rho \Longrightarrow \exists l, \ \pi \cap \rho = l$$

is not dimension-free.

In the present project this fact is not left to geometric intuition. It is encoded in the spatial incidence architecture through the principle that two distinct planes with one common point admit a second common point, from which the intersection line is then constructed.

This is the first Book XI proposition whose formal meaning strongly depends on our decision to work in synthetic 3-space rather than in an arbitrary ambient dimension.

3.6 Formal Statement

The production theorem is:

```
theorem euclid_proposition_11_3
    [H : HilbertIncidence Geo]
    [HilbertPlaneIncidence Geo]
    [S : HilbertSpacePrimitive Geo]
    [HSI : HilbertSpaceIncidence Geo]
    (pi rho : S.Plane)
    (hneq : Ne pi rho)
    (A : Geo.Point)
    (hApi : S.OnPlane A pi)
    (hArho : S.OnPlane A rho) :
    exists l : Geo.Line,
      H.OnLine A l /\
      HilbertLineInPlane Geo l pi /\
```

```
      HilbertLineInPlane Geo l rho /\
      forall X : Geo.Point,
        (S.OnPlane X pi /\ S.OnPlane X rho) <->
        H.OnLine X l := by

  exact
    hilbert_plane_intersection_line
      (Geo := Geo)
      pi rho hneq
      A hApi hArho
```

This is a deliberately faithful formalization of the phrase "their common section is a straight line". The theorem does not stop at

$$\exists l,\ l \subset \pi,\ l \subset \rho.$$

It also includes the extensional characterization

$$X \in \pi \cap \rho \Longleftrightarrow X \in l.$$

3.7 The Stronger Intersection Characterization

The theorem used in the proof is not proposition-local infrastructure. It is the general spatial theorem

```
theorem hilbert_plane_intersection_line
    [H : HilbertIncidence Geo]
    [HilbertPlaneIncidence Geo]
    [S : HilbertSpacePrimitive Geo]
    [HilbertSpaceIncidence Geo]
    (pi rho : S.Plane)
    (hneq : Ne pi rho)
    (A : Geo.Point)
    (hApi : S.OnPlane A pi)
    (hArho : S.OnPlane A rho) :
    exists l : Geo.Line,
      H.OnLine A l /\
      HilbertLineInPlane Geo l pi /\
      HilbertLineInPlane Geo l rho /\
      forall X : Geo.Point,
        (S.OnPlane X pi /\ S.OnPlane X rho) <->
        H.OnLine X l
```

Thus XI.3 is a thin Euclid wrapper over a reusable theorem whose conclusion already has exactly the right extensional strength.

3.8 How the Hilbert Theorem Works

The proof of `hilbert_plane_intersection_line` reveals the real incidence structure hidden behind Euclid's short argument.

Starting from the common point A, the theorem first uses the spatial incidence clause

```
plane_second_common_point
```

to obtain a second common point B such that

$$B \neq A, \qquad B \in \pi, \qquad B \in \rho.$$

Then the ordinary two-point line theorem produces the line l through A and B. Since both points lie in π, Hilbert I.6 implies

$$l \subset \pi.$$

The same argument gives

$$l \subset \rho.$$

The remaining task is extensionality. Suppose X lies in both planes. If X were not on l, then A, B, X would be noncollinear, so by plane uniqueness the plane through A, B, X would have to be both π and ρ. This would contradict $\pi \neq \rho$. Therefore every common point X lies on l.

Conversely, if $X \in l$, then the two line-in-plane containments immediately imply $X \in \pi$ and $X \in \rho$.

3.9 Formal Dependency Chain

The real formal dependency pattern is therefore

$$
\begin{gathered}
\pi \neq \rho, \quad A \in \pi, \quad A \in \rho \\
\Downarrow \\
\text{second common point } B \\
\Downarrow \\
A \neq B, \quad A, B \in \pi, \quad A, B \in \rho \\
\Downarrow \\
\text{line } l = AB \\
\Downarrow \\
\text{Hilbert I.6} \\
\Downarrow \\
l \subset \pi, \quad l \subset \rho \\
\Downarrow \\
\text{if } X \in \pi \cap \rho \text{ and } X \notin l, \text{ then } A, B, X \text{ noncollinear} \\
\Downarrow \\
\text{Hilbert I.5} \\
\Downarrow \\
\pi = \rho, \text{ contradiction} \\
\Downarrow \\
X \in l \\
\Downarrow \\
\pi \cap \rho = l.
\end{gathered}
$$

This is the real geometric content behind the short public theorem.

3.10 Spatial Architecture

The theorem uses only the incidence-side spatial structure:

```
HilbertIncidence Geo
HilbertPlaneIncidence Geo
HilbertSpacePrimitive Geo
HilbertSpaceIncidence Geo
```

It does not require

```
HilbertSpaceOrder
HilbertSpaceCongruence
HilbertSpaceEuclidean
PlaneGeo
```

as theorem parameters.

Thus XI.3 still lies fully inside the Group I incidence layer. What makes it subtle is not order or congruence, but the need to express a genuinely three-dimensional intersection theorem at the incidence level.

3.11 Ambient Space and the Absence of PlaneGeo

No induced planar geometry is created in the proof.

All objects remain ambient:

```
pi rho : S.Plane
A X    : Geo.Point
l      : Geo.Line
```

and all relations are ambient incidence relations. The theorem does not enter a slice PlaneGeo(π) or PlaneGeo(ρ) because the result is not a planar theorem inside one chosen plane. It is a statement about the intersection pattern of two ambient planes.

3.12 Wyler-Style Flat Reconstruction

XI.3 is naturally a meet theorem. Once $\pi \neq \rho$ and a common point are known, the flat layer produces a line l whose carrier is exactly the set-theoretic intersection of the two plane carriers:

$$\mathrm{carrier}(\pi) \cap \mathrm{carrier}(\rho) = \mathrm{carrier}(l).$$

This identity is stronger than merely returning one line lying in both planes: it says that every common point is on the same line and that every point of the line is common to both planes.

The proposition is therefore recovered by a single extensional passage:

```
pi != rho,  A in pi cap rho
        |
        v
Meet(pi,rho) = carrier(l)
        |
        v
X in pi and X in rho  <->  X in l
```

In the Wyler-style vocabulary the meet identity is the normal form of XI.3. It also records why the proposition is specifically three-dimensional: in higher rank, the meet of two codimension-one flats need not have the same rank-one form without additional rank information.

3.13 Hidden Formal Data

Several facts that are visually obvious in a traditional diagram must be present as proof data in Lean.

3.13.1 The planes must be distinct

The hypothesis

```
hneq : Ne pi rho
```

is essential. If $\pi = \rho$, then the common section is the whole plane, not a line.

3.13.2 The common point must be explicit

The theorem takes a witness

```
A : Geo.Point
hApi : S.OnPlane A pi
hArho : S.OnPlane A rho
```

rather than a primitive relation saying merely that the planes intersect.

3.13.3 The second common point is nontrivial

Euclid's picture encourages one to imagine the common line immediately. Lean must first prove that a second common point exists. This is the hidden three-dimensional step.

3.13.4 The line must capture all common points

It is not enough to produce one line lying in both planes. The formal conclusion contains the equivalence

```
(S.OnPlane X pi /\ S.OnPlane X rho) <-> H.OnLine X l
```

which states that the line is exactly the whole common section.

3.14 Relationship with XI.1 and XI.2

Classically XI.3 depends on the coplanarity infrastructure established in XI.1–XI.2.

In the formal development the immediate public theorem does not call `euclid_proposition_11_1` or `euclid_proposition_11_2`. Instead, it uses the stronger general theorem `hilbert_plane_intersection_li` whose proof internally consumes the corresponding Hilbert incidence principles.

The conceptual relation is nevertheless clear:

- XI.1 supplies the line-in-plane closure pattern;
- XI.2 supplies the idea that noncollinear or intersecting data determine a plane;
- XI.3 turns from plane determination to plane intersection.

Thus the classical sequence XI.1–XI.3 is reflected formally in the spatial incidence layer, even though later public theorems may call stronger reusable interface results directly.

3.15 Relationship with Later Book XI Propositions

XI.3 is a major infrastructure proposition for the remainder of Book XI.

Whenever a later argument introduces an auxiliary plane and then needs its section with another plane, XI.3 is the conceptual bridge. This becomes especially important in propositions such as XI.13, where the proof passes to the auxiliary plane determined by two lines and then studies its intersection with a reference plane.

In the Lean development, later propositions typically consume the reusable theorem `hilbert_plane_intersecti` rather than the Euclid wrapper `euclid_proposition_11_3`. The mathematical content, however, is exactly XI.3.

3.16 Classical DAG versus Formal DAG

The difference between the historical and formal organizations is again instructive.

Euclid.

$$\begin{array}{c} \pi, \rho \text{ meet} \\ \Downarrow \\ \text{take another common point} \\ \Downarrow \\ \text{join the two common points} \\ \Downarrow \\ \text{any further common point lies on the same join} \\ \Downarrow \\ \text{XI.3.} \end{array}$$

Lean / Hilbert.

$$\begin{array}{c} \pi \neq \rho, \quad A \in \pi \cap \rho \\ \Downarrow \\ \texttt{plane\_second\_common\_point} \\ \Downarrow \\ B \neq A, \quad B \in \pi \cap \rho \\ \Downarrow \\ \texttt{line\_through}\ A, B \\ \Downarrow \\ \texttt{line\_in\_plane}\text{ twice} \\ \Downarrow \\ l \subset \pi, \quad l \subset \rho \\ \Downarrow \\ \texttt{plane\_unique} \\ \Downarrow \\ \forall X,\ X \in \pi \cap \rho \leftrightarrow X \in l \\ \Downarrow \\ \texttt{euclid\_proposition\_11\_3}. \end{array}$$

The formal DAG is longer because Euclid's phrase "common section" has been made fully extensional.

3.17 Structural Lesson

XI.3 illustrates a basic principle of spatial formalization: once several planes are present, their intersections cannot remain visual background. They must be represented by explicit proof objects and characterized extensionally.

This proposition also shows that "working synthetically" does not mean leaving dimensional constraints implicit. On the contrary, the synthetic reconstruction isolates exactly the point where 3-dimensionality matters.

The theorem is still incidence-only, but it is no longer merely elementary bookkeeping. It identifies one of the structural laws that make Book XI possible as a theory of solid geometry rather than a collection of planar pictures drawn in space.

3.18 Audit Status

This documentation was checked against the current production declaration `Geometry.euclid_proposition_11_` and the reusable plane intersection theorem on which it is based. The extensional description of the common section and the classical/formal dependency comparison in this note reflect the current production architecture.

3.19 Conclusion

Proposition XI.3 states that two distinct planes with a common point have a common section which is a straight line.

In the present reconstruction this is expressed in its strongest natural incidence form:

$$\pi \neq \rho, \quad A \in \pi \cap \rho \Longrightarrow \exists l,\ A \in l,\ l \subset \pi,\ l \subset \rho,$$

together with the extensional characterization

$$X \in \pi \cap \rho \Longleftrightarrow X \in l.$$

The proof is short at the public level, but the underlying helper theorem reveals the real 3D structure: from one common point one obtains a second, from two common points one constructs the section line, and plane uniqueness forces every further common point onto that same line.

Thus XI.3 is not just a visual fact about two drawn sheets. It is one of the first genuinely structural theorems of synthetic 3-space in the Book XI development.

Chapter 4

Proposition XI.4

4.1 Introduction

Euclid XI.4 gives the basic synthetic criterion for perpendicularity to a plane: if a line through the common point of two intersecting lines of a plane is perpendicular to each of those two lines, then it is perpendicular to the plane itself.

The proposition is the first major test of the spatial Hilbert architecture. The classical picture suggests a local statement about two right angles, but the conclusion is universal: the candidate line must be perpendicular to *every* line of the plane through the common point. The formal proof therefore has to reconstruct Euclid's arbitrary-transversal argument, keep planar reasoning inside explicit plane slices, and compare triangles that may lie in different planes without introducing coordinates or an ambient vector metric.

The production reconstruction remains synthetic. Its main structural features are the balanced cross, the arbitrary transversal selected by a Pasch argument, spatial SSS/SAS, and the final conversion of the resulting right angle into the universal line–plane perpendicularity predicate.

4.2 Classical Configuration

Let m, n be two distinct straight lines of a plane π, meeting at O, and let l be a line through O perpendicular to both.

The desired conclusion is that l is perpendicular to every line $q \subset \pi$ through O.

4.3 Statement

Theorem 1 (Euclid XI.4)**.** *Let m and n be two distinct lines of a plane π, meeting at O. If a line l through O is perpendicular to both m and n, then l is perpendicular to the plane π.*

Equivalently, the conclusion says that for every line $q \subset \pi$ through O, the line l is perpendicular to q at O. This is exactly the universal content of Euclid XI.Def.3.

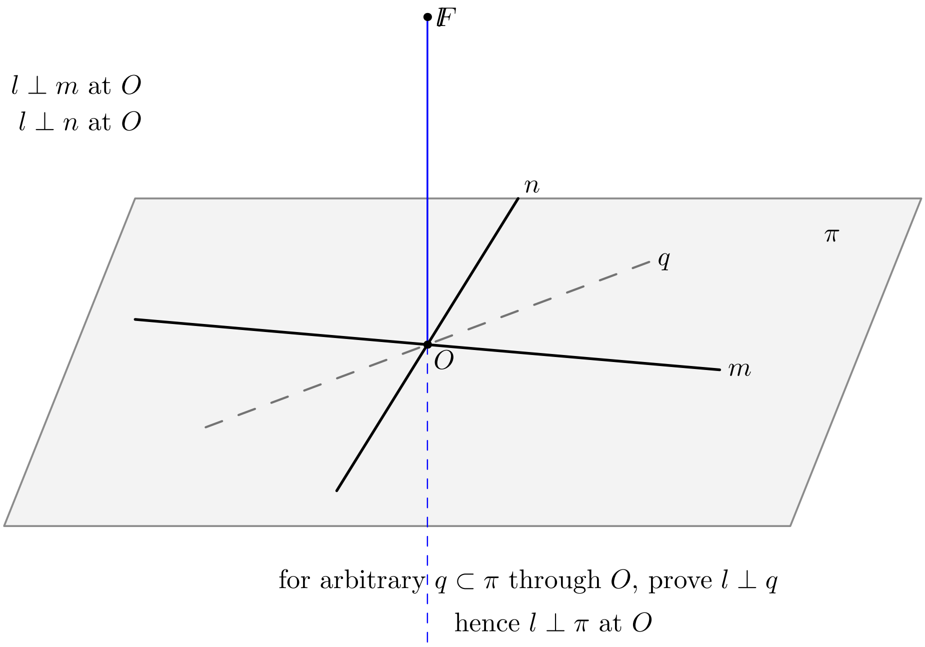


Figure 4.1: Classical configuration for Euclid XI.4. The ambient line l passes through O and is perpendicular at O to two distinct lines m, n lying in the plane π. The dashed line q represents an arbitrary line of π through O, which is the universal target of the conclusion.

4.4 Classical Proof

Euclid makes the argument constructive.

Take equal segments on the two given plane lines:

$$AE = EB = CE = ED,$$

where E is the common point of the two given lines. Draw an arbitrary straight line GEH through E in the plane, choose an arbitrary point F on the elevated perpendicular, and join the auxiliary segments.

The classical dependency chain is:

1. Vertical angles at E are equal (I.15). Together with the four equal radial segments, I.4 gives
$$AD = CB$$
and a corresponding angle equality.

2. Applying I.26 to the planar transversal gives
$$GE = EH, \qquad AG = BH.$$

3. Since FE is perpendicular to both original plane lines, I.4 applied to two right triangles gives
$$FA = FB, \qquad FC = FD.$$

4. I.8 compares two spatial triangles and supplies the angle relation needed to transport the planar data to the arbitrary transversal.

5. A further I.4 gives
$$FG = FH.$$

6. Finally, from
$$GE = EH, \qquad FE = FE, \qquad FG = FH,$$
I.8 yields
$$\angle GEF = \angle HEF.$$
Since $G - E - H$, the adjacent equal angles are right angles.

7. The transversal GEH was arbitrary. Hence FE makes a right angle with every line through E in the plane. By XI.Def.3, $FE \perp \pi$.

This is a genuinely three-dimensional use of Book I congruence theory: several compared triangles are not assumed to lie in one common plane.

4.5 What is genuinely three-dimensional here?

The balanced-cross and transversal constructions live in the reference plane π, but the decisive point F lies on the candidate line l outside that plane. Euclid then compares triangles such as those formed from F and points of the balanced cross. These triangles are not assumed to share one carrier plane.

Accordingly, the reconstruction alternates between two kinds of reasoning: ordinary Hilbert plane geometry inside explicit `PlaneGeo` slices, and spatial congruence arguments comparing triangles carried by different planes. This alternation is the genuinely solid-geometric content of XI.4.

4.6 Formal Target: Euclid XI.Def.3

Book XI, Definition 3 says that a straight line is perpendicular to a plane when it makes right angles with every straight line which meets it and lies in that plane. The project encodes exactly this universal synthetic notion.

For two incidence lines:

```
def HilbertLinesPerpendicularAt
    [H : HilbertIncidence Geo]
    (l m : Geo.Line)
    (O : Geo.Point) : Prop :=
  H.OnLine O l /\
  H.OnLine O m /\
  exists A B : Geo.Point,
    Ne A O /\
    Ne B O /\
    H.OnLine A l /\
    H.OnLine B m /\
    Not (PrimCollinear Geo A O B) /\
    HilbertRightAngle Geo A O B
```

The explicit noncollinearity condition is important. It prevents a degenerate "self-perpendicular" witness and yields the reusable theorem

```
hilbert_linesPerpendicularAt_ne
```

showing that perpendicular lines are distinct.

For a line and a plane:

```
def HilbertLinePerpendicularPlaneAt
    [H : HilbertIncidence Geo]
    [S : HilbertSpacePrimitive Geo]
    (l : Geo.Line)
    (pi : S.Plane)
    (O : Geo.Point) : Prop :=
  H.OnLine O l /\
  S.OnPlane O pi /\
  forall m : Geo.Line,
    HilbertLineInPlane Geo m pi ->
    H.OnLine O m ->
    HilbertLinesPerpendicularAt Geo l m O
```

Thus the conclusion of XI.4 is not a metric abbreviation. It is literally the universal quantification required by Euclid XI.Def.3.

4.7 Formal Statement

The production theorem is:

```
theorem euclid_proposition_11_4
    [H : HilbertIncidence Geo]
    [HilbertPlaneIncidence Geo]
    [S : HilbertSpacePrimitive Geo]
    [HSI : HilbertSpaceIncidence Geo]
    [_HSO : HilbertSpaceOrder
      (Geo := Geo) (H := H) (S := S)]
    [HSC : HilbertSpaceCongruence
      (Geo := Geo) (H := H) (S := S)]
    (pi : S.Plane)
    (m n : PlaneLine Geo pi)
    (l : Geo.Line)
    (O : PlanePoint Geo pi)
    (hmn : Ne m n)
    (hPerpM :
      HilbertLinesPerpendicularAt Geo l m.1 O.1)
    (hPerpN :
      HilbertLinesPerpendicularAt Geo l n.1 O.1) :
    HilbertLinePerpendicularPlaneAt Geo l pi O.1
```

The theorem does not assume separately that $l \neq m$ or $l \neq n$. Those facts follow from the strengthened nondegenerate line–line perpendicularity predicate.

4.8 Spatial / Planar Architecture Used

4.8.1 No ambient planar type-class instance

The ambient geometry is three-dimensional. The proof deliberately does *not* install a global instance

```
HilbertCongruence Geo
```

or

```
HilbertOrder Geo
```

on the whole space. Planar reasoning is performed only in explicit plane slices

$$\mathrm{PlaneGeo}(\pi).$$

The corresponding points and lines are subtypes:

```
PlanePoint Geo pi
PlaneLine  Geo pi
```

and the interface provides bridges between induced planar facts and ambient facts, for example

```
planeGeo_congruent
planeGeo_angleCongruent_iff_ambient
planeGeo_sameRay_iff_ambient
planeGeo_rightAngle_iff_ambient
planeGeo_linesPerpendicularAt_iff_ambient
```

This separation is one of the central architectural features of the proof.

4.8.2 Spatial congruence

Euclid freely applies I.4 and I.8 to triangles in different planes. The project reconstructs this through the spatial congruence layer rather than by coordinates or by an ambient Euclidean metric.

The general spatial infrastructure includes:

```
hilbert_space_congruent_reflexive
hilbert_space_congruent_symmetry
hilbert_space_sas_remaining_angles
hilbert_space_sas_third_side_and_angle
hilbert_space_sss_angleA_in_plane
```

The target triangle may be placed in its own explicit plane and compared with an ambient source triangle. This is the formal counterpart of Euclid's unqualified use of plane triangle congruence inside solid geometry.

4.9 The Balanced Cross

The first construction is the formal analogue of Euclid's

$$AE = EB = CE = ED.$$

The helper

```
planeGeo_opposite_points_on_line_congruent_to
```

constructs opposite equal points on one line. Applying it twice yields

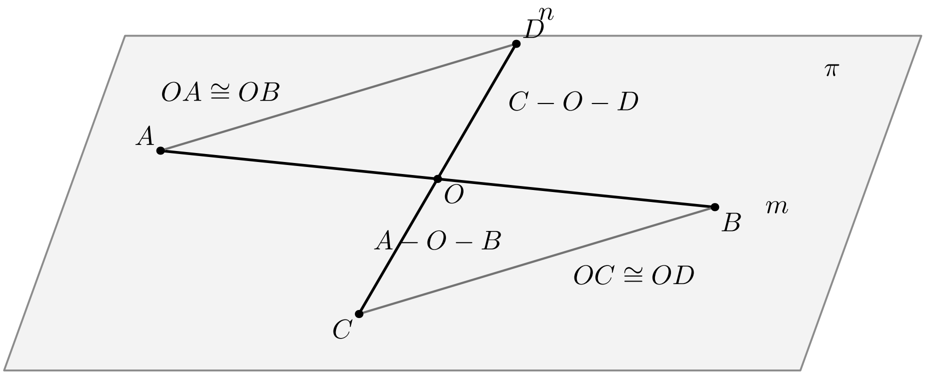
```
hilbert_XI4_balanced_cross_exists
```

and points A, B, C, D satisfying

$$A - O - B, \qquad C - O - D$$

and

$$OA \cong OR, \quad OB \cong OR, \quad OC \cong OR, \quad OD \cong OR.$$



all objects in this figure lie in π

Figure 4.2: Balanced cross in π. Points A, B lie on m, points C, D lie on n, with $A - O - B$, $C - O - D$, $OA \cong OB$, and $OC \cong OD$. The connector segments AD and BC are the first pair used by Euclid's congruence argument.

The first triangle comparison is isolated by

```
planeGeo_XI4_cross_noncollinear
hilbert_XI4_balanced_cross_first_SAS
```

and proves the ambient consequences

$$AD \cong BC, \qquad \angle OAD \cong \angle OBC.$$

This is the formal Hilbert version of Euclid's first use of I.4.

4.10 Fixed-Transversal Prototype

Before the proof was generalized to every transversal, a fixed connector configuration was developed. Its role is mathematically transparent: given

$$A - G - D, \qquad B - J - C, \qquad G - O - J,$$

the balanced cross implies

$$OG \cong OJ, \qquad AG \cong BJ.$$

The relevant planar ingredients are

```
planeGeo_XI4_transversal_noncollinear
hilbert_XI4_transversal_ASA
```

corresponding to Euclid's I.26 step.

The spatial point $F \in l$, together with the two original perpendicularities, then gives

$$AF \cong BF, \qquad CF \cong DF.$$

The proof packages this in

```
hilbert_XI4_spatial_perpendicular_bisector_equidistant
hilbert_XI4_spatial_opposite_pairs_equidistant
```

The spatial SSS/SAS chain is then:

```
hilbert_XI4_spatial_SSS_angle
hilbert_XI4_space_angle_sameRay_first_congruent
hilbert_XI4_spatial_SSS_angle_normalized
hilbert_XI4_spatial_SAS_transversal_side
hilbert_XI4_spatial_transversal_F_equidistant
hilbert_XI4_second_spatial_SSS_angle
hilbert_XI4_right_angle_of_adjacent_equal
```

Its mathematical content is

$$
\begin{gathered}
AD \cong BC, \quad AF \cong BF, \quad CF \cong DF \\
\Downarrow \\
\angle DAF \cong \angle CBF \\
\Downarrow \\
A - G - D,\ B - J - C \\
\Downarrow \\
\angle GAF \cong \angle JBF \\
\Downarrow \\
AG \cong BJ,\ AF \cong BF \\
\Downarrow \\
GF \cong JF \\
\Downarrow \\
OG \cong OJ,\ OF \cong OF,\ GF \cong JF \\
\Downarrow \\
\angle GOF \cong \angle JOF \\
\Downarrow \\
G - O - J \\
\Downarrow \\
\mathrm{RightAngle}(G, O, F).
\end{gathered}
$$

The last implication is exactly Euclid's definition of a right angle as two adjacent equal angles formed by a straight line.

4.11 The Arbitrary Transversal: the Real Formal Difficulty

The public conclusion quantifies over *every* line of π through O. A conventional drawing suggests that such a transversal meets the two connector segments AD and BC, but that is false in general. This was the key positional issue in the formal proof.

The correct argument uses two possible connector families:

$$AD/BC \qquad \text{or} \qquad BD/AC.$$

For a transversal $q \neq m, n$, apply Pasch to the triangle ABD. The line q enters through AB at O, hence it exits either through AD or through BD.

This is isolated as

```
hilbert_XI4_transversal_pasch_first_exit
```

Next,

```
hilbert_XI4_opposite_point_on_transversal
```

constructs the point J opposite G on q with

$$G - O - J, \qquad OG \cong OJ.$$

Two neutral landing lemmas then prove:

```
hilbert_XI4_first_exit_opposite_lands_on_BC
hilbert_XI4_second_exit_opposite_lands_on_AC
```

so that the opposite point lands on the correct paired connector. The combined package is

```
hilbert_XI4_arbitrary_transversal_connector_package
```

and returns

$$G, J \in q, \quad G - O - J, \quad OG \cong OJ,$$
$$\text{and either}$$
$$A - G - D,\ B - J - C,$$
$$\text{or}$$
$$B - G - D,\ A - J - C.$$

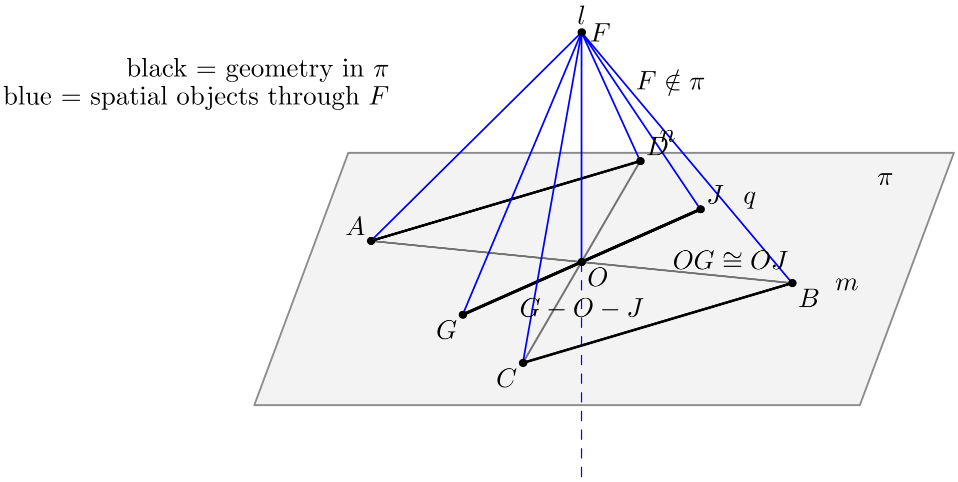


Figure 4.3: Arbitrary-transversal stage. For $q \subset \pi$ through O, Pasch selects one of the two connector families. The opposite point J is constructed with $G - O - J$ and $OG \cong OJ$. The spatial SSS/SAS chain then proves that $\angle GOF$ is right.

This entire stage uses incidence, order, Pasch, and congruence. It does not use the Euclidean parallel axiom.

4.12 Right Angle on Every Transversal

With the connector package available, both branches reduce to the same spatial SSS/SAS scheme. The theorem

```
hilbert_XI4_arbitrary_transversal_right_angle
```

states, in substance:

$$q \subset \pi, \quad O \in q, \quad q \neq m, n,$$
$$F \notin \pi, \quad AF \cong BF, \quad CF \cong DF$$

imply the existence of $G \in q$, $G \neq O$, such that

$$\text{RightAngle}(G, O, F).$$

The two exceptional cases $q = m$ and $q = n$ are discharged directly from the hypotheses of XI.4.

4.13 Why the Point F Is Outside the Plane

The core proof chooses an arbitrary nonvertex point F on l. To use spatial noncollinearity cleanly, it proves

$$F \notin \pi.$$

This is the theorem

```
hilbert_XI4_perpendicular_point_off_plane
```

The proof is neutral and conceptually important.

Assume $F \in \pi$. Then the line l, determined by O and F, is itself a line of π. Transfer the two ambient perpendicularities to `PlaneGeo pi`. Choose nonvertex points $A \in m$ and $C \in n$. Then both

$$\angle FOA \quad \text{and} \quad \angle FOC$$

are right.

The helper

```
hilbert_XI4_two_right_angles_same_first_arm_collinear
```

uses Hilbert III.4 uniqueness to prove that A, O, C are collinear. Line uniqueness then forces $m = n$, contradicting the hypothesis.

No parallel postulate is used.

4.14 Closing the Universal Line–Plane Statement

The core theorem

```
hilbert_XI4_line_perpendicular_plane_core
```

does the following:

1. choose $F \neq O$ on l;
2. construct the balanced cross A, B, C, D in π;
3. prove $F \notin \pi$;
4. prove $AF \cong BF$ and $CF \cong DF$;
5. let q be an arbitrary line of π through O;
6. if $q = m$ or $q = n$, use the hypotheses;

7. otherwise invoke the arbitrary-transversal theorem and obtain a point $G \in q$ such that $\angle GOF$ is right;

8. construct the witness required by `HilbertLinesPerpendicularAt Geo l q O`.

There is one subtle final representation step. The right angle obtained from the transversal theorem is oriented as

$$\angle GOF.$$

The line-line perpendicularity witness is more convenient with the order

$$\angle FOG.$$

The proof does not use an ambient planar congruence instance to swap the arms. Since l and q intersect at O, they determine an explicit plane ρ. Inside `PlaneGeo rho` the right angle is swapped using the ordinary planar congruence API, and the result is transported back to the ambient geometry.

This preserves the 3D architecture: all planar congruence reasoning occurs inside an explicit plane slice.

4.15 Wyler-Style Flat Reconstruction

XI.4 is the first proposition in which the flat language supplies only the incidence skeleton and the metric layer performs the decisive step. The two distinct plane lines m, n through O are rank-two generators of the reference plane. Conceptually,

$$\mathrm{Join}(\mathrm{carrier}(m), \mathrm{carrier}(n)) = \mathrm{carrier}(\pi).$$

The theorem then says that perpendicularity to two independent generating directions propagates to the whole generated plane.

The proof architecture can therefore be compressed to

```
m,n distinct generators of pi
l perp m at O
l perp n at O
        |
        +-- normalize l != m, l != n
        +-- normalize O in l,m,n
        |
        v
XI.4 metric core
        |
        v
l perp pi at O
```

This is not a theorem of flats alone: join tells us which plane is generated, but no closure axiom turns line–line perpendicularity into line–plane perpendicularity. The proposition therefore marks the first clean interface between the incidence calculus and the spatial congruence layer.

4.16 Formalization Notes

4.16.1 The proof is synthetic

There are no coordinates, vectors, scalar products, normal vectors, trigonometric functions, angle measures, or real-valued lengths.

All claims are expressed through:

- incidence of points, lines, and planes;
- strict betweenness;
- same-ray and side-of-line relations;
- segment and angle congruence;
- Hilbert right angles;
- Pasch;
- planar and spatial SAS/SSS/ASA arguments.

4.16.2 The point F is not a normal-vector surrogate

The point F is merely a nonvertex point chosen on the candidate line l. It is used to form ordinary synthetic triangles such as AOF, BOF, GOF, and JOF. The proof never assigns coordinates to F and never interprets OF as a vector.

4.16.3 Why `PlaneGeo` is essential

Some arguments are intrinsically planar: Pasch, same-side uniqueness of an angle copy, same-ray normalization, and ordinary Hilbert angle construction. Other arguments compare triangles in different planes. The proof therefore alternates explicitly between:

$$\text{ambient 3D geometry} \quad \longleftrightarrow \quad \text{PlaneGeo}(\pi) \quad \text{or} \quad \text{PlaneGeo}(\rho).$$

This is not implementation noise. It records exactly where Euclid is using plane geometry inside a solid-geometric proof.

4.16.4 Hidden positional data

The formal proof makes explicit all of the following facts:

- the two given plane lines are genuinely distinct;
- perpendicularity witnesses are nondegenerate;
- the four balanced points lie on the intended carriers;
- $A - O - B$ and $C - O - D$ are strict orders;

- the auxiliary point F is outside π;
- a general transversal need not meet the fixed pair AD, BC;
- Pasch selects one of two connector families;
- the opposite point on the transversal lies on the paired connector;
- every spatial triangle used by SSS or SAS is noncollinear;
- the final arm swap is performed in the explicit plane through l and the arbitrary transversal.

These facts are precisely the information supplied visually, but not logically, by a conventional diagram.

4.16.5 Why the two connector families matter

A tempting but incorrect formalization is to assume that every line through O meets both segments AD and BC. The diagram can make this appear plausible, but it is not true.

The final proof instead analyzes the first exit of the transversal from triangle ABD. This produces exactly the dichotomy required by the geometry:

$$(AD, BC) \quad \text{or} \quad (BD, AC).$$

This is one of the most significant proof-design corrections in the XI.4 development.

4.17 Reusable Infrastructure Produced by XI.4

The proposition required one general correction to the 3D interface: line-line perpendicularity now carries an explicit noncollinearity witness, so that distinctness of the carriers is a theorem rather than an external hypothesis.

The general 3D interface also exposes the bridges

```
HilbertLinesPerpendicularAt
hilbert_linesPerpendicularAt_ne
planeGeo_linesPerpendicularAt_iff_ambient
HilbertLinePerpendicularPlaneAt
HilbertLinePerpendicularPlaneAt.perpendicular_to_line
HilbertLinePerpendicularPlaneAt.incidence
```

The remaining declarations beginning with `hilbert_XI4_` are kept in `Proposition11_4.lean` because they package the concrete Euclidean XI.4 configuration.

4.18 Classical Dependencies versus Lean Dependencies

Role	Euclid	Lean reconstruction
Four equal radial segments	cut off AE, EB, CE, ED equal	opposite-point construction on each of the two lines inside `PlaneGeo pi`
Vertical angles	I.15	developed vertical-angle and angle-congruence infrastructure
First triangle congruence	I.4	balanced-cross SAS helper
Transversal equalities	I.26	Pasch, transversal ASA, and the two landing branches
Distances from F	I.4 on right triangles	spatial perpendicular-bisector/equidistance helper
Cross-plane triangle comparison	I.8 and I.4	general spatial SSS/SAS API through explicit carrier planes
Arbitrary line in the plane	"drawn at random"	universal q; cases $q = m$, $q = n$, or the Pasch branch
Line perpendicular to plane	XI.Def.3	the universal line–plane perpendicularity predicate

The Lean DAG is longer because it must prove all incidence, order, noncollinearity, carrier-plane, and orientation facts that the classical diagram leaves implicit. The geometric argument itself remains Euclidean XI.4.

4.19 Complete Formal DAG

The proof architecture can be summarized as follows.

```
two distinct plane lines m,n through O
+ l perpendicular to m and n at O
-> balanced cross A-O-B, C-O-D
-> AD  = BC and angle OAD  = angle OBC
-> choose F != O on l
-> F notin pi
-> AF  = BF, CF  = DF
-> arbitrary q in pi through O
-> if q=m or q=n:  hypothesis
-> otherwise Pasch in triangle ABD
-> exit through AD or BD
-> opposite J on q, G-O-J, OG  = OJ
-> landing on BC or AC
-> ASA: AG  = BJ (or BG  = AJ)
-> spatial SSS/SAS: GF  = JF
-> spatial SSS: angle GOF  = angle JOF
-> G-O-J -> RightAngle GOF
-> l perpendicular q at O
-> forall q in pi through O, l perpendicular q
-> HilbertLinePerpendicularPlaneAt Geo l pi O.
```

4.20 Dependency Summary

$$\begin{gathered}
\text{Hilbert spatial incidence + order + congruence} \\
\Downarrow \\
\text{PlaneGeo slices and ambient/plane bridges} \\
\Downarrow \\
\text{nondegenerate line-line perpendicularity} \\
\Downarrow \\
\text{balanced cross } A - O - B,\ C - O - D \\
\Downarrow \\
AD \cong BC,\ \angle OAD \cong \angle OBC \\
\Downarrow \\
F \in l,\ F \neq O,\ F \notin \pi \\
\Downarrow \\
AF \cong BF,\ CF \cong DF \\
\Downarrow \\
\text{Pasch for arbitrary } q \subset \pi \text{ through } O \\
\Downarrow \\
\text{two connector branches + opposite point } J \\
\Downarrow \\
GF \cong JF, \quad \angle GOF \cong \angle JOF \\
\Downarrow \\
\text{RightAngle}(G, O, F) \\
\Downarrow \\
l \perp q \text{ for every } q \subset \pi \text{ through } O \\
\Downarrow \\
\texttt{euclid_proposition_11_4}.
\end{gathered}$$

4.21 Relationship with the Foundations

The public theorem uses spatial incidence, order, and congruence together with induced Hilbert geometry on explicit plane slices. It does not assume Hilbert Group IV: no Euclidean parallel axiom is used in the proof of XI.4. No continuity principle is needed either.

This is mathematically significant. The arbitrary-transversal argument is closed by Pasch and congruence, not by parallel-line theory. The theorem is therefore neutral in the sense relevant to the Hilbert hierarchy used by the project.

4.22 Audit Status

This documentation was checked against the current production theorem `Geometry.euclid_proposition_11_4`, the spatial perpendicularity predicates used in its statement, and the current Book XI figures. The classical/formal dependency comparison and the description of the two connector branches reflect the current production architecture.

4.23 Conclusion

Euclid XI.4 is the first major test of the project's spatial Hilbert architecture. The proposition begins with only two line-line perpendicularities at a common point and must end with a universal statement about every line of a plane through that point.

The formal proof shows that this transition can be reconstructed synthetically. The essential tools are not coordinates or vector orthogonality, but the same ingredients visible in Euclid's

Lean exposes two pieces of structure that the classical diagram conceals. First, the arbitrary transversal requires a genuine Pasch argument with two connector families. Second, the repeated uses of Book I congruence in solid geometry require an explicit mechanism for comparing triangles carried by different planes.

Once these points are made explicit, the final statement is exactly Euclid XI.Def.3: the candidate line is perpendicular to every line of the given plane through the common point. In that sense the proof is not a modern replacement of XI.4 but a detailed synthetic reconstruction of its logical content.

Chapter 5

Proposition XI.5

5.1 Introduction

Proposition XI.5 is the first Book XI proposition whose conclusion is a genuinely spatial coplanarity statement forced by perpendicularity.

A line l passes through a point O and is perpendicular there to three lines m, n, p. Two of these lines, m and n, are assumed distinct. The conclusion is that m, n, p lie in one plane.

The formal reconstruction is not a line-by-line transcription of Euclid's proof. The current spatial interface makes it possible to use Proposition XI.4 twice and thereby obtain a short contradiction argument.

5.2 Euclid's Statement

Euclid's Proposition XI.5 states:

> If a straight line is set up at right angles to three straight lines which meet one another at their common point of section, then the three straight lines lie in one plane.

In Euclid's notation the common perpendicular is AB, and the three lines through the common point B are BC, BD, and BE.

5.3 Classical Configuration

Euclid assumes, for contradiction, that two of the three lines, say BD and BE, lie in a reference plane while BC lies in a more elevated plane.

He then considers the plane through AB and BC. Its intersection with the reference plane is a line BF. Thus AB, BC, BF lie in one plane.

At the same time, since AB is perpendicular to BD and BE, Proposition XI.4 implies that AB is perpendicular to the reference plane. Therefore AB is also perpendicular to BF.

The contradiction is that, in the plane through AB and BC, the lines BC and BF would both form a right angle with AB at the same point.

5.4 Classical Proof

The local classical dependency pattern is:

$$\text{XI.3} \longrightarrow \text{intersection line } BF,$$

$$\text{XI.4} \longrightarrow AB \perp (\text{reference plane}),$$

followed by Euclid's Definition XI.3, which turns line–plane perpendicularity into perpendicularity to every line of the plane meeting the perpendicular.

Thus the classical proof is a contradiction proof based on the intersection of two planes and on the impossibility of two distinct perpendiculars to the same line at the same point in one plane.

5.5 What is genuinely three-dimensional here?

The essential spatial content is not the elementary fact about right angles. It is the management of planes.

The proof must establish:

- a plane through two intersecting lines;
- a second plane through another pair of intersecting lines;
- that these two planes are distinct;
- a line of intersection of the two planes;
- containment of relevant lines in the appropriate planes.

The current Lean proof does not open a new planar `PlaneGeo` argument of its own. Instead, its planar geometric content is inherited through the already proved Proposition XI.4.

5.6 Formal Statement

The production theorem is:

```
theorem euclid_proposition_11_5
    [H : HilbertIncidence Geo]
    [HilbertPlaneIncidence Geo]
    [S : HilbertSpacePrimitive Geo]
    [HSI : HilbertSpaceIncidence Geo]
    [_HSO : HilbertSpaceOrder
      (Geo := Geo) (H := H) (S := S)]
```

```
    [_HSC : HilbertSpaceCongruence
      (Geo := Geo) (H := H) (S := S)]
    (l m n p : Geo.Line)
    (O : Geo.Point)
    (hmn : Ne m n)
    (hPerpM :
      HilbertLinesPerpendicularAt Geo l m O)
    (hPerpN :
      HilbertLinesPerpendicularAt Geo l n O)
    (hPerpP :
      HilbertLinesPerpendicularAt Geo l p O) :
    exists pi : S.Plane,
      HilbertLineInPlane Geo m pi /\
      HilbertLineInPlane Geo n pi /\
      HilbertLineInPlane Geo p pi
```

The conclusion is constructive at the level of incidence: Lean returns an explicit plane π containing all three lines.

The hypothesis `hmn : Ne m n` makes explicit a point that is usually left implicit in the classical diagram. The two intersecting lines m, n must be distinct in order to determine the reference plane.

5.7 Spatial / Planar Architecture Used

The proposition uses the ambient three-dimensional layer:

- `HilbertSpacePrimitive` for planes and point–plane incidence;
- `HilbertSpaceIncidence` for spatial plane constructions;
- `HilbertLineInPlane` for line–plane containment;
- `HilbertLinesPerpendicularAt` for spatial line–line perpendicularity;
- `HilbertLinePerpendicularPlaneAt` for line–plane perpendicularity.

The ambient geometry does not install the planar order and congruence classes as global instances on the spatial object. The architectural distinction is:

```
-- not global ambient instances:
HilbertOrder Geo
HilbertCongruence Geo

-- spatial layers:
HilbertSpaceOrder
HilbertSpaceCongruence
```

Planar order and congruence enter only through the established `PlaneGeo` machinery used inside Proposition XI.4.

5.8 Proof Architecture of the Reconstruction

Let m and n be the first two lines perpendicular to l at O.

5.8.1 Step 1: construct the reference plane

Since $m \neq n$ and both pass through O, the theorem

```
hilbert_plane_through_two_intersecting_lines
```

constructs a plane π containing m and n.

5.8.2 Step 2: apply Proposition XI.4

The lines m and n are packaged as `PlaneLine`s of π, and O as a `PlanePoint` of π.

Proposition XI.4 then yields

$$l \perp \pi \quad \text{at } O.$$

In Lean this is the call to

```
euclid_proposition_11_4
```

with the two given perpendicularities $l \perp m$ and $l \perp n$.

5.8.3 Step 3: split on whether p already lies in π

If

$$p \subset \pi,$$

the theorem is finished immediately.

Assume therefore

$$p \not\subset \pi.$$

Since $l \perp p$, the lines l and p are distinct. They meet at O, so they determine another plane σ.

5.8.4 Step 4: intersect the two planes

The planes π and σ must be distinct. If $\pi = \sigma$, then $p \subset \sigma$ would imply $p \subset \pi$, contrary to the case assumption.

The theorem

```
hilbert_plane_intersection_line
```

therefore gives a line q through O contained in both planes:

$$q \subset \pi, \qquad q \subset \sigma.$$

Since $l \perp \pi$ and q is a line of π through O,

$$l \perp q.$$

This is obtained directly from

```
HilbertLinePerpendicularPlaneAt.perpendicular_to_line
```

5.8.5 Step 5: the second use of XI.4

Now p and q both lie in σ, and

$$l \perp p, \qquad l \perp q.$$

Suppose $p \neq q$. Applying Proposition XI.4 inside σ gives

$$l \perp \sigma.$$

But $l \subset \sigma$ by construction. This is impossible.

Hence

$$p = q.$$

Since $q \subset \pi$, it follows that $p \subset \pi$, and therefore

$$m, n, p \subset \pi.$$

5.9 Wyler-Style Flat Reconstruction

XI.5 is the first proposition whose complete synthetic structure is naturally expressed by both join and meet. The three lines m, n, p are all perpendicular to the same line l at O, and $m \neq n$.

First m, n generate the reference plane:

$$\mathrm{Join}(m, n) = \pi.$$

XI.4 gives $l \perp \pi$. If $p \subseteq \pi$, the proof is finished. In the nontrivial branch, l and p generate a second plane σ, and π and σ meet in a line q through O. Since $q \subseteq \pi$, one has $l \perp q$. If $p \neq q$,

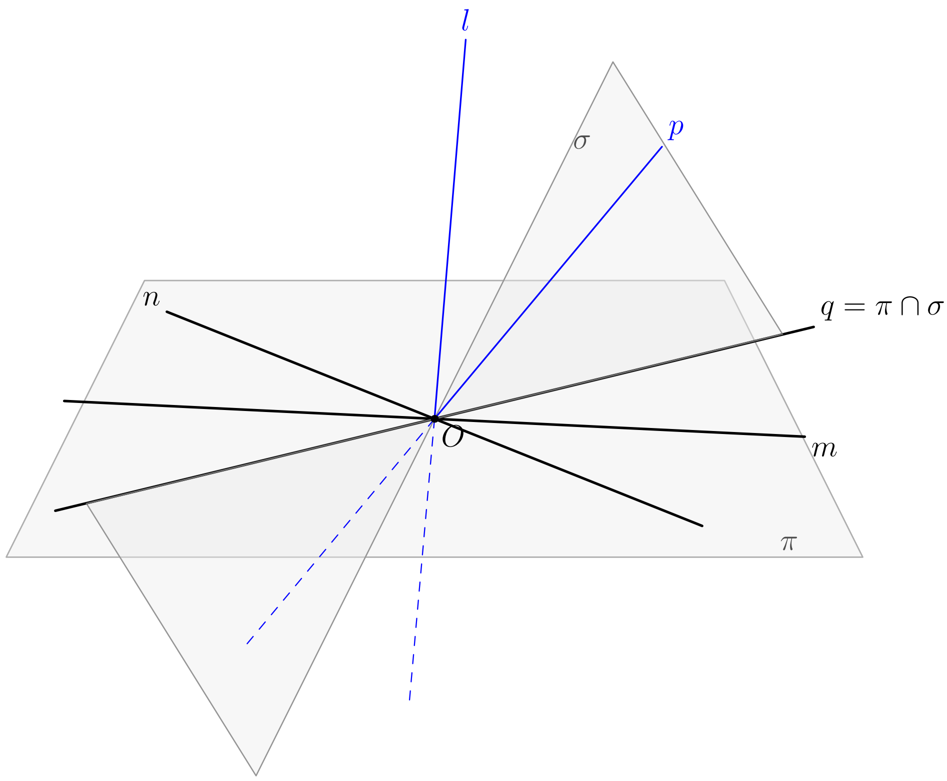


Figure 5.1: The formal contradiction argument for Proposition XI.5. The black lines m, n, q lie in the reference plane π. The spatial lines l, p are blue. Under the temporary assumption $p \not\subset \pi$, the lines l, p determine the second plane σ, and $q = \pi \cap \sigma$. If $p \neq q$, a second application of XI.4 would make l perpendicular to σ, although $l \subset \sigma$.

then the two directions p, q generate σ, and XI.4 would force $l \perp \sigma$; this contradicts $l \subseteq \sigma$. Hence $p = q$, so $p \subseteq \pi$.

The complete structural graph is

```
m,n --join--> pi
 |             |
 |             +-- XI.4 --> l perp pi
 |
p not subset pi
        |
l,p --join--> sigma
        |
pi,sigma --meet--> q
        |
l perp q
        |
assume p != q
        |
XI.4 in sigma --> l perp sigma
        |
l subset sigma
        |
contradiction
        |
p = q subset pi
```

The point of the reformulation is not merely shorter code. The auxiliary planes and the section line become canonical generated objects rather than anonymous witnesses inside a long incidence argument.

5.10 Lean Formalization

The small proposition-local helper is:

```
theorem hilbert_linePerpendicularPlaneAt_not_in_plane
    [H : HilbertIncidence Geo]
    [S : HilbertSpacePrimitive Geo]
    (l : Geo.Line)
    (pi : S.Plane)
    (O : Geo.Point)
    (hPerp :
      HilbertLinePerpendicularPlaneAt Geo l pi O) :
    Not (HilbertLineInPlane Geo l pi) := by

  intro hlpi

  have hSelf :
      HilbertLinesPerpendicularAt Geo l l O :=
    HilbertLinePerpendicularPlaneAt.perpendicular_to_line
      (Geo := Geo)
      hPerp
      hlpi
      hPerp.1
```

```
exact
  (hilbert_linesPerpendicularAt_ne
    Geo l l O hSelf) rfl
```

This helper is purely definitional in geometric content. If a line perpendicular to a plane were itself contained in the plane, then the universal line–plane perpendicularity condition would make the line perpendicular to itself. The predicate `HilbertLinesPerpendicularAt` is nondegenerate, so this is impossible.

The theorem body then consists of two plane constructions, one plane-intersection construction, and two calls to XI.4.

5.11 Hidden Formal Data

Several facts visible in a classical spatial diagram must be supplied explicitly in Lean.

- O lies in the plane determined by m, n.
- The ambient lines m, n must be packaged as `PlaneLine Geo pi`.
- Their distinctness must be transported to the subtype level.
- If $p \not\subset \pi$, the plane through l, p must be constructed explicitly.
- The two planes must be proved distinct before their intersection line can be used.
- The intersection line q must carry proofs that it lies in both π and σ.
- For the second XI.4 call, p, q and O are repackaged as `PlaneLine` and `PlanePoint` objects of σ.

These are mathematical incidence obligations, not merely syntactic casts. They encode information that Euclid's diagram leaves implicit.

5.12 Relationship with Earlier Book XI Propositions

The classical proof explicitly uses XI.3 and XI.4.

The Lean proof imports Proposition XI.4 and uses the current spatial intersection theorem

```
hilbert_plane_intersection_line
```

instead of importing a separate formal Proposition XI.3.

Thus the two dependency graphs are not identical.

The actual Lean Book XI dependency is:

$$\boxed{\text{XI.4}} \longrightarrow \boxed{\text{XI.5}}.$$

Proposition XI.5 is then the coplanarity tool required by the classical route to Proposition XI.6.

5.13 Relationship with Book I / Book Zero / Hilbert Infrastructure

No Book I proposition is called directly by the XI.5 theorem body.

The inherited planar Hilbert and Book Zero infrastructure enters transitively through Proposition XI.4. In particular, XI.4 internally uses induced plane geometry, while XI.5 itself is primarily a spatial incidence argument.

The new helper `hilbert_linePerpendicularPlaneAt_not_in_plane` does not add a geometric axiom; it follows from the existing definitions and nondegeneracy of line–line perpendicularity.

5.14 Relationship with the Foundations

XI.5 is neutral in the current reconstruction. Its proof uses spatial incidence, order, congruence, Proposition XI.4, and explicit plane intersection arguments, but no Hilbert Group IV assumption and no continuity principle.

The important foundational point is therefore architectural rather than axiomatic: coplanarity is never inferred from the picture. The proof must construct or identify the relevant plane and carry the corresponding line-in-plane witnesses explicitly.

5.15 Dependency Summary

The actual formal dependency chain is:

Hilbert incidence / order / congruence infrastructure
↓
spatial primitive / incidence / order / congruence
↓
PlaneGeo and plane–space bridges used by XI.4
↓
`euclid_proposition_11_4`
↓
plane through intersecting lines
intersection line of two planes
`hilbert_linePerpendicularPlaneAt_not_in_plane`
↓
`euclid_proposition_11_5`.

At the current checkpoint the production theorem compiles locally and its axiom audit has been run successfully. A full repository `lake build` is a separate checkpoint and should be recorded only after it is run.

5.16 Conclusion

Proposition XI.5 identifies the plane perpendicular to a given line at a given point as the unique carrier of all lines through that point perpendicular to the line, in the precise three-line form stated by Euclid.

The formal proof is shorter than the classical local proof graph because the present 3D interface already contains an intersection-line theorem for two planes and because XI.4 can be applied a second time to exclude the hypothetical second plane.

No coordinates, vectors, scalar products, continuity principle, or new spatial axiom enter the proof.

Chapter 6

Proposition XI.6

6.1 Introduction

Proposition XI.6 is the first proposition in the present Book XI development whose conclusion is a genuinely spatial parallelism statement. Two ambient lines l and m are perpendicular to the same plane π, at distinct feet B and D. Euclid concludes that the two lines are parallel.

In three-dimensional geometry this conclusion contains two logically different assertions. The lines must first be shown to be *coplanar*; only then can planar geometry prove that they do not meet. This distinction is essential in the formal reconstruction, because two disjoint ambient lines may be skew.

The current Lean proof follows the metric core of Euclid's construction closely:

$$\text{auxiliary perpendicular and equal segment} \longrightarrow \text{SAS} \longrightarrow \text{SSS} \longrightarrow \text{XI.5}.$$

After XI.5 has supplied the missing common plane, the proof changes deliberately to an induced planar geometry and closes the disjointness argument by the neutral alternate-angle criterion.

6.2 Euclid's Statement

Euclid's Proposition XI.6 states:

> If two straight lines be at right angles to the same plane, the straight lines will be parallel.

In the classical configuration the two perpendiculars are AB and CD, meeting the reference plane at B and D. The formal theorem makes the nondegenerate case explicit by assuming

$$B \neq D.$$

6.3 Classical Configuration

Join the feet B and D. In the reference plane construct through D a line DE perpendicular to BD, and choose E so that

$$DE \cong AB.$$

Join BE, AE, and AD.

Because AB is perpendicular to the reference plane, it is perpendicular to every line of that plane through B, in particular to BD and BE. Hence

$$\angle ABD, \qquad \angle ABE$$

are right angles.

Likewise CD is perpendicular to the reference plane at D, so it is perpendicular to both DB and DE.

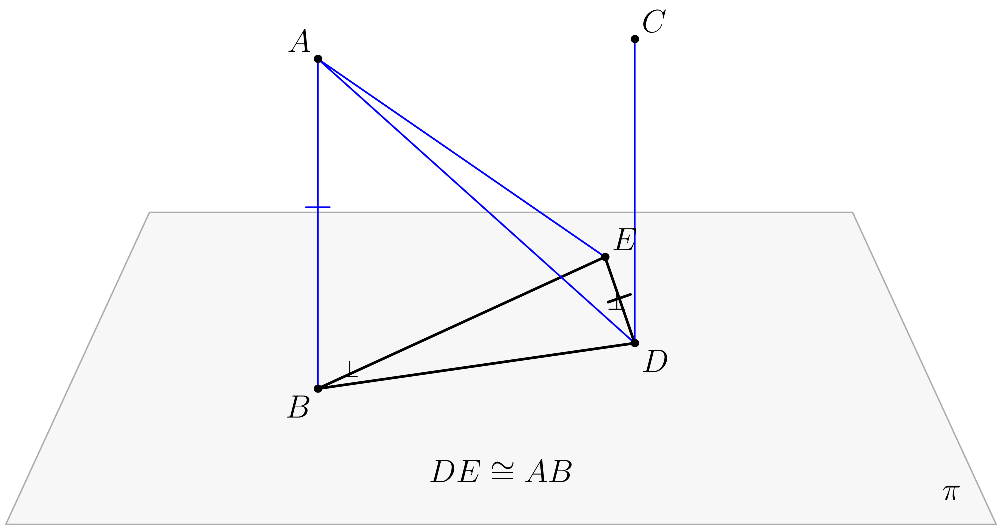


Figure 6.1: The classical configuration of Proposition XI.6. The black geometry lies in the reference plane π: B, D, E, the transversal BD, and the auxiliary perpendicular DE. The blue lines and connectors leave π. The construction $DE \cong AB$ prepares the SAS comparison of BAD and DEB; the resulting equality $AD \cong EB$ prepares the SSS comparison of DEA and BAE.

6.4 Classical Proof

The classical metric part consists of two triangle-congruence steps.

First,

$$AB \cong DE, \qquad BD \cong DB,$$

and the included angles

$$\angle ABD, \qquad \angle EDB$$

are both right. Proposition I.4 therefore gives

$$AD \cong EB.$$

Second, the triangles DEA and BAE have

$$DE \cong BA, \qquad EA \cong AE, \qquad DA \cong BE.$$

Proposition I.8 gives

$$\angle EDA \cong \angle ABE.$$

Since $\angle ABE$ is right, $\angle EDA$ is right; hence

$$ED \perp DA.$$

Now at the point D the line ED is perpendicular to the three lines

$$DB, \qquad DA, \qquad DC.$$

Proposition XI.5 therefore places those three lines in one plane. Proposition XI.2 identifies the triangle plane determined by the intersecting lines, so AB is brought into the same plane as BD and DC. Finally

$$\angle ABD = \angle BDC = 90^\circ,$$

and Proposition I.28 yields

$$AB \parallel CD.$$

A compact classical dependency graph is therefore

$$\begin{gathered}
\text{construct } DE \perp BD, \quad DE \cong AB \\
\Downarrow \\
\text{I.4} \\
\Downarrow \\
AD \cong EB \\
\Downarrow \\
\text{I.8} \\
\Downarrow \\
ED \perp DA \\
\Downarrow \\
\text{XI.5} \\
\Downarrow \\
DB, DA, DC \text{ coplanar} \\
\Downarrow \\
\text{XI.2} \\
\Downarrow \\
AB, BD, DC \text{ coplanar} \\
\Downarrow \\
\text{I.28} \\
\Downarrow \\
AB \parallel CD.
\end{gathered}$$

The construction of the auxiliary perpendicular and equal segment is part of the earlier Book I construction machinery. The substantive local congruence steps are I.4 and I.8.

6.5 What is genuinely three-dimensional here?

The difficult point is not the planar theorem that two perpendiculars to one transversal are parallel. That statement cannot even be applied until one knows that the two candidate parallel lines lie in one plane.

In ambient three-space the information

$$l \perp BD, \qquad m \perp BD$$

does not by itself constitute a parallelism proof: without a coplanarity theorem the lines could still be skew.

The formal proof therefore separates XI.6 into

$$\boxed{\text{coplanarity}} \quad + \quad \boxed{\text{planar disjointness}}.$$

The first box is the genuinely spatial content and is exactly where Proposition XI.5 is used.

6.6 Formal Statement

The production theorem is:

```
theorem euclid_proposition_11_6
    [H : HilbertIncidence Geo]
    [HilbertPlaneIncidence Geo]
    [S : HilbertSpacePrimitive Geo]
    [HSI : HilbertSpaceIncidence Geo]
    [_HSO : HilbertSpaceOrder
      (Geo := Geo) (H := H) (S := S)]
    [HSC : HilbertSpaceCongruence
      (Geo := Geo) (H := H) (S := S)]
    (pi : S.Plane)
    (l m : Geo.Line)
    (B D : Geo.Point)
    (hBD : Ne B D)
    (hLperp :
      HilbertLinePerpendicularPlaneAt
        Geo l pi B)
    (hMperp :
      HilbertLinePerpendicularPlaneAt
        Geo m pi D) :
    HilbertSpaceLinesParallel Geo l m
```

The target relation is part of the reusable spatial interface `Hilbert3DInterface`:

```
def HilbertSpaceLinesParallel
    [H : HilbertIncidence Geo]
    [S : HilbertSpacePrimitive Geo]
    (l m : Geo.Line) : Prop :=
  exists sigma : S.Plane,
    HilbertLineInPlane Geo l sigma /\
    HilbertLineInPlane Geo m sigma /\
    HilbertLinesDisjoint Geo l m
```

Thus "parallel in space" means exactly:

1. there is a plane containing both lines;
2. the two lines are disjoint.

The first clause is not redundant. If spatial parallelism were defined only by ambient disjointness, skew lines would incorrectly satisfy the definition.

6.7 Spatial / Planar Architecture Used

The proof uses three layers which must remain distinct.

6.7.1 Ambient three-space

The ambient objects and relations include

```
HilbertSpacePrimitive
HilbertSpaceIncidence
HilbertSpaceOrder
HilbertSpaceCongruence
HilbertLineInPlane
HilbertLinesPerpendicularAt
HilbertLinePerpendicularPlaneAt
```

In particular, segment congruence between configurations carried by different planes is handled through `HilbertSpaceCongruence`.

6.7.2 Induced plane geometry

Whenever a planar theorem is needed, the relevant ambient plane σ or π is converted into

```
PlanePoint
PlaneLine
PlaneGeo
```

and ordinary planar incidence, order, right-angle, side-of-line, and triangle theorems are used there.

There is deliberately no global ambient instance

```
HilbertOrder Geo
HilbertCongruence Geo
```

for the three-dimensional space.

6.7.3 Plane–space bridges

The proof repeatedly uses bridges such as

```
planeGeo_linesPerpendicularAt_iff_ambient
planeGeo_rightAngle_iff_ambient
planeGeo_angleCongruent_iff_ambient
planeGeo_primCollinear_to_ambient
planeGeo_not_primCollinear_to_ambient
planeGeo_sameRay_iff_ambient
```

These are not cosmetic casts. They state which geometric fact is being used in which plane slice.

6.8 Proof Architecture of the Reconstruction

6.8.1 Step 1: Euclid's auxiliary construction in the reference plane

Choose a point $A \neq B$ on the first normal l. Inside the reference plane π, construct the line

$$d = BD.$$

Through D construct a line e perpendicular to d, and lay off on e a point E such that

$$DE \cong AB.$$

This is packaged by

```
hilbert_XI6_auxiliary_perpendicular_equal_segment
```

The line d and the auxiliary line e are constructed in `PlaneGeo pi`. The segment copied onto DE, however, may have the spatial endpoint A; that copy is therefore performed by `HilbertSpaceCongruence.segment_construction`.

6.8.2 Step 2: the first spatial SAS

Since $l \perp \pi$ at B and $d \subset \pi$ passes through B,

$$l \perp d.$$

The auxiliary construction gives

$$e \perp d$$

at D.

The helper

```
hilbert_XI6_space_linesPerpendicularAt_right_angle_of_points
```

normalizes these line–line perpendicularities to the concrete right angles

$$\angle ABD, \qquad \angle EDB.$$

The spatial right-angle comparison then supplies

$$\angle ABD \cong \angle EDB.$$

With

$$BA \cong DE, \qquad BD \cong DB,$$

the spatial SAS theorem gives

$$AD \cong EB.$$

This whole block is packaged by

```
hilbert_XI6_first_SAS
```

and uses

```
hilbert_space_sas_third_side_and_angle
```

for the actual triangle comparison.

6.8.3 Step 3: the second spatial SSS and the new right angle

The next comparison is between the triangles DEA and BAE. The three side relations are

$$DE \cong BA, \qquad EA \cong AE, \qquad DA \cong BE.$$

The spatial SSS theorem gives

$$\angle EDA \cong \angle ABE.$$

Because $AB \perp \pi$ and $BE \subset \pi$, the angle ABE is right. The rightness is transported to EDA, yielding

$$ED \perp DA.$$

This block is packaged by

```
hilbert_XI6_second_SSS_right_angle
```

The proof must create explicit carrier planes for the relevant triangles. The right-angle transport is handled by

```
hilbert_XI6_space_right_angle_transport_in_plane
```

rather than by installing a forbidden ambient `HilbertCongruence Geo` instance.

6.8.4 Step 4: Proposition XI.5 supplies coplanarity

At D, the line $e = DE$ is now perpendicular to

$$d = DB, \qquad a = DA, \qquad m.$$

The last perpendicularity follows because $m \perp \pi$ and $e \subset \pi$.

Proposition XI.5 is applied exactly here:

```
euclid_proposition_11_5
```

It produces a plane σ containing

$$d, \qquad a, \qquad m.$$

Since $A \in a$ and $B \in d$, both A and B lie in σ. The line $l = AB$ therefore also lies in σ.

The result is isolated as

```
hilbert_XI6_normals_to_same_plane_coplanar
```

with conclusion

```
exists sigma : S.Plane,
  HilbertLineInPlane Geo l sigma /\
  HilbertLineInPlane Geo m sigma
```

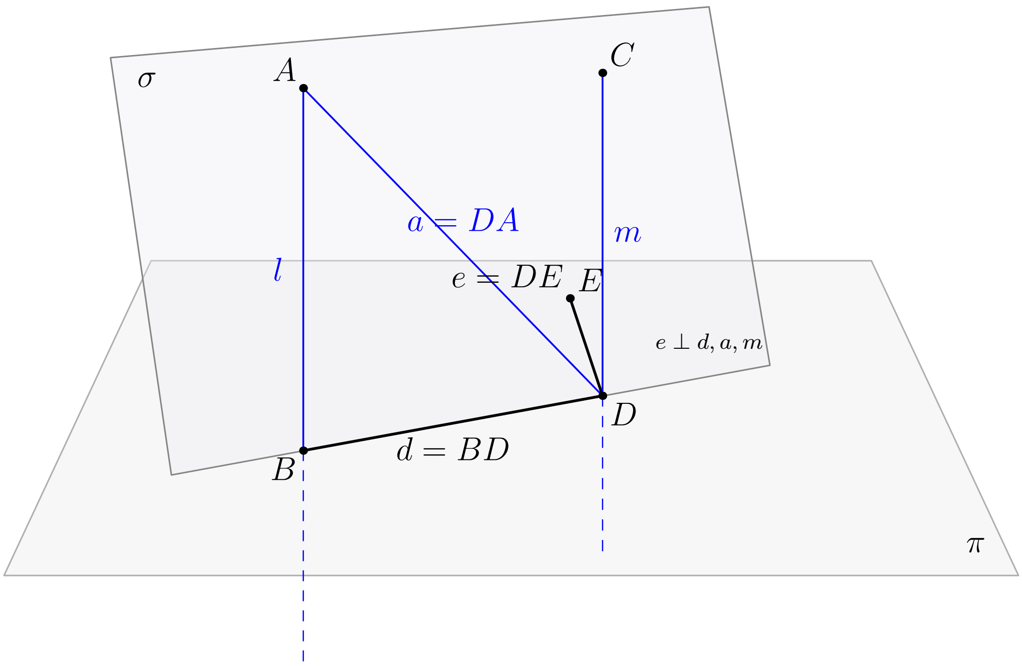


Figure 6.2: The genuinely spatial step in the formal XI.6 proof. The black lines $d = BD$ and $e = DE$ lie in the reference plane π. The blue lines $a = DA$, $m = DC$, and $l = AB$ leave π. At D, the line e is perpendicular to d, a, m, so XI.5 produces the second plane σ containing d, a, m. Because σ contains A and B, it also contains l. The lines l and m are therefore coplanar.

6.8.5 Step 5: the remaining problem is planar

After Step 4, both l and m lie in σ. The common transversal $d = BD$ lies in σ as well, and

$$l \perp d \text{ at } B, \qquad m \perp d \text{ at } D.$$

The rest of the proof is carried out entirely inside

`PlaneGeo Geo sigma`.

The helper

```
hilbert_XI6_coplanar_normals_disjoint
```

chooses nonfoot points on l and m, establishes the required side-of-line orientation relative to d, compares the two right angles, and applies

```
hilbert_parallel_of_alternate_angles_oppositeSide_lines
```

to obtain planar parallelism.

The final conversion proves that the actual ambient carriers l and m are disjoint.

6.8.6 Step 6: assemble spatial parallelism

The public theorem now has exactly the two pieces required by `HilbertSpaceLinesParallel`:

$$l, m \subset \sigma$$

and

$$\text{HilbertLinesDisjoint}(l, m).$$

Therefore

$$\text{HilbertSpaceLinesParallel}(l, m).$$

6.9 Wyler-Style Flat Reconstruction

The target notion of spatial parallelism already decomposes into two geometric components: common-plane incidence and disjointness. XI.6 is therefore read in the flat calculus as a two-stage theorem about two normals to the same plane at distinct feet.

The first stage produces a plane σ containing both normals:

$$l \subseteq \sigma, \qquad m \subseteq \sigma.$$

Equivalently, once $l \neq m$ is known, their join is the carrier of σ. The second stage proves

$$l \cap m = \varnothing.$$

The public parallelism object is exactly the package of these two facts.

In the Wyler-style second pass the theorem is therefore read as

```
l perp pi at B
m perp pi at D
B != D
         |
         +--> common carrier plane sigma
         |
         +--> l,m disjoint
         |
         v
HilbertSpaceLinesParallel l m
```

The deeper metric work that establishes coplanarity and disjointness remains synthetic and is not replaced by lattice theory. The flat language identifies the correct output object: a rank-two carrier containing the two rank-one carriers, together with the separate order/Euclidean fact that they do not meet.

6.10 Lean Formalization

The final theorem body is short because the geometric work has been isolated into two major helpers. First,

```
hilbert_XI6_normals_to_same_plane_coplanar
```

returns a plane `sigma` containing both ambient normals. Then

```
hilbert_XI6_coplanar_normals_disjoint
```

proves that their ambient carriers are disjoint. The public theorem packages exactly these three fields of `HilbertSpaceLinesParallel`: containment of l, containment of m, and disjointness.

The production module imports

```
import CGJteamLab.Proposition11_5
import CGJteamLab.Proposition27
```

The first import supplies the decisive spatial coplanarity theorem. The second supplies the neutral alternate-angle machinery used in the final induced plane.

6.11 Hidden Formal Data

The classical diagram suppresses several obligations which the Lean proof must make explicit.

- The feet B, D are distinct.
- The point A selected on l is outside the reference plane π.
- The auxiliary point E is nondegenerate and lies both on e and in π.
- Every triangle used in SAS or SSS carries an explicit noncollinearity proof.
- Spatial triangle comparisons require an explicit target carrier plane.
- Right-angle transport between ambient configurations is performed through an explicit `PlaneGeo`, not through an ambient planar congruence instance.
- Before XI.5 can be applied, the three lines d, a, m must be proved distinct in the required places and must all pass through D.
- The plane σ returned by XI.5 must be shown to contain both A and B, so that the whole carrier $l = AB$ lies in σ.
- In the final planar proof, the chosen points on l and m must be oriented on opposite sides of the transversal d. The proof performs an explicit SameSide/OppositeSide case split.
- Point-pair parallelism inside `PlaneGeo sigma` must finally be converted into disjointness of the original ambient carriers.

These obligations are geometric. They are the incidence, order, and plane-membership data that a conventional three-dimensional drawing usually leaves implicit.

6.12 Relationship with Earlier Book XI Propositions

Proposition XI.5 is a direct formal dependency:

$$\boxed{\text{XI.5}} \longrightarrow \boxed{\text{XI.6}}.$$

XI.5 itself depends on XI.4, and XI.6 also reuses several XI.4 normalization helpers for perpendicular lines and right angles. Consequently the effective Book XI chain is

$$\boxed{\text{XI.4}} \longrightarrow \boxed{\text{XI.5}} \longrightarrow \boxed{\text{XI.6}}.$$

The classical proof also invokes XI.2 when it moves from the three coplanar lines through D to the plane containing the triangle ABD. The Lean proof does not need a separate formal call to Proposition XI.2: the spatial incidence API directly constructs and tracks the relevant carrier plane.

6.13 Relationship with Book I / Book Zero / Hilbert Infrastructure

The metric part of the formal proof reproduces the roles of Euclid I.4 and I.8 through the spatial SAS and SSS infrastructure:

```
hilbert_space_sas_third_side_and_angle
hilbert_space_sss_angleA_in_plane
```

Right angles and their transport are handled synthetically; no angle measure is introduced.

The classical final step is I.28. The production Lean file instead imports `Proposition27` and uses the established neutral alternate-angle theorem

```
hilbert_parallel_of_alternate_angles_oppositeSide_lines
```

inside the explicitly constructed plane σ. Thus the formal module dependency and the classical Euclid proposition-number dependency are not identical, even though they close the same planar configuration.

Book Zero enters only transitively through the established Hilbert geometric infrastructure. No new Book Zero axiom or theorem is added by XI.6.

6.14 Relationship with the Foundations

XI.6 is also neutral in the current reconstruction. The proof uses spatial incidence, order, congruence, the induced geometry of explicit plane slices, and the earlier perpendicularity results, but it does not invoke Hilbert Group IV or continuity.

A reusable spatial parallelism predicate is introduced at this stage for later propositions. Its role is representational: it packages coplanarity and disjointness of ambient lines without turning the ambient three-space into a globally planar geometry.

6.15 Dependency Summary

The classical local chain is

$$
\begin{array}{c}
\text{Book I perpendicular/equal-segment construction} \\
\downarrow \\
\text{I.4 (SAS)} \\
\downarrow \\
\text{I.8 (SSS)} \\
\downarrow \\
\text{XI.5} \\
\downarrow \\
\text{XI.2} \\
\downarrow \\
\text{I.28} \\
\downarrow \\
\text{XI.6.}
\end{array}
$$

The actual formal chain is

$$
\begin{array}{c}
\text{Hilbert incidence / order / congruence infrastructure} \\
\downarrow \\
\text{spatial primitive / incidence / order / congruence} \\
\downarrow \\
\text{PlaneGeo slices and plane–space bridges} \\
\downarrow \\
\text{XI.4 right-angle and perpendicular normalizers} \\
\downarrow \\
\text{auxiliary } d = BD,\ e \perp d,\ DE \cong AB \\
\downarrow \\
\text{spatial SAS: } AD \cong EB \\
\downarrow \\
\text{spatial SSS: } ED \perp DA \\
\downarrow \\
\texttt{euclid\_proposition\_11\_5} \\
\downarrow \\
l, m \text{ coplanar in } \sigma \\
\downarrow \\
\text{PlaneGeo}(\sigma) \text{ side/order + alternate angles} \\
\downarrow \\
\text{ambient carrier disjointness} \\
\downarrow \\
\texttt{HilbertSpaceLinesParallel} \\
\downarrow \\
\texttt{euclid\_proposition\_11\_6}.
\end{array}
$$

6.16 Conclusion

Proposition XI.6 is a useful boundary theorem for the three-dimensional architecture. Its metric core is recognizably Euclidean: construct an auxiliary perpendicular and an equal segment, use SAS, then SSS, and obtain a third perpendicular at D.

The genuinely spatial step is XI.5. It converts three perpendicularities through one point into the common plane in which the final parallelism argument is allowed to take place. Lean makes this logical boundary explicit rather than allowing the diagram to hide it.

The final definition of spatial parallelism records the same distinction: coplanarity is proved first and disjointness second. No coordinates, vectors, scalar products, numerical angle measure, continuity principle, new spatial axiom, or Euclidean parallel axiom is introduced.

Chapter 7

Proposition XI.7

7.1 Introduction

Proposition XI.7 is an incidence proposition about spatial parallelism. Two ambient straight lines are parallel, points are chosen on the two lines, and Euclid asserts that the straight line joining the chosen points lies in the same plane as the parallels.

The formal reconstruction is substantially shorter than XI.4–XI.6. The reason is structural. In the current spatial interface, `HilbertSpaceLinesParallel` already contains an explicit common carrier plane. Once the two chosen points are shown to lie in that plane, Hilbert's spatial incidence axiom I.6 places their joining line in the same plane.

Thus the formal proof is not a line-by-line translation of Euclid's reductio. It proves the same geometric target through the stronger and more explicit incidence organization adopted by Hilbert.

7.2 Euclid's Statement

Euclid's Proposition XI.7 states:

> If two straight lines be parallel and points be taken at random on each of them, the straight line joining the points is in the same plane with the parallel straight lines.

Let AB and CD be parallel straight lines. Choose E on AB and F on CD. The claim is that the joining straight line EF lies in the plane of AB and CD.

7.3 Classical Configuration

The classical configuration begins with the plane containing the two parallel lines AB and CD. The points E and F are arbitrary points on the two lines.

The proposition is genuinely spatial not because the final figure contains a line outside the plane, but because the assertion excludes that possibility.

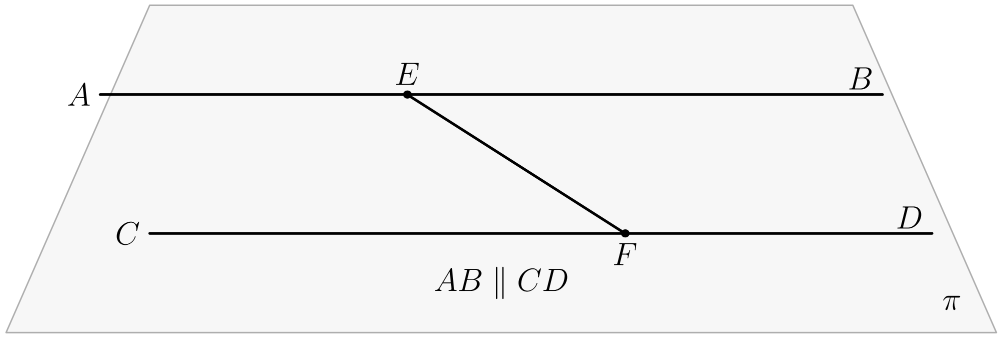


Figure 7.1: The configuration of Proposition XI.7. The two parallel lines AB and CD lie in the reference plane π, with $E \in AB$ and $F \in CD$. The proposition asserts that the joining line EF also lies in π. All final geometric objects in the figure are therefore drawn in black; no object in the proved configuration leaves the plane.

7.4 Classical Proof

Euclid argues by contradiction. Suppose that the straight line joining E and F does not lie in the plane of the parallels, but in a more elevated plane. Draw a plane through that alleged elevated straight line.

The new plane cuts the plane of the parallel lines in a straight line, by Proposition XI.3. Since both E and F belong to both planes, the common section passes through E and F. Euclid then obtains two distinct straight lines through the same two points, described in the text as two straight lines enclosing an area, which is impossible.

Hence the joining line EF cannot leave the plane of the parallels.

The local classical DAG is therefore

$$\begin{gathered} AB \parallel CD, \quad E \in AB, \quad F \in CD \\ \Downarrow \\ \text{suppose the join } EF \text{ leaves the plane of the parallels} \\ \Downarrow \\ \text{draw a second plane through the alleged elevated join} \\ \Downarrow \\ \text{XI.3} \\ \Downarrow \\ \text{the two planes have a straight-line section through } E, F \\ \Downarrow \\ \text{two distinct straight lines through } E, F \\ \Downarrow \\ \text{impossible} \\ \Downarrow \\ EF \text{ lies in the plane of } AB, CD. \end{gathered}$$

The only explicit Book XI proposition cited in the proof is XI.3.

7.5 What is genuinely three-dimensional here?

The genuinely spatial datum is the carrier plane of the two parallel ambient lines. In ordinary plane geometry this datum is invisible, because every line under discussion already belongs to the unique ambient plane.

In three-space, disjointness alone is insufficient for parallelism: two disjoint lines may be skew. The formal relation `HilbertSpaceLinesParallel` therefore records both

$$\text{coplanarity} \qquad \text{and} \qquad \text{disjointness.}$$

XI.7 uses the coplanarity component directly. No metric geometry is needed.

7.6 Formal Statement

The production theorem is:

```
theorem euclid_proposition_11_7
    [H : HilbertIncidence Geo]
    [HilbertPlaneIncidence Geo]
    [S : HilbertSpacePrimitive Geo]
    [HSI : HilbertSpaceIncidence Geo]
    (l m : Geo.Line)
    (E F : Geo.Point)
    (hParallel : HilbertSpaceLinesParallel Geo l m)
    (hEl : H.OnLine E l)
    (hFm : H.OnLine F m) :
    exists sigma : S.Plane,
      HilbertLineInPlane Geo l sigma /\
      HilbertLineInPlane Geo m sigma /\
      exists n : Geo.Line,
        H.OnLine E n /\
        H.OnLine F n /\
        HilbertLineInPlane Geo n sigma
```

The conclusion returns both the common carrier plane `sigma` and an explicit joining line `n` through E and F, together with the proof that `n` lies in `sigma`.

7.7 Spatial / Planar Architecture Used

XI.7 uses only the ambient incidence layer:

```
HilbertIncidence
HilbertPlaneIncidence
HilbertSpacePrimitive
HilbertSpaceIncidence
HilbertLineInPlane
HilbertLinesDisjoint
HilbertSpaceLinesParallel
```

No induced `PlaneGeo` is constructed. There is no planar betweenness, SameRay, SameSide/OppositeSide, angle congruence, right angle, SAS, SSS, or plane–space congruence transport.

This is an important contrast with XI.4–XI.6: although XI.7 is a spatial proposition, its formal content is pure incidence.

7.8 Proof Architecture of the Reconstruction

7.8.1 Step 1: unpack spatial parallelism

From

$$\mathrm{HilbertSpaceLinesParallel}(l, m)$$

the proof obtains a plane σ such that

$$l \subset \sigma, \qquad m \subset \sigma,$$

together with

$$\mathrm{HilbertLinesDisjoint}(l, m).$$

The definition is now part of the general `Hilbert3DInterface` rather than a proposition-local declaration.

7.8.2 Step 2: place the chosen points in the common plane

Since $E \in l$ and $l \subset \sigma$,

$$E \in \sigma.$$

Likewise $F \in m$ and $m \subset \sigma$, hence

$$F \in \sigma.$$

These are direct applications of the line-in-plane predicates supplied by the parallelism witness.

7.8.3 Step 3: prove that the chosen points are distinct

The joining line construction requires $E \neq F$. Euclid's diagram does not state this separately, but the formal proof derives it from the disjointness of the two parallel carriers.

If $E = F$, then the common point would lie on both l and m, contradicting

$$\mathrm{HilbertLinesDisjoint}(l, m).$$

Thus

$$E \neq F.$$

7.8.4 Step 4: construct the joining line

The ordinary incidence structure supplies a line through the two distinct points:

```
HilbertPlaneIncidence.line_through E F hEF
```

This yields a line n with

$$E \in n, \qquad F \in n.$$

7.8.5 Step 5: apply Hilbert I.6

Now E and F are two distinct points of n, and both lie in σ. The decisive step is:

```
HilbertSpaceIncidence.line_in_plane
```

which gives

$$n \subset \sigma.$$

This is the direct formal counterpart of Hilbert's incidence axiom I.6: two points of a line lying in a plane force the whole line into that plane.

7.9 Wyler-Style Flat Reconstruction

XI.7 isolates a particularly reusable generator statement. If $E \in l$ and $F \in m$, then independently of parallelism one has

$$\mathrm{Span}\{E, F\} \subseteq \mathrm{Join}(l, m).$$

This is pure monotonicity of generated flats: the two point generators already belong to the join of the two line carriers.

Parallelism enters only afterwards. It supplies a common plane σ for l, m, identifies their join with the carrier of σ, and guarantees that $E \neq F$. The span of the two points is therefore the carrier of the connector line $n = EF$, and the generic inclusion becomes $n \subseteq \sigma$.

```
E in l, F in m
      |
      v
Span{E,F} subset Join(l,m)
      |
l || m gives Join(l,m)=carrier(sigma)
      |
E != F gives Span{E,F}=carrier(EF)
      |
      v
EF subset sigma
```

This is a useful example of abstraction at the correct level: the central flat lemma is more general than XI.7, but the public proposition still records the classical parallel-line configuration.

7.10 Lean Formalization

The proof DAG is extremely short:

$$\begin{gathered}
\texttt{HilbertSpaceLinesParallel}\ l\ m \\
\Downarrow \\
\exists\sigma, \quad l \subset \sigma, \quad m \subset \sigma, \quad l \cap m = \varnothing \\
\Downarrow \\
E \in \sigma, \quad F \in \sigma \\
\Downarrow \\
E \neq F \\
\Downarrow \\
\texttt{line_through}\ E\ F \\
\Downarrow \\
n \text{ through } E, F \\
\Downarrow \\
\texttt{HilbertSpaceIncidence.line_in_plane} \\
\Downarrow \\
n \subset \sigma \\
\Downarrow \\
\texttt{euclid_proposition_11_7}.
\end{gathered}$$

There are no proposition-local helper theorems.

7.11 Hidden Formal Data

XI.7 has very little hidden data, but two obligations must be explicit.

- The arbitrary points E and F are distinct. This follows from disjointness of the two parallel carriers.
- The phrase "the straight line joining E and F" is represented by an explicit existential line witness produced by `HilbertPlaneIncidence.line_through`.

Once those two points are available, Hilbert I.6 closes the spatial incidence problem directly.

7.12 Relationship with Earlier Book XI Propositions

The classical proof explicitly uses Proposition XI.3, because Euclid constructs a second plane and studies the straight line in which the two planes intersect.

The current Lean theorem does *not* import or invoke XI.3, XI.4, XI.5, or XI.6. Its production file imports only

```
import CGJteamLab.Hilbert3DInterface
```

This difference is structural rather than geometric. Euclid proves the plane containment by contradiction from a plane-intersection theorem. The formal spatial incidence interface instead provides the line-in-plane closure principle directly.

XI.6 is nevertheless conceptually adjacent. It produces `HilbertSpaceLinesParallel`, which is exactly the hypothesis consumed by XI.7. The theorem XI.7 itself, however, is independent of the proof of XI.6.

7.13 Relationship with Book I / Spatial Incidence Infrastructure

Euclid's definition of parallel straight lines includes coplanarity. The current formal definition makes this explicit:

```
def HilbertSpaceLinesParallel
    [H : HilbertIncidence Geo]
    [S : HilbertSpacePrimitive Geo]
    (l m : Geo.Line) : Prop :=
  exists sigma : S.Plane,
    HilbertLineInPlane Geo l sigma /\
    HilbertLineInPlane Geo m sigma /\
    HilbertLinesDisjoint Geo l m
```

For XI.7, the disjointness field is used only to derive $E \neq F$; the common-plane fields provide the main spatial information.

The decisive formal ingredient is the spatial line-in-plane closure principle used by `HilbertSpaceIncidence.line_in_plane`. Thus the proof lands entirely in the spatial incidence layer and nowhere higher in the hierarchy. The source-level identification of this clause inside the classical incidence systems is collected in the Book XI source appendices.

7.14 Relationship with the Foundations

XI.7 is incidence-only at the geometric level. The proof does not use spatial order, spatial congruence, induced `PlaneGeo`, Hilbert Group IV, or continuity. Once the two parallel lines are unpacked into a common carrier plane, the joining line is obtained by ordinary spatial incidence and the line-closure principle.

This makes XI.7 a useful contrast with XI.6 and XI.8: its content is almost entirely about retaining the correct carrier plane through the construction.

7.15 Dependency Summary

The classical local chain is

$$\begin{gathered} AB \parallel CD, \quad E \in AB, \quad F \in CD \\ \downarrow \\ \text{contradictory elevated joining line} \\ \downarrow \\ \text{XI.3} \\ \downarrow \\ \text{common section of two planes is a straight line} \\ \downarrow \\ \text{impossible double join through } E, F \\ \downarrow \\ \text{XI.7.} \end{gathered}$$

The actual formal chain is

$$\begin{gathered} \text{Hilbert incidence} \\ \downarrow \\ \text{Hilbert spatial incidence I.6} \\ \downarrow \\ \texttt{HilbertSpaceLinesParallel} \\ \downarrow \\ \text{common carrier plane } \sigma \\ \downarrow \\ E, F \in \sigma, \quad E \neq F \\ \downarrow \\ \texttt{HilbertPlaneIncidence.line\_through} \\ \downarrow \\ \texttt{HilbertSpaceIncidence.line\_in\_plane} \\ \downarrow \\ \texttt{euclid\_proposition\_11\_7}. \end{gathered}$$

7.16 Conclusion

Proposition XI.7 is a useful contrast with the preceding Book XI formalizations. Its statement is spatial, but its proof requires no order, congruence, right angles, induced plane geometry, or earlier Book XI theorem.

Euclid proves the result by a reductio involving a second plane and Proposition XI.3. The formal reconstruction exposes the same geometric content at the incidence level: spatial parallelism already supplies the common plane, and `HilbertSpaceIncidence.line_in_plane` closes the line under two distinct points of that plane.

The resulting Lean theorem is correspondingly small. Its axiom audit reports no global axiomatic dependencies, while the required spatial incidence assumptions remain explicit theorem parameters. No coordinates, vectors, scalar products, numerical angle measures, continuity principles, or Euclidean parallel axiom are introduced.

Chapter 8

Proposition XI.8

8.1 Introduction

Proposition XI.8 is the Euclidean converse companion to XI.6. Proposition XI.6 proved that two straight lines perpendicular to the same plane are parallel. Proposition XI.8 starts with two parallel spatial lines and assumes that one of them is perpendicular to a reference plane; the conclusion is that the second parallel is perpendicular to the same plane.

The proposition is also a structural boundary in the reconstruction. The formalizations XI.4–XI.7 did not require Hilbert Group IV. XI.8 does. The reason is already visible in Euclid's classical proof: the decisive transversal step uses Proposition I.29, the converse direction of the neutral parallel-angle criterion.

The Lean proof preserves the synthetic geometry but makes the Euclidean dependency more explicit. Instead of calling a planar I.29 wrapper, it separates the argument into

$$\text{neutral I.27} \quad + \quad \text{uniqueness of the parallel through a point.}$$

The second component is precisely the plane-local Group IV assumption.

A second feature is genuinely three-dimensional. Euclid begins by saying that the two parallel lines meet the reference plane at points B and D. The first intersection is given by the hypothesis that the first line is perpendicular to the plane. The second intersection is not automatic in an abstract incidence space. The formal proof derives it.

8.2 Euclid's Statement

Euclid's Proposition XI.8 states:

> If two straight lines be parallel, and one of them be at right angles to any plane, the remaining one will also be at right angles to the same plane.

Let AB and CD be parallel straight lines, and suppose that AB is perpendicular to the reference plane π at B. The claim is that CD meets π at a point D and is perpendicular to π there.

8.3 Classical Configuration

Euclid lets the parallel lines AB and CD meet the reference plane at B and D. He joins BD, draws DE in the reference plane perpendicular to BD, makes $DE = AB$, and joins BE, AE, and AD.

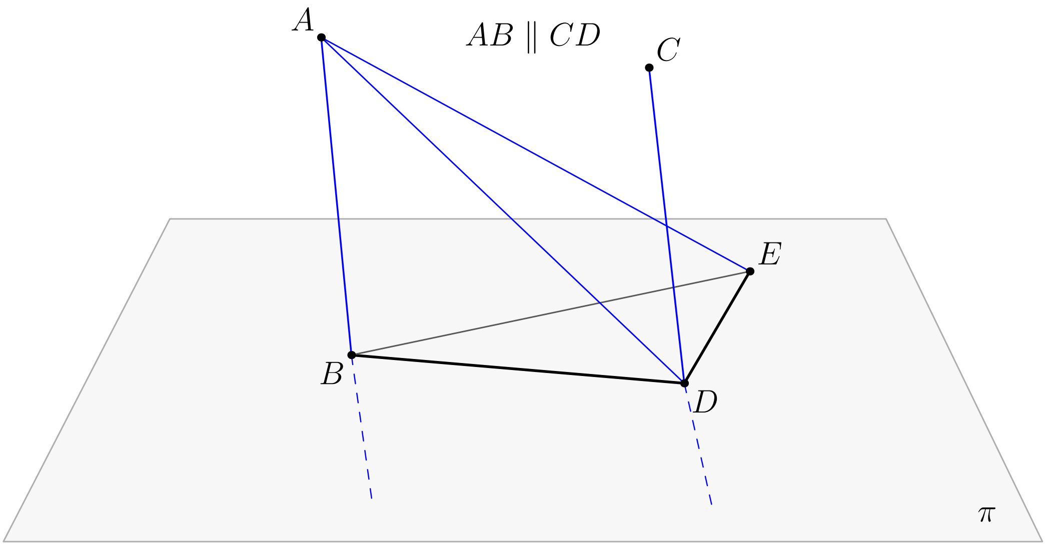


Figure 8.1: The classical configuration of Proposition XI.8. Black lines lie in the reference plane π. Blue lines or segments leave π. The hidden continuations of the two spatial parallel carriers are shown blue and dashed. The auxiliary construction reproduces the metric core already used in XI.6: $DE \perp BD$ and $DE = AB$.

The line BD plays two roles. It is a line of the reference plane and also lies in the common plane of the spatial parallels. This is the transversal on which the classical I.29 step acts.

8.4 Classical Proof

The local classical proof has five main stages.

First, since $AB \parallel CD$, Proposition XI.7 places AB, CD, and the joining line BD in one plane. The opening phrase that the parallel lines meet the reference plane at B, D is diagrammatic in the classical text.

Second, Euclid constructs DE in the reference plane with

$$DE \perp BD, \qquad DE = AB.$$

Since $AB \perp \pi$, the angles

$$\angle ABD, \qquad \angle ABE$$

are right.

Third, BD falls on the parallel lines AB and CD. Proposition I.29 says that the appropriate interior angles sum to two right angles. Since $\angle ABD$ is right, $\angle CDB$ is right. Hence

$$CD \perp BD.$$

Fourth, Euclid repeats the metric triangle argument familiar from XI.6. From

$$AB = DE, \qquad BD = DB,$$

and equality of the included right angles, Proposition I.4 gives

$$AD = BE.$$

Then

$$AB = DE, \qquad BE = AD, \qquad AE = EA,$$

so Proposition I.8 gives

$$\angle ABE = \angle EDA.$$

The former is right, hence the latter is right:

$$ED \perp AD.$$

Now ED is perpendicular to the two intersecting lines DB and DA. Proposition XI.4 therefore gives

$$ED \perp \text{plane}(BDA).$$

The line CD lies in this plane, so

$$ED \perp CD.$$

Finally, in the reference plane π the two lines DB and DE meet at D, and CD is perpendicular to both:

$$CD \perp DB, \qquad CD \perp DE.$$

A second application of XI.4 yields

$$CD \perp \pi.$$

The classical local dependency chain is therefore

$$\begin{gathered} AB \parallel CD, \quad AB \perp \pi \\ \Downarrow \\ \text{XI.7} \quad \text{and the joining line } BD \\ \Downarrow \\ DE \perp BD, \quad DE = AB \quad (\text{I.11, I.3}) \\ \Downarrow \\ AB \perp \pi \Rightarrow \angle ABD, \angle ABE \text{ right} \\ \Downarrow \\ \text{I.29} \\ \Downarrow \\ CD \perp BD \\ \Downarrow \\ \text{I.4, I.8} \\ \Downarrow \\ ED \perp AD \\ \Downarrow \\ \text{XI.4} \\ \Downarrow \\ ED \perp CD \\ \Downarrow \\ \text{XI.4} \\ \Downarrow \\ CD \perp \pi. \end{gathered}$$

The incidence assertion that CD belongs to the plane through BD, DA also uses the fact that the triangle ABD is planar and that AB, CD, BD share the plane supplied by XI.7. In a formal proof this plane identification cannot be left to the printed diagram.

8.5 What is genuinely three-dimensional here?

There are two distinguished planes.

The first is the reference plane π, to which AB is perpendicular. The second is the common plane σ containing the parallel spatial lines AB and CD. Since AB is not contained in π, the planes are distinct.

The points B and D lie in both planes. Consequently their joining line

$$d = BD$$

lies in both π and σ; in the formal proof it is the relevant line of intersection of the two plane slices.

The auxiliary line

$$e = DE$$

lies in π, while the line

$$a = DA$$

lies in σ. The proof therefore alternates between two induced plane geometries.

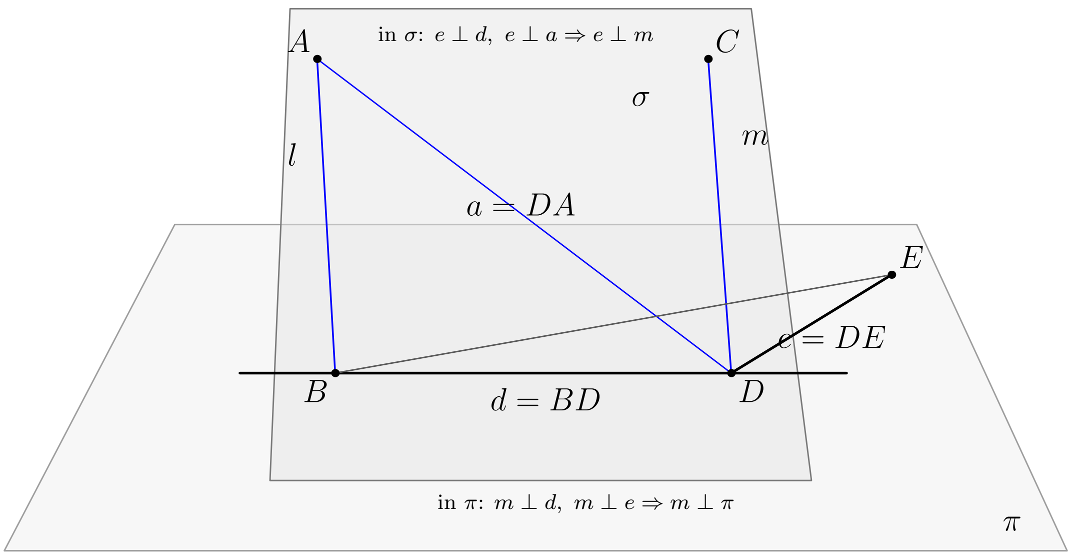


Figure 8.2: The two plane slices used by the reconstruction. The common line $d = BD$ lies in both π and σ. The first XI.4 application uses the two lines $d, a \subset \sigma$ to prove $e \perp \sigma$, hence $e \perp m$. The final XI.4 application uses $d, e \subset \pi$ to prove $m \perp \pi$.

This change of plane is not bookkeeping. It is the spatial geometry of the proposition.

8.6 Formal Statement

The production theorem is:

```
theorem euclid_proposition_11_8
    [H : HilbertIncidence Geo]
    [HilbertPlaneIncidence Geo]
    [S : HilbertSpacePrimitive Geo]
```

```
    [HSI : HilbertSpaceIncidence Geo]
    [_HSO : HilbertSpaceOrder
      (Geo := Geo) (H := H) (S := S)]
    [HSC : HilbertSpaceCongruence
      (Geo := Geo) (H := H) (S := S)]
    [HSE : HilbertSpaceEuclidean Geo]
    (l m : Geo.Line)
    (pi : S.Plane)
    (B : Geo.Point)
    (hParallel : HilbertSpaceLinesParallel Geo l m)
    (hPerp : HilbertLinePerpendicularPlaneAt Geo l pi B) :
    exists D : Geo.Point,
      HilbertLinePerpendicularPlaneAt Geo m pi D
```

The point D is deliberately not an input. Its existence is part of the theorem. This is stronger as a formal interface than assuming in advance that the second parallel already meets the plane.

8.7 Spatial / Planar Architecture Used

The ambient spatial layer consists of

```
HilbertSpacePrimitive
HilbertSpaceIncidence
HilbertSpaceOrder
HilbertSpaceCongruence
HilbertSpaceEuclidean
HilbertSpaceLinesParallel
HilbertLinePerpendicularPlaneAt
HilbertLinesPerpendicularAt
```

The proof then uses induced planar geometry in two different fixed planes:

$$\mathrm{PlaneGeo}(Geo, \sigma) \qquad \text{and} \qquad \mathrm{PlaneGeo}(Geo, \pi).$$

Inside σ, the proof uses neutral planar order, right angles, and the I.27 alternate-angle criterion. Inside π, the final configuration supplies two distinct intersecting lines d, e for XI.4.

The plane–space bridges used in the argument include

```
planeGeo_linesPerpendicularAt_iff_ambient
planeGeo_rightAngle_iff_ambient
planeGeo_angleCongruent_iff_ambient
planeGeo_not_primCollinear_to_ambient
planeGeo_linesDisjoint_iff_ambient
```

The last bridge was added with the Group IV layer. It identifies disjointness of two lines in a fixed `PlaneGeo` with ambient disjointness of their underlying spatial carriers.

8.8 The Euclidean Layer: Group IV

XI.8 is the first proposition in the present Book XI sequence whose formal proof requires the Euclidean parallel axiom.

The spatial axiom is not installed as a global `HilbertEuclideanPlaneGeo` instance. Such an instance would incorrectly treat the ambient three-space as a plane. Instead the axiom is stated plane-locally:

```
class HilbertSpaceEuclidean
    (Geo : Geometry.Geo)
    [H : HilbertIncidence Geo]
    [S : HilbertSpacePrimitive Geo]
    [HilbertSpaceIncidence Geo]
    [HilbertSpaceOrder Geo]
    [HilbertSpaceCongruence Geo] : Prop where

  parallel_unique_in_plane :
    forall pi : S.Plane,
      forall l : Geo.Line,
        HilbertLineInPlane Geo l pi ->
        forall A : Geo.Point,
          S.OnPlane A pi ->
          Not (H.OnLine A l) ->
          forall b c : Geo.Line,
            HilbertLineInPlane Geo b pi ->
            HilbertLineInPlane Geo c pi ->
            H.OnLine A b ->
            HilbertLinesDisjoint Geo b l ->
            H.OnLine A c ->
            HilbertLinesDisjoint Geo c l ->
            b = c
```

For every fixed ambient plane, the bridge instance

```
instance planeGeoHilbertEuclidean
    ...
    (pi : S.Plane) :
    HilbertEuclideanPlane (PlaneGeo Geo pi)
```

recovers the existing planar Group IV API. The architectural rule is therefore preserved:

Euclidean parallelism is planar, even inside three-space.

8.9 Proof Architecture of the Reconstruction

8.9.1 Step 1: derive the hidden intersection point D

The helper

```
hilbert_XI8_parallel_meets_perpendicular_plane
```

proves that the second parallel m meets the reference plane.

Unpacking

$$\text{HilbertSpaceLinesParallel}(l, m)$$

gives a common plane σ containing l, m, with the two lines disjoint. The foot B of $l \perp \pi$ lies in both π and σ.

The planes are distinct. If $\sigma = \pi$, then $l \subset \pi$, contradicting the theorem

```
hilbert_linePerpendicularPlaneAt_not_in_plane
```

The general plane-intersection theorem then supplies a line

$$q = \pi \cap \sigma$$

through B.

Suppose m did not meet q. Then both l and q would be lines through B in σ disjoint from m. Group IV uniqueness would force $l = q$, but $q \subset \pi$, again contradicting $l \not\subset \pi$.

Hence m meets q at some point D, and therefore

$$D \in m \cap \pi.$$

This is the formal proof of a fact Euclid leaves implicit at the opening of XI.8.

8.9.2 Step 2: reuse the XI.6 metric core

With D fixed, the proof reuses

```
hilbert_XI6_second_SSS_right_angle
```

from Proposition XI.6.

The helper constructs points and lines corresponding to Euclid's auxiliary configuration and returns, in particular,

$$d = BD, \qquad e = DE,$$

with

$$d, e \subset \pi, \qquad e \perp d,$$

and the right angle

$$\angle EDA.$$

Thus the SAS/SSS part of XI.8 is not duplicated. The earlier XI.6 development is used as reusable spatial congruence infrastructure.

8.9.3 Step 3: first application of XI.4 in σ

Let a be the line DA. The proof shows

$$d, a \subset \sigma, \qquad d \neq a.$$

The inequality is geometrically meaningful: A lies on the spatial line l but outside the reference plane π, while $d \subset \pi$.

The XI.6 metric result gives

$$e \perp d, \qquad e \perp a.$$

Therefore XI.4, applied with the induced geometry of σ, gives

$$e \perp \sigma.$$

Since $m \subset \sigma$ and $D \in m$,

$$e \perp m.$$

This is the formal counterpart of Euclid's first XI.4 step.

8.9.4 Step 4: replace I.29 by neutral I.27 plus Group IV

The remaining task is to prove

$$m \perp d.$$

The proof does not call a public I.29 theorem. Instead, inside PlaneGeo(Geo, σ), it constructs a line p through D perpendicular to d.

The neutral helper

```
hilbert_XI8_coplanar_perpendiculars_to_same_line_disjoint
```

shows that p and l are disjoint. Its proof is an I.27 argument: two right angles are normalized to equal alternate angles and the neutral alternate-angle theorem produces nonintersection.

But m is also disjoint from l, because $l \parallel m$. Thus m and p are two lines of the same plane σ, both passing through D, both disjoint from l. Group IV uniqueness forces

$$m = p.$$

Since $p \perp d$,

$$m \perp d.$$

This factorization isolates the exact Euclidean content:

$$\underbrace{\text{right angles} \Rightarrow \text{nonintersection}}_{\text{neutral I.27}} \quad + \quad \underbrace{\text{unique parallel through } D}_{\text{Group IV}} \quad \Longrightarrow \quad m \perp d.$$

It is the formal counterpart of Euclid's I.29 step.

8.9.5 Step 5: final application of XI.4 in π

At this point

$$d, e \subset \pi, \qquad m \perp d, \qquad m \perp e.$$

The lines d and e are distinct because $e \perp d$.

The final call to

```
euclid_proposition_11_4
```

uses d, e as the two distinct intersecting lines of the reference plane. It returns

$$m \perp \pi$$

at D, which is the required conclusion.

8.10 Wyler-Style Flat Reconstruction

XI.8 shows the exact boundary of the flat calculus. The common-plane and join structure organizes the auxiliary configuration, but incidence alone cannot transfer perpendicularity from one parallel line to the other. The Euclidean parallel axiom is essential.

The second-pass proof separates three layers:

```
flat / incidence layer
    construct the auxiliary planes and lines at m
        |
        v
Euclidean layer (Group IV)
    transfer the fixed-transversal perpendicularity
        |
        v
metric layer
    m perp d, m perp e
        |
        v
    XI.4 metric core
        |
        v
    m perp pi
```

Thus XI.8 is not absorbed into join and meet. Rather, join and carrier language make the incidence skeleton explicit, Group IV provides the genuinely Euclidean step, and XI.4 converts two independent perpendicular directions into perpendicularity to the reference plane. This separation is one of the most important architectural checks in Book XI.

8.11 Lean Formalization

The proposition-local proof is organized around four named helpers:

```
hilbert_XI8_parallel_meets_perpendicular_plane

hilbert_XI8_auxiliary_line_perpendicular_to_parallel

hilbert_XI8_coplanar_perpendiculars_to_same_line_disjoint

hilbert_XI8_parallel_line_perpendicular_to_given_BD
```

The actual Lean DAG is

$$
\begin{gathered}
l \parallel m, \quad l \perp \pi \text{ at } B \\
\Downarrow \\
\texttt{hilbert\_XI8\_parallel\_meets\_perpendicular\_plane} \\
\Downarrow \\
D \in m \cap \pi \\
\Downarrow \\
\texttt{hilbert\_XI6\_second\_SSS\_right\_angle} \\
\Downarrow \\
d = BD,\ e = DE, \quad e \perp d, \quad \angle EDA \text{ right} \\
\Downarrow \\
\text{XI.4 in } \sigma \\
\Downarrow \\
e \perp m \\
\Downarrow \\
\text{I.27 in } \text{ PlaneGeo}(\sigma) + \text{ Group IV} \\
\Downarrow \\
m \perp d \\
\Downarrow \\
\text{XI.4 in } \pi \\
\Downarrow \\
\texttt{euclid\_proposition\_11\_8}.
\end{gathered}
$$

The production file imports XI.4, XI.5, and XI.6. It does not import XI.7. The common plane that Euclid obtains through XI.7 is already contained in the formal hypothesis `HilbertSpaceLinesParallel`.

8.12 Hidden Formal Data

XI.8 contains substantially more hidden incidence information than its classical diagram suggests.

- The second parallel must be proved to meet the reference plane. The point D is an existential output, not an assumed input.
- The common plane σ of the parallels must be proved distinct from π.
- The intersection line $q = \pi \cap \sigma$ is constructed explicitly.
- The points B, D must be proved distinct from disjointness of the parallel carriers.
- The line $d = BD$ must be shown to lie both in π and in σ.
- The point A selected on l must be proved to lie outside π; otherwise the line l would lie in π, contrary to $l \perp \pi$.
- The line $a = DA$ must be constructed and shown distinct from d.
- Right-angle and perpendicularity statements must be transported between ambient geometry and `PlaneGeo`.
- The neutral I.27 argument requires explicit SameSide / OppositeSide and SameRay normalization that Euclid suppresses in the diagram.

- The two disjointness orientations required by Group IV are supplied explicitly.
- Before the final XI.4 application, $d \neq e$ is derived from $e \perp d$.

Subtype projections and record fields occur in the Lean code, but they are implementation details. The mathematically significant hidden data are the plane memberships, distinctness statements, side orientations, and the derived intersection point D.

8.13 Relationship with Earlier Book XI Propositions

The classical and formal dependency graphs differ in a controlled way.

Classically, XI.7 is an explicit opening dependency: it places AB, CD, BD in one plane. XI.4 is then used twice. The classical plane-identification step also relies on the ordinary fact from XI.2 that a triangle lies in one plane.

The production Lean file imports

```
import CGJteamLab.Proposition11_4
import CGJteamLab.Proposition11_5
import CGJteamLab.Proposition11_6
```

but does not import Proposition XI.7.

XI.4 is used twice exactly as in the classical architecture: first in the common plane σ, then in the reference plane π.

XI.5 contributes established line–plane perpendicularity infrastructure, including the theorem that a line perpendicular to a plane is not contained in that plane.

XI.6 contributes the reusable metric construction and the spatial SAS/SSS right-angle machinery. This reuse is substantial: the central auxiliary configuration of XI.8 is almost the same as the configuration built for XI.6.

XI.7 is absent from the Lean import graph because `HilbertSpaceLinesParallel` already stores the common carrier plane as part of the definition of spatial parallelism. Thus the formal proof consumes the conclusion-like data that XI.7 exposes, without invoking the proposition theorem itself.

8.14 Relationship with Book I / Book Zero / Hilbert Infrastructure

The classical proof uses I.11 and I.3 to construct the auxiliary perpendicular and equal segment. In the Lean reconstruction these operations are already packaged in the XI.6 helper chain over the Hilbert/Book Zero construction interface.

The metric part corresponds to I.4 and I.8 and is likewise inherited from XI.6.

The parallel step is the important logical boundary. The direction represented by I.27,

$$\text{equal alternate angles} \Longrightarrow \text{nonintersection},$$

belongs to the neutral congruence/order layer. The reverse use of parallelism represented by I.29 requires the Euclidean parallel axiom. The formal proof displays this by using the neutral I.27 theorem first and applying Group IV only to identify the unique line through D disjoint from l.

No coordinate system, vector space, scalar product, distance formula, trigonometric function, or numerical angle measure is used.

8.15 Relationship with the Foundations

XI.8 is the first proposition in this local sequence whose current Euclid-faithful reconstruction genuinely uses Hilbert Group IV. The public theorem assumes `HilbertSpaceEuclidean Geo`, and the proof consumes that structure through the induced Euclidean geometry of an explicit plane slice.

This use of Group IV is already analyzed in the section "The Euclidean Layer: Group IV" above. The essential point is mathematical: the proof needs the uniqueness/existence behavior of parallels in the relevant plane; no global ambient planar instance is installed. No continuity principle is used.

8.16 Dependency Summary

The classical local chain is

$$\begin{array}{c}
\text{parallel } AB, CD, \quad AB \perp \pi \\
\downarrow \\
\text{XI.7} \\
\downarrow \\
BD \text{ lies in the common plane of the parallels} \\
\downarrow \\
\text{I.11, I.3} \\
\downarrow \\
DE \perp BD, \quad DE = AB \\
\downarrow \\
\text{I.29} \\
\downarrow \\
CD \perp BD \\
\downarrow \\
\text{I.4, I.8} \\
\downarrow \\
ED \perp AD \\
\downarrow \\
\text{XI.4} \\
\downarrow \\
ED \perp CD \\
\downarrow \\
\text{XI.4} \\
\downarrow \\
CD \perp \pi.
\end{array}$$

The actual formal chain is

$$\begin{gathered}\text{Hilbert spatial incidence/order/congruence}\\ \downarrow\\ \texttt{HilbertSpaceEuclidean}\ \text{(Group IV)}\\ \downarrow\\ \texttt{HilbertSpaceLinesParallel}\\ \downarrow\\ D \in m \cap \pi\\ \downarrow\\ \text{XI.6 metric helper chain}\\ \downarrow\\ d = BD, \quad e = DE, \quad e \perp d, \quad e \perp a\\ \downarrow\\ \text{XI.4 in } \sigma\\ \downarrow\\ e \perp m\\ \downarrow\\ \text{I.27 in } \text{ PlaneGeo}(\sigma) + \text{ Group IV uniqueness}\\ \downarrow\\ m \perp d\\ \downarrow\\ \text{XI.4 in } \pi\\ \downarrow\\ \texttt{euclid\_proposition\_11\_8}.\end{gathered}$$

The dependency classification is therefore:

New proposition-specific global axiom	no
New spatial axiom class	yes: `HilbertSpaceEuclidean`
Continuity	no
Group IV / Euclidean parallelism	yes
Earlier Book XI production imports	XI.4, XI.5, XI.6
Classical explicit Book XI dependency	XI.7, XI.4
New PlaneGeo Euclidean bridge	yes
Coordinates / vectors / scalar product	no
`sorry/admit`	no

8.17 Conclusion

Proposition XI.8 is the first point in the present Book XI reconstruction where Euclidean parallel uniqueness is mathematically necessary. The formal proof makes that boundary sharper than the classical text: neutral I.27 proves nonintersection, while Group IV supplies the uniqueness needed to recover the I.29 direction.

At the same time, the proof remains fully synthetic. It reuses the metric construction of XI.6, applies XI.4 in two different plane slices, and derives the hidden intersection point of the second parallel with the reference plane instead of assuming it from the diagram.

The result therefore tests three parts of the spatial architecture at once: plane-local Euclidean parallelism, transport between ambient three-space and induced plane geometry, and the universal line–plane perpendicularity predicate. No continuity or analytic representation is introduced.

Chapter 9

Proposition XI.9

9.1 Introduction

Proposition XI.9 is the spatial counterpart of Euclid I.30. It asserts that if two lines are each parallel to a third line, then they are parallel to one another, provided that the three lines are not all contained in one plane.

This extra non-coplanarity clause is the essential new feature. In a plane, Euclid I.30 already proves transitivity of parallelism. In space, however, the theorem must distinguish two cases:

- the planar case, where the three lines lie in one plane and I.30 is the relevant model;
- the genuinely spatial case, where the three lines are distributed across different planes and the conclusion must be rebuilt from 3D incidence.

The current formal proof treats XI.9 as a theorem of synthetic spatial incidence plus one plane-local use of the Euclidean parallel axiom. Contrary to the historical proof, it does not proceed through a chain of perpendicular constructions.

The production file is

```
CGJteamLab/Proposition11_9.lean
```

and XI.9 introduces two proposition-local helpers:

```
hilbert_XI9_point_on_second_line_off_first
hilbert_XI9_planes_eq_of_common_line_and_external_point
```

9.2 Euclid's Statement

Euclid states:

> If two straight lines are parallel to the same straight line, then they are parallel to one another, if they do not lie in the same plane with it.

In the present library, spatial parallelism is represented by

```
HilbertSpaceLinesParallel Geo l m
```

which packages:

- a plane containing both lines, and
- ambient disjointness of the two lines.

Thus the formal conclusion of XI.9 is not merely "coplanar with the same direction" but the full structured statement that l and m are parallel lines in the project's spatial sense.

The phrase "if they do not lie in the same plane with it" is formalized as

```
Not (exists omega : S.Plane,
  HilbertLineInPlane Geo l omega /\
  HilbertLineInPlane Geo m omega /\
  HilbertLineInPlane Geo n omega)
```

That is: there is no single plane containing all three lines l, m, n. This is the exact ambient reading of Euclid's proviso.

9.3 Classical Configuration

We are given three lines l, m, n such that

$$l \parallel n, \qquad m \parallel n,$$

and the three are not all coplanar.

Let σ be a plane containing l and n, and let τ be a plane containing m and n. Because all three lines are not contained in one plane, these two carrier planes are distinct:

$$\sigma \neq \tau.$$

The formal proof then chooses a point B on m not lying on l, forms the plane ω through l and B, and studies the intersection

$$r = \omega \cap \tau.$$

The line r passes through B. Inside τ, both r and m pass through B and are parallel to n. Euclidean uniqueness in the plane τ forces

$$r = m.$$

Hence both l and m lie in ω. The final step proves they are disjoint, so

$$l \parallel m.$$

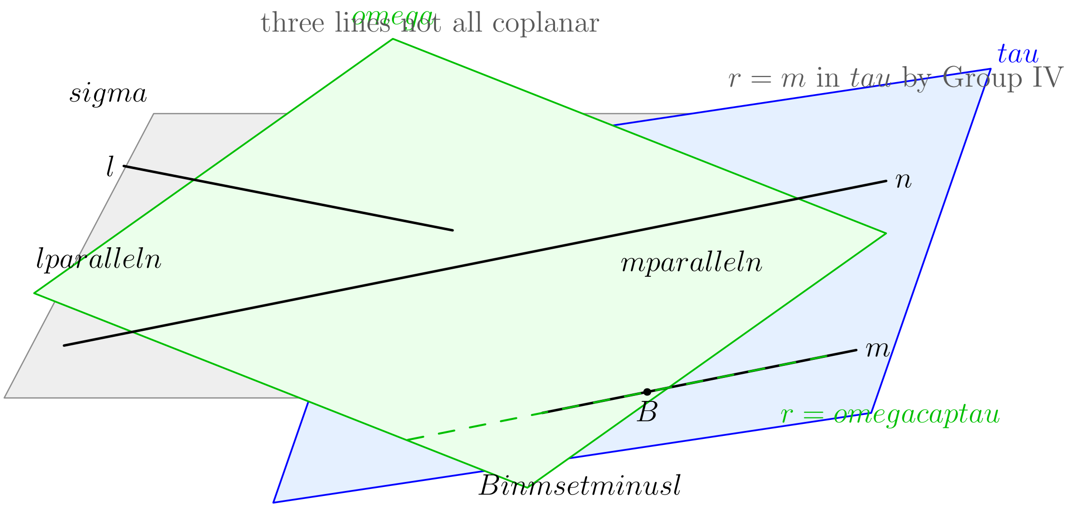


Figure 9.1: A schematic picture for Proposition XI.9. The line n lies in both carrier planes σ and τ. The line l lies in σ, the line m lies in τ, and the three lines are not all coplanar. Choosing $B \in m \setminus l$, the plane ω through l and B intersects τ in a line r. Since both r and m pass through B and are parallel to n inside τ, Group IV gives $r = m$.

9.4 Euclid's Classical Proof Pattern

Euclid's printed proof is not organized in this affine way. Its characteristic strategy is to convert the parallel problem into a perpendicular problem. Schematically, the classical route is:

$$\text{I.11 in two planes} \longrightarrow \text{common perpendiculars} \longrightarrow \text{XI.4} \longrightarrow \text{XI.8} \longrightarrow \text{XI.6} \longrightarrow \text{XI.9.}$$

In more explicit local terms:

1. in suitable planes, erect lines perpendicular to the common line;
2. use XI.4 to pass from right-angle data in one plane to perpendicularity to the plane;
3. use XI.8 to transport plane-perpendicularity across parallels;
4. use XI.6 to conclude that the resulting perpendiculars are parallel;
5. conclude that the original lines are parallel.

This is entirely reasonable in Euclid's sequence, since Book XI is gradually building a theory of perpendicularity and parallelism for lines and planes.

Nevertheless, from the standpoint of the current formal library, this is not the most efficient reconstruction. The actual theorem can be proved more directly from spatial incidence and planar parallel uniqueness.

9.5 Why the Formal Reconstruction Takes a Different Route

The project already contains a precise synthetic notion of ambient line parallelism, a theorem on the intersection of two planes, and a plane-local Euclidean parallel uniqueness axiom.

Once these are available, XI.9 can be reconstructed with a cleaner structure:

- first create the target coplanarity of l and m;
- then prove their disjointness;
- package both facts as `HilbertSpaceLinesParallel Geo l m`.

This proof is still fully synthetic. It uses no coordinates, vectors, or scalar products. The difference is conceptual, not analytic: the formal proof exposes XI.9 as a theorem about how distinct planes and parallel lines fit together in 3-space.

9.5.1 First local incidence lemma

The first local helper is

```
theorem hilbert_XI9_point_on_second_line_off_first
    ...
    (l m : Geo.Line)
    (hlm : Ne l m) :
    exists P : Geo.Point,
      H.OnLine P m /\
      Not (H.OnLine P l)
```

Its content is elementary but useful: if two lines are distinct, then the second line has a point not lying on the first.

The proof uses two points on m. If both were also on l, line uniqueness would force $m = l$, contradiction.

This helper is a pure incidence lemma and abstracts away a routine but repeatedly needed argument.

9.5.2 Plane uniqueness from a line and an external point

The second helper is

```
theorem hilbert_XI9_planes_eq_of_common_line_and_external_point
    ...
    (l : Geo.Line)
    (P : Geo.Point)
    (hPl : Not (H.OnLine P l))
    (pi rho : S.Plane)
    (hlpi : HilbertLineInPlane Geo l pi)
    (hlrho : HilbertLineInPlane Geo l rho)
    (hPpi : S.OnPlane P pi)
    (hPrho : S.OnPlane P rho) :
```

```
    pi = rho
```

This is the uniqueness clause of the theorem "a line and an external point determine a unique plane," extracted into a convenient reusable form.

Its role in XI.9 is structural. Whenever two planes both contain the same line and the same point outside that line, the planes coincide. This is the key contradiction engine used twice in the proof:

- first to show that the auxiliary intersection line r is disjoint from n;
- then to show that the final target lines l and m are disjoint.

9.6 Formal Statement

The public theorem is:

```
theorem euclid_proposition_11_9
    [H : HilbertIncidence Geo]
    [HilbertPlaneIncidence Geo]
    [S : HilbertSpacePrimitive Geo]
    [HSI : HilbertSpaceIncidence Geo]
    [_HSO : HilbertSpaceOrder (Geo := Geo) (H := H) (S := S)]
    [HSC : HilbertSpaceCongruence (Geo := Geo) (H := H) (S := S)]
    [HSE : HilbertSpaceEuclidean Geo]
    (l m n : Geo.Line)
    (hParallelLN : HilbertSpaceLinesParallel Geo l n)
    (hParallelMN : HilbertSpaceLinesParallel Geo m n)
    (hNoCommonPlane :
      Not (exists omega : S.Plane,
        HilbertLineInPlane Geo l omega /\
        HilbertLineInPlane Geo m omega /\
        HilbertLineInPlane Geo n omega)) :
    HilbertSpaceLinesParallel Geo l m
```

The assumptions separate clearly into three layers:

- spatial incidence and plane structure;
- spatial order/congruence, needed through the Group IV interface;
- the Euclidean parallel uniqueness axiom `HilbertSpaceEuclidean Geo`.

9.7 Proof Architecture of the Reconstruction

The proof has six main steps.

Step 1: unpack the two given parallels

From

$$l \parallel n, \qquad m \parallel n,$$

we obtain planes σ and τ such that

$$l, n \subset \sigma, \qquad m, n \subset \tau,$$

with the corresponding disjointness facts

$$l \cap n = \varnothing, \qquad m \cap n = \varnothing.$$

Step 2: prove $\sigma \neq \tau$ and $l \neq m$

If $\sigma = \tau$, then the three lines would lie in one plane, contrary to the explicit hypothesis `hNoCommonPlane`. Likewise, if $l = m$, then again all three would lie in one plane.

Step 3: choose $B \in m \setminus l$ and form ω

Using `hilbert_XI9_point_on_second_line_off_first`, choose a point B on m which does not lie on l. Then the theorem

```
hilbert_plane_through_line_and_external_point
```

produces a plane ω containing l and B.

Step 4: intersect ω and τ

The planes ω and τ are distinct, and both contain B. By XI.3 in its strengthened Hilbert form, they intersect in a line r through B:

$$r = \omega \cap \tau.$$

Step 5: identify r with m

The crucial point is that r is disjoint from n. For if a point P belonged to both r and n, then P would lie in both ω and σ, while being outside l. Since both planes also contain l, helper 2 would force

$$\sigma = \omega,$$

contradiction.

Thus in the plane τ, both lines r and m:

- pass through the point B, and

- are disjoint from the line n.

By plane-local Euclidean uniqueness of a parallel through a point,

$$r = m.$$

Hence $m \subset \omega$, and the target coplanarity of l and m is established.

Step 6: prove l and m are disjoint

Suppose a point P lay on both l and m. Since l is disjoint from n, this point is outside n. But P lies in σ (because $P \in l \subset \sigma$) and in τ (because $P \in m \subset \tau$). Since both planes also contain n, helper 2 forces

$$\sigma = \tau,$$

again contradicting the non-coplanarity assumption. Therefore

$$l \cap m = \varnothing.$$

Since l and m lie in the common plane ω and are disjoint, the packaged conclusion follows:

$$l \parallel m.$$

9.8 Formal DAG

The actual dependency chain of the current reconstruction is:

```
hParallelLN, hParallelMN, hNoCommonPlane
        |
        +-- hilbert_XI9_point_on_second_line_off_first
        |
        +-- hilbert_plane_through_line_and_external_point
        |
        +-- hilbert_plane_intersection_line   (XI.3 infrastructure)
        |
        +-- HilbertSpaceEuclidean.parallel_unique_in_plane
        |
        +-- hilbert_XI9_planes_eq_of_common_line_and_external_point
        |
        v
euclid_proposition_11_9
```

This should be contrasted with the historical local chain:

```
I.11
  -> XI.4
  -> XI.8
  -> XI.6
  -> XI.9
```

The formal DAG is not less synthetic. It simply exposes the spatial incidence content more directly.

9.9 Wyler-Style Flat Reconstruction

The production reconstruction above is affine and synthetic. The Wyler-style second pass changes the way the auxiliary planes are identified: instead of a proposition-local uniqueness lemma, two planes are shown equal because they are the same generated join of a line and an external point.

The resulting structural graph is

```
sigma contains l,n
 tau  contains m,n
 sigma != tau
        |
        v
 B in m, B notin l
 omega = Join(l,{B})
        |
        v
 r = Meet(omega,tau)
        |
        v
 r disjoint n
        |
        v
 Group IV in tau: r = m
        |
        v
 m subset omega
 l,m disjoint
        |
        v
 l || m
```

The key normalization is the generated plane

$$\omega = \mathrm{Join}(l, \{B\}).$$

Whenever another plane contains the same line l and the same point $B \notin l$, equality is detected by equality with this join. Likewise the section line r is treated as the exact meet of ω and τ. Only the identification $r = m$ requires the plane-local Euclidean uniqueness of a parallel through B.

This graph is proposition-specific and therefore belongs here rather than in the general Wyler appendix.

9.10 Spatial / Planar Architecture Used

The theorem depends on the following ambient structure:

```
HilbertSpacePrimitive
HilbertSpaceIncidence
HilbertSpaceOrder
HilbertSpaceCongruence
HilbertSpaceEuclidean
HilbertSpaceLinesParallel
```

It also reuses the previously developed spatial theorems:

```
hilbert_plane_through_line_and_external_point
hilbert_plane_intersection_line
```

No coordinates, vectors, affine spaces, or scalar products are introduced. No induced `PlaneGeo` slice is required in the public proof, even though the decisive Euclidean axiom is conceptually plane-local.

9.11 Role of Group IV

XI.9 uses the Euclidean parallel axiom exactly once, in the form

```
HilbertSpaceEuclidean.parallel_unique_in_plane
```

applied in the plane τ.

This is the precise place where Euclidean strength enters. Everything before that point is incidence-only: choose a point off a line, construct a plane, intersect two planes, and compare two candidate planes by uniqueness.

The proof therefore gives a clean conceptual reading of XI.9:

$$\text{3D incidence geometry} + \text{one planar uniqueness step} \Longrightarrow \text{spatial transitivity of parallelism under Euclid's prov}$$

9.12 Hidden Formal Data

Several facts that are implicit in the classical picture must be made explicit in Lean.

The carrier planes are proof objects

The hypotheses `hParallelLN` and `hParallelMN` do not merely say "parallel" in a bare relational sense. They supply witness planes σ and τ.

The non-coplanarity clause is genuinely global

The condition `hNoCommonPlane` is not a statement about one chosen pair of lines; it quantifies over all candidate planes ω.

The point B is not diagrammatic decoration

The point off l but on m is the key to constructing the new plane ω that will eventually contain both target lines.

Disjointness must be proved twice

The proof must explicitly certify both

$$r \cap n = \varnothing \qquad \text{and} \qquad l \cap m = \varnothing.$$

In a traditional diagram these facts often look visually obvious, but in a synthetic proof they are independent obligations.

9.13 Audit Status

This documentation was checked against the current production theorem `Geometry.euclid_proposition_11_9` and its two proposition-local incidence lemmas. The note records the current synthetic route through coplanarity creation, plane intersection, disjointness, and plane-local parallel uniqueness. The role of Hilbert Group IV is described separately above because it is part of the mathematics of the reconstruction, not a build-status report.

9.14 Mathematical Significance of the Reconstruction

XI.9 is a good example of the difference between the order of exposition in the *Elements* and the most transparent dependency structure of the formal theory.

Euclid's proof moves through perpendicularity because Book XI is organized as a gradual synthetic development of the geometry of lines and planes. The modern formal system, however, already has explicit infrastructure for planes through lines, intersections of planes, and plane-local uniqueness of parallels.

Once those pieces are isolated, the theorem turns out to be structurally about *coplanarity creation* and *disjointness control*. The core problem is not to manufacture right angles, but to produce the unique plane in which the two target lines must lie.

In this sense XI.9 is one of the first propositions of Book XI where the formal reconstruction becomes conceptually more informative than the classical proof: it reveals the theorem as a hybrid of spatial incidence and planar Euclidean uniqueness.

9.15 Conclusion

Proposition XI.9 states that if two lines are each parallel to a third line, and the three are not all contained in one plane, then the first two are parallel to one another.

The present Lean proof establishes this in a direct synthetic way:

$$l \parallel n, \quad m \parallel n, \quad \text{no common plane for } l, m, n \Longrightarrow l \parallel m.$$

The proof proceeds by choosing a point on m outside l, constructing a plane through l and that point, intersecting it with the plane containing m and n, and invoking Euclidean uniqueness of a parallel through a point. A second plane-uniqueness argument then supplies the required disjointness of l and m.

Thus XI.9 is formalized not as a proposition about perpendicular gadgets, but as a structural theorem about three lines distributed across several planes. It remains fully synthetic, uses Group IV at exactly one point, and achieves a very clean axiom profile.

Chapter 10

Proposition XI.10

10.1 Introduction

Proposition XI.10 transfers an angle from one plane to another by means of two pairs of corresponding parallel lines. In Euclid's configuration, AB and BC meet at B, while DE and EF meet at E. The first pair is parallel to the second pair, and the two angle configurations are not all contained in one plane. The conclusion is

$$\angle ABC \cong \angle DEF.$$

The proposition is easy to underestimate. Ordinary, unoriented parallelism is not sufficient to determine the correct angle. Reversing exactly one of the four rays changes the desired angle to its supplement. The formal reconstruction therefore has to retain directional information which Euclid leaves to the diagram and to the phrase "parallel to" in a particular direction.

The production proof remains fully synthetic. It uses no coordinates, vectors, scalar product, normal vectors, trigonometric functions, or numerical angle measure. Its central ingredients are:

- Euclid I.33 in explicitly chosen plane slices;
- Euclid XI.9 for spatial parallelism;
- a plane-separation / section-line bridge;
- spatial SSS;
- a final SameRay transport used by the public wrapper.

The proof also illustrates a distinction that is especially important in Book XI. The note intentionally contains several schematic figures: a main spatial overview, the I.33–XI.9 connector state, the directed-parallel slice, the section-line / plane-side bridge, the public-wrapper normalization, and the final SameRay/SSS states. Each figure is generated from an auditable Asymptote source.

$$\text{unoriented carrier parallelism} \neq \text{directed parallel segments.}$$

The former is enough for XI.9. The latter is needed to recover exactly the angle intended by XI.10.

10.2 Euclid's Statement

Euclid's Proposition XI.10 states:

> If two straight lines meeting one another be parallel to two straight lines meeting one another not in the same plane, they will contain equal angles.

Let AB and BC meet at B, and let DE and EF meet at E. Assume that the corresponding straight lines are parallel and that the two intersecting pairs are not contained in one common plane. Euclid's claim is

$$\angle ABC = \angle DEF.$$

The phrase "parallel to" here carries directional content. If one ray is reversed while the other three are kept fixed, the conclusion is generally not equality of the two displayed angles. This directional issue is part of the mathematical statement, not merely a formalization detail.

10.3 Classical Configuration

Euclid first cuts four equal segments:

$$BA = BC = ED = EF.$$

He then joins

$$AD, \qquad CF, \qquad BE, \qquad AC, \qquad DF.$$

The resulting configuration contains three successive parallelogram-type steps.

First,

$$BA \parallel ED, \qquad BA = ED$$

implies by I.33 that

$$AD \parallel BE, \qquad AD = BE.$$

Second,

$$BC \parallel EF, \qquad BC = EF$$

implies, again by I.33,

$$CF \parallel BE, \qquad CF = BE.$$

Third, XI.9 gives

$$AD \parallel CF,$$

and equality follows through the common segment BE:

$$AD = CF.$$

A final use of I.33 yields

$$AC \parallel DF, \qquad AC = DF.$$

The triangles

$$\triangle ABC, \qquad \triangle DEF$$

therefore have three corresponding equal sides, and I.8 gives

$$\angle ABC = \angle DEF.$$

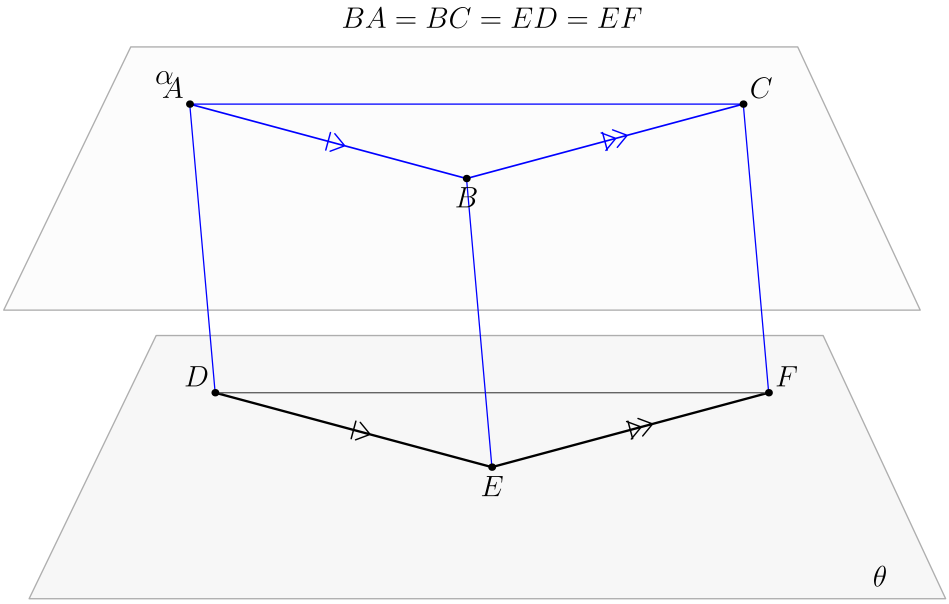


Figure 10.1: Classical spatial configuration for Proposition XI.10. The two lines meeting at B are compared with the two lines meeting at E; the two angle configurations are not contained in one common plane. This figure is the global reference picture for the proposition.

10.4 Classical Proof

The classical local chain is

$$\begin{gathered} BA = BC = ED = EF \\ \Downarrow \\ BA \parallel ED, \quad BC \parallel EF \\ \Downarrow \\ \text{I.33 twice} \\ \Downarrow \\ AD \parallel BE, \quad AD = BE, \\ CF \parallel BE, \quad CF = BE \\ \Downarrow \\ \text{XI.9} \\ \Downarrow \\ AD \parallel CF, \quad AD = CF \\ \Downarrow \\ \text{I.33} \\ \Downarrow \\ AC \parallel DF, \quad AC = DF \\ \Downarrow \\ \text{I.8} \\ \Downarrow \\ \angle ABC = \angle DEF. \end{gathered}$$

Thus the explicit classical proposition dependencies are

$$\text{I.3}, \qquad \text{I.33}, \qquad \text{XI.9}, \qquad \text{I.8}.$$

The first step, I.3, is a normalization: equal pieces are laid off on the four rays. The geometric core begins after those equal segments have been chosen.

10.5 Why Ordinary Spatial Parallelism Is Not Enough

The project definition

```
HilbertSpaceLinesParallel Geo l m
```

records that two carrier lines are coplanar and disjoint. This is exactly the correct spatial analogue of ordinary line parallelism: skew disjoint lines are excluded.

XI.10 needs more. I.33 is directional: if two equal parallel segments are taken in corresponding directions, then the joining segments are equal and parallel in the corresponding direction. The same carriers with one endpoint order reversed describe the supplementary configuration.

For this reason XI.10 introduces

```
def HilbertSpaceDirectedParallelSegments
    [H : HilbertIncidence Geo]
    [S : HilbertSpacePrimitive Geo]
    (A B C D : Geo.Point) : Prop :=
  exists pi : S.Plane,
    exists l m t : Geo.Line,
      H.OnLine A l /\
      H.OnLine B l /\
      H.OnLine C m /\
      H.OnLine D m /\
      HilbertLineInPlane Geo l pi /\
      HilbertLineInPlane Geo m pi /\
      HilbertLinesDisjoint Geo l m /\
      H.OnLine B t /\
      H.OnLine D t /\
      HilbertLineInPlane Geo t pi /\
      HilbertSameSideInPlane Geo A C t pi
```

The ordered segments are $A \to B$ and $C \to D$. The connector t passes through the terminal points B, D. The initial points A, C must lie on the same side of that connector in the common plane of the two parallel carriers.

This is a synthetic orientation condition. It contains no coordinates and no selected global orientation of space.

10.6 Formal Statement

The production theorem is:

```
theorem euclid_proposition_11_10
    [H : HilbertIncidence Geo]
    [HilbertPlaneIncidence Geo]
    [S : HilbertSpacePrimitive Geo]
    [HSI : HilbertSpaceIncidence Geo]
    [HSO : HilbertSpaceOrder
      (Geo := Geo) (H := H) (S := S)]
    [HSC : HilbertSpaceCongruence
```

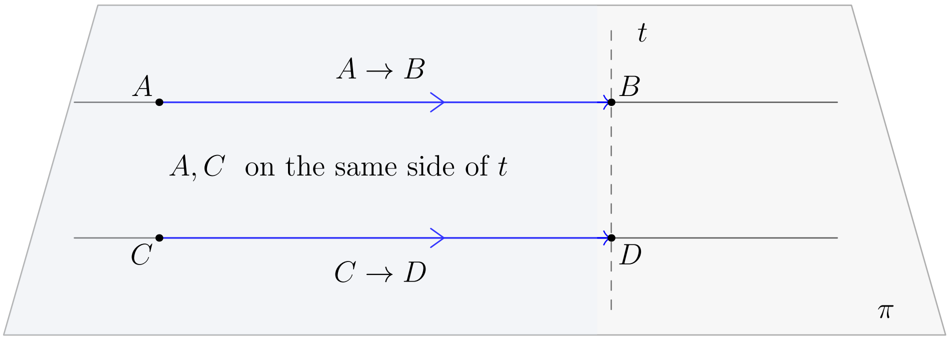


Figure 10.2: Directed parallel segments in a common plane slice. The carriers are parallel, the connector joins the terminal endpoints, and the initial endpoints lie on the same side of the connector. This is the synthetic orientation datum encoded by `HilbertSpaceDirectedParallelSegments`.

```
      (Geo := Geo) (H := H) (S := S)]
    [HSE : HilbertSpaceEuclidean Geo]
    (A B C D E F : Geo.Point)
    (hDirAB_DE :
      HilbertSpaceDirectedParallelSegments Geo A B D E)
    (hDirCB_FE :
      HilbertSpaceDirectedParallelSegments Geo C B F E)
    (hABC : Not (PrimCollinear Geo A B C))
    (hDEF : Not (PrimCollinear Geo D E F))
    (hNoCommonPlane :
      Not (exists omega : S.Plane,
        S.OnPlane A omega /\
        S.OnPlane B omega /\
        S.OnPlane C omega /\
        S.OnPlane D omega /\
        S.OnPlane E omega /\
        S.OnPlane F omega)) :
    Geo.AngleCongruent A B C D E F
```

The hypotheses separate four kinds of information.

1. `hDirAB_DE` and `hDirCB_FE` encode the two corresponding directed parallel pairs.
2. `hABC` and `hDEF` ensure that both displayed angles are nondegenerate.
3. `hNoCommonPlane` states the genuinely spatial condition that the six selected points do not all belong to one plane.
4. The Hilbert typeclass hypotheses provide ambient incidence, spatial order, congruence, and plane-local Euclidean parallel uniqueness.

The conclusion is directly:

```
Geo.AngleCongruent A B C D E F
```

10.7 Spatial / Planar Architecture Used

The ambient layer is:

```
HilbertIncidence
HilbertPlaneIncidence
HilbertSpacePrimitive
HilbertSpaceIncidence
HilbertSpaceOrder
HilbertSpaceCongruence
HilbertSpaceEuclidean
```

The proof never assumes

```
HilbertOrder Geo
HilbertCongruence Geo
```

globally on ambient space.

Whenever an ordinary planar theorem is needed, an explicit plane slice is used:

```
PlanePoint Geo pi
PlaneLine Geo pi
PlaneGeo Geo pi
```

The principal bridges used by XI.10 include:

```
planeGeo_sameSide_iff_space
planeGeo_sameRay_iff_ambient
planeGeo_angleCongruent_iff_ambient
planeGeo_primCollinear_to_ambient
planeGeoAngleToAmbient_angle
hilbert_onPlane_of_primCollinear_with_two_on_plane
hilbert_plane_intersection_line
```

The formalization therefore alternates between ambient three-space and ordinary planar Hilbert geometry, but it never conflates the two.

10.8 Proof Architecture of the Reconstruction

10.8.1 The normalized XI.10 core

The proof was first closed in a normalized form, in which the four selected arms already have the equal lengths used by Euclid:

$$BA \cong ED, \qquad BC \cong EF.$$

At this stage the first two I.33 applications are represented by

```
hilbert_XI10_I33_directed_output
```

and packaged by

```
hilbert_XI10_first_two_I33_directed_package
```

The package supplies explicit carriers AD, CF, and BE, together with

$$AD \parallel BE, \qquad AD \cong BE,$$

$$CF \parallel BE, \qquad CF \cong BE,$$

and, crucially, the directed versions

$$BE \longrightarrow AD, \qquad BE \longrightarrow CF.$$

The ordinary carrier information is enough for XI.9. The directed information is retained for the later orientation step.

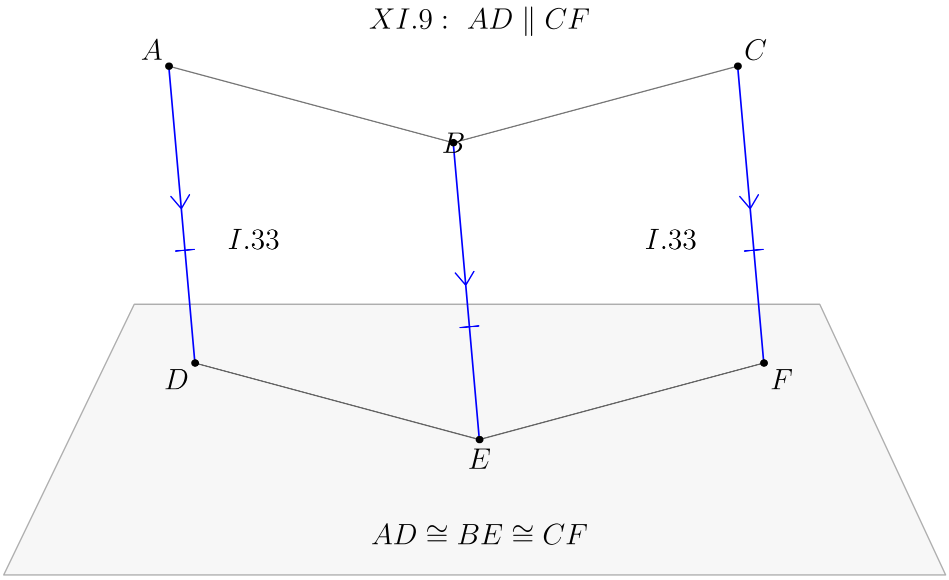


Figure 10.3: The connector state after the first two I.33 applications. The three joining segments AD, BE, and CF are congruent, while XI.9 supplies the carrier relation $AD \parallel CF$. This is the ordinary-parallel part of the normalized core.

10.8.2 The XI.9 step

The two first I.33 applications give two lines parallel to the same line BE. The hypothesis that the six selected points do not belong to one common plane supplies the noncoplanarity clause needed by XI.9.

Thus

```
euclid_proposition_11_9
```

gives

$$AD \parallel CF.$$

Segment congruence transitivity through BE gives

$$AD \cong CF.$$

At this point the unoriented part of the classical proof has been recovered. What remains is to prove that AD and CF have the correct *relative direction* for the third I.33 application.

10.8.3 Plane separation and the section-line bridge

This is the main formal point which is almost invisible in Euclid's printed proof.

Let θ be the plane of the second angle, containing D, E, F. The first two directed I.33 steps imply, in two auxiliary planes, that A and B, and respectively B and C, lie on the appropriate sides of the section lines ED and EF.

To combine these data in ambient space the proof introduces:

```
def HilbertSegmentMeetsPlane ...
def HilbertSameSideOfPlane ...
```

The essential bridge is

```
hilbert_XI10_borsuk_szmielew_52
```

Its content is the following. Suppose two planes π, θ meet along a line t, and $P, Q \in \pi$. Then the segment PQ crosses θ exactly when it crosses the section line t inside π. Equivalently, under the usual off-boundary conditions, same-side in the plane π with respect to t is the same as same-half-space in ambient space with respect to θ.

The source-level identification of this bridge is collected in the Book XI source appendices. Here only its formal geometric role in XI.10 is needed.

No topological separation axiom is added. The bridge is derived from the existing spatial incidence and order structure.

10.8.4 Transitivity of same side of a plane

The next helper is

```
hilbert_XI10_sameSideOfPlane_trans_noncollinear
```

It proves the transitivity needed for three noncollinear points:

$$A \sim_\theta B, \qquad B \sim_\theta C \quad \Longrightarrow \quad A \sim_\theta C.$$

The proof has two geometric cases.

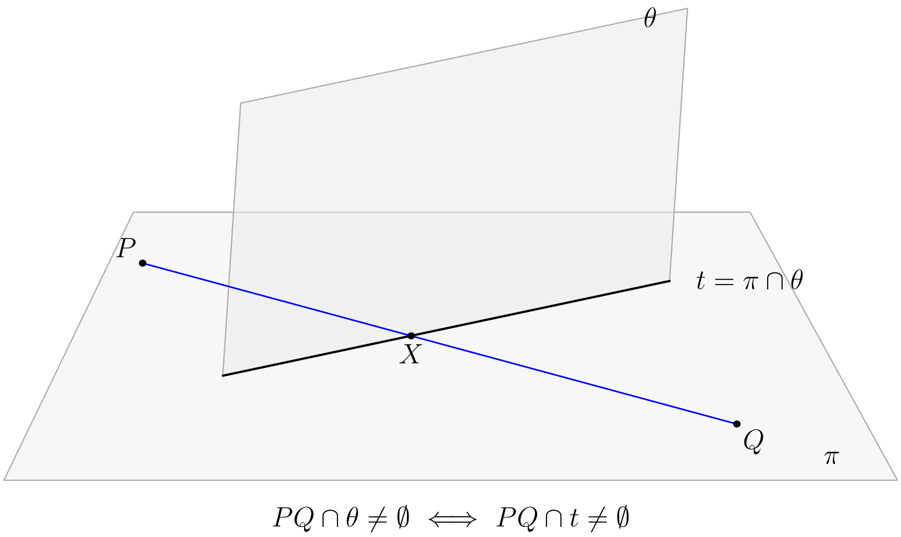


Figure 10.4: A segment PQ in a plane slice crosses the ambient plane θ exactly when it crosses the section line $t = \pi \cap \theta$ inside the slice plane π. This is the geometric bridge between plane-side and half-space information.

If the plane $\alpha = \text{plane}(A, B, C)$ meets θ, their intersection is a line. The section-line / plane-side bridge converts both half-space relations into ordinary same-side relations in α; the planar same-side paths compose there, and the result is converted back.

If α and θ have no common point, every point of the open segment AC remains in α, so that segment cannot meet θ. Hence A and C are automatically on the same side of θ.

This disjoint-plane branch is essential. One cannot assume that the two angle planes intersect.

10.8.5 Recovering the direction $AD \to CF$

The central orientation theorem is

```
hilbert_XI10_directed_AD_CF_from_common_BE
```

It combines

$$BE \longrightarrow AD, \qquad BE \longrightarrow CF,$$

the ordinary conclusion

$$AD \parallel CF,$$

and the noncoplanarity of the two angle configurations.

The proof lifts the two local same-side statements to same-side-of-plane relations with respect to $\theta = \text{plane}(D, E, F)$, composes them there, and then cuts the result back down in the plane containing AD and CF. The section line is DF.

The conclusion is the exact directed statement

$$AD \longrightarrow CF.$$

This is the missing orientation datum which ordinary XI.9 cannot provide.

10.8.6 The third I.33 and spatial SSS

Now both ingredients required by the third I.33 are available:

$$AD \longrightarrow CF, \qquad AD \cong CF.$$

The helper

```
hilbert_XI10_third_I33_AC_DF
```

applies the directed I.33 package and obtains

$$AC \cong DF.$$

The three side congruences are therefore

$$BA \cong ED, \qquad BC \cong EF, \qquad AC \cong DF.$$

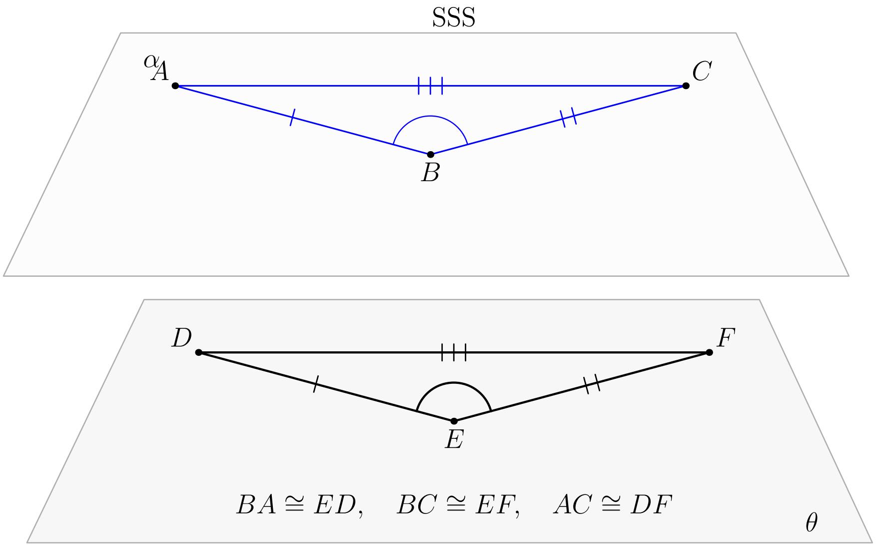


Figure 10.5: The final normalized SSS state. The three pairs of corresponding sides are marked separately: $BA \cong ED$, $BC \cong EF$, and $AC \cong DF$. The two triangles lie in distinct parallel plane slices.

The final normalized step is spatial SSS:

```
hilbert_space_sss_angleA_in_plane
```

Applied to the triangles BAC and EDF, it yields exactly

$$\angle ABC \cong \angle DEF.$$

The target triangle is represented inside the explicit slice PlaneGeo(plane(D, E, F)), while the first triangle remains ambient. The SSS bridge is designed precisely for this cross-plane comparison.

10.8.7 Fixed-connector normalization

For the public theorem a slightly different endpoint orientation is more convenient:

$$AB \longrightarrow DE, \qquad CB \longrightarrow FE.$$

This is the theorem

```
euclid_proposition_11_10_normalized_fixed_connector
```

The advantage is that the connector in both directed-parallel relations is the same fixed line BE. When the outer endpoints are moved along their rays during segment construction, BE does not change.

Two small transport results support this form:

```
hilbert_XI10_directedParallelSegments_swap_pairs
hilbert_XI10_directedParallelSegments_transport_first_endpoints
```

This is a representation choice only. It does not change the geometry of Euclid's proposition.

$$ED_1 \cong AB, \quad EF_1 \cong CB$$

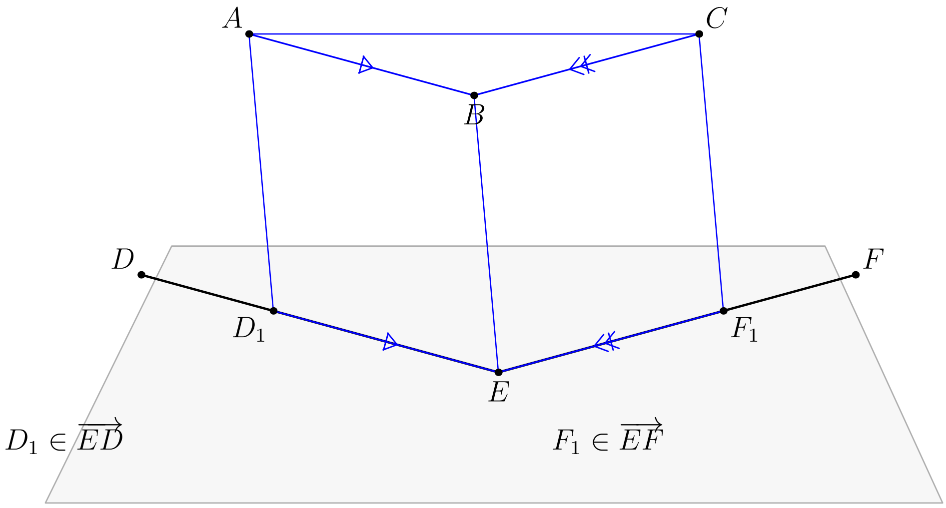


Figure 10.6: Wrapper-level segment construction. Only the points D_1 and F_1 are laid off on the rays ED and EF, with $ED_1 \cong AB$ and $EF_1 \cong CB$. This yields the normalized configuration used by the internal core theorem.

10.8.8 The public wrapper: Euclid's I.3 normalization

The public theorem does not assume equal selected arms.

Instead, spatial segment construction lays off on the rays ED and EF new points D_1, F_1 such that

$$ED_1 \cong AB, \qquad EF_1 \cong CB.$$

Only the second angle is normalized; the points A, B, C are left unchanged. This is formally more economical than reproducing all four layoffs of the classical prose.

Because

$$\mathrm{SameRay}(E, D, D_1), \qquad \mathrm{SameRay}(E, F, F_1),$$

the directed parallel hypotheses transport to

$$AB \longrightarrow D_1E, \qquad CB \longrightarrow F_1E.$$

The fixed-connector normalized theorem gives

$$\angle ABC \cong \angle D_1EF_1.$$

Finally, the SameRay angle bridge is applied inside the genuine plane plane(D, E, F). This is an important architectural point: the proof does *not* install a global

```
[HilbertOrder Geo]
```

on ambient three-space. The ray-angle transport occurs legally in the induced planar geometry and is then mapped back to ambient angle data.

Thus

$$\angle D_1EF_1 = \angle DEF,$$

and the public conclusion follows.

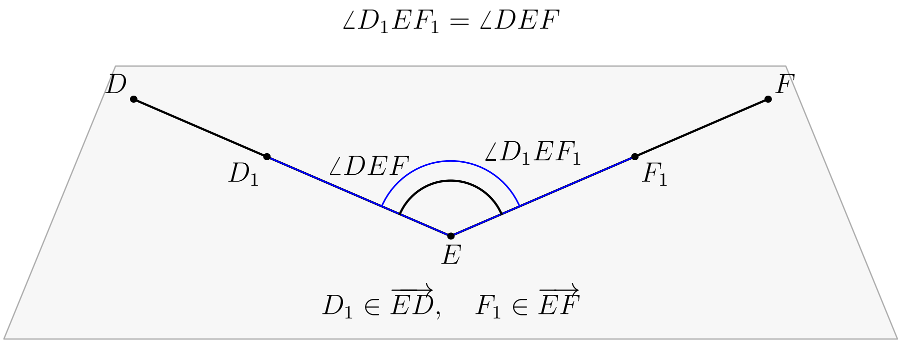


Figure 10.7: Final SameRay angle transport in the plane of DEF. After the normalized proof gives $\angle ABC \cong \angle D_1EF_1$, the facts that D_1 lies on the ray ED and F_1 lies on the ray EF identify the copied angle with the original target angle $\angle DEF$.

10.8.9 Normalized proof spine

The following diagram is intentionally kept in ASCII form. It is the actual proof spine of the normalized Lean reconstruction and should not be replaced by a cosmetically simplified diagram.

```
BA ~= ED, BC ~= EF
directed parallels
        |
   I.33 twice
        |
AD ~= BE, CF ~= BE
        |
       XI.9
        |
```

```
AD || CF
        |
Borsuk-Szmielew 52
        |
AD directed-parallel CF
        |
       I.33
        |
     AC ~= DF
        |
  spatial SSS
        |
angle ABC ~= angle DEF
```

This DAG is not merely a restatement of Euclid's local theorem references. The additional section-line / plane-side stage is the formal replacement for directional plane-separation information that is carried silently by the classical spatial diagram.

10.9 Wyler-Style Flat Reconstruction

XI.10 requires particular care because the flat statement is only a spatial certificate inside a theorem whose actual conclusion is angle congruence. The public target remains

$$\angle ABC \cong \angle DEF.$$

The noncoplanarity assumptions first produce two rank-two carriers:

$$\mathrm{Span}\{A, B, C\} = \mathrm{carrier}(\alpha), \qquad \mathrm{Span}\{D, E, F\} = \mathrm{carrier}(\theta).$$

If $\alpha = \theta$, all six points would be coplanar, contrary to the public hypothesis. Hence

$$\alpha \neq \theta, \qquad \mathrm{Join}(\alpha, \theta) = E^3.$$

This is the rank-three certificate of the configuration.

```
ABC noncollinear      DEF noncollinear
      |                      |
      v                      v
 Span ABC = alpha       Span DEF = theta
          \              /
           alpha != theta
                 |
                 v
        Join(alpha,theta)=E^3
                 |
                 v
     full directed/metric XI.10 proof
                 |
                 v
        AngleCongruent ABC DEF
```

The final two lines are essential. Rank three explains why the configuration is genuinely spatial, but it does not prove the directed-parallel transport or the angle congruence. XI.10 is therefore the main warning against confusing a useful flat invariant with the theorem itself.

10.10 Hidden Formal Data

The classical diagram suppresses several obligations which are explicit in Lean.

10.10.1 Direction on each parallel pair

Carrier parallelism does not determine whether the selected ordered segments point in corresponding directions. XI.10 must store this information explicitly.

10.10.2 The connector line

Each directed parallel pair contains an explicit connector through the terminal endpoints. For the fixed-connector wrapper this line is BE.

10.10.3 The planes of the parallel pairs

The common plane of each pair of spatial parallels is an explicit witness. Disjointness alone would also admit skew lines and is therefore insufficient.

10.10.4 The plane of the second angle

The points D, E, F determine a genuine plane because

$$\neg\,\mathrm{PrimCollinear}(D, E, F).$$

This plane is needed both for the half-space argument and for the final SameRay angle transport.

10.10.5 The intersection line in the section bridge

When two auxiliary planes meet, their common line must be constructed by

```
hilbert_plane_intersection_line
```

and characterized pointwise. It is never read off from a perspective drawing.

10.10.6 Disjoint angle planes are possible

The proof of same-side-of-plane transitivity cannot assume that the two angle planes meet. Write

$$\alpha = \mathrm{plane}(A, B, C), \qquad \theta = \mathrm{plane}(D, E, F).$$

The branch $\alpha \cap \theta = \varnothing$ is a genuine part of the formal proof.

10.10.7 Noncoplanarity after segment layoff

The public wrapper replaces D, F by D_1, F_1 on the same rays. It must prove that a common plane containing A, B, C, D_1, E, F_1 would also contain the original D, F, contradicting the input hypothesis.

10.10.8 No ambient planar order instance

Several old SameRay utilities require `HilbertOrder Geo`. XI.10 does not introduce such an instance. The final ray-angle transport is performed inside the plane DEF, where the induced `PlaneGeo` legitimately has planar Hilbert order.

10.11 Relationship with Earlier Propositions

The direct Book XI dependency is XI.9:

$$AD \parallel BE, \qquad CF \parallel BE \implies AD \parallel CF.$$

The proof also uses I.33 three times in mathematical substance:

1. BA, ED produce AD, BE;
2. BC, EF produce CF, BE;
3. AD, CF produce AC, DF.

The final triangle argument is I.8 / spatial SSS.

Euclid explicitly starts with I.3. In the current public Lean wrapper this role is played directly by

```
HilbertSpaceCongruence.segment_construction
```

rather than by importing a numbered I.3 theorem.

Thus there are two useful dependency descriptions:

Euclid's local DAG	current Lean route
I.3	`segment_construction`
I.33	directed I.33 helper
XI.9	`euclid_proposition_11_9`
I.33	directed I.33 helper
I.8	`hilbert_space_sss_angleA_in_plane`

The additional plane-separation stage has no separate Euclid proposition number. It exposes the section-line content needed to make the directional I.33 chain formally valid.

10.12 Relationship with the Foundations

The production theorem uses spatial incidence, order, congruence, and the Euclidean layer. Hilbert Group IV enters through the directed I.33 machinery and Proposition XI.9, both of which require plane-local uniqueness of parallels. No continuity principle is used.

The additional section-line / plane-side bridge does not introduce a new separation axiom. It is derived from the existing spatial incidence and order structure and serves only to make explicit the directional information which the classical three-dimensional diagram carries visually.

No coordinates, vectors, scalar products, global orientation of space, or numerical angle measures are used.

10.13 Dependency Summary

The classical proof spine is:

```
I.3
 |
I.33 twice
 |
XI.9
 |
I.33
 |
I.8
 |
angle ABC = angle DEF
```

The actual normalized Lean spine is:

```
BA ~= ED, BC ~= EF
directed parallels
        |
   I.33 twice
        |
AD ~= BE, CF ~= BE
        |
       XI.9
        |
AD || CF
        |
Borsuk-Szmielew 52
        |
AD directed-parallel CF
        |
       I.33
        |
     AC ~= DF
        |
  spatial SSS
        |
angle ABC ~= angle DEF
```

The public-wrapper layer is:

```
original rays ED, EF
        |
segment construction
        |
ED1 ~= AB
EF1 ~= CB
```

```
        |
SameRay transport
        |
AB -> D1E
CB -> F1E
        |
normalized XI.10
        |
angle ABC ~= angle D1EF1
        |
SameRay angle transport
        |
angle ABC ~= angle DEF
```

These three diagrams should be read together. The first is Euclid's local theorem DAG. The second is the mathematical core actually realized by the Lean reconstruction. The third records the normalization wrapper needed to expose a public theorem with no prior equal-length hypotheses.

10.14 Conclusion

Proposition XI.10 is the first proposition in this part of Book XI where orientation of spatial parallels becomes unavoidable. Ordinary coplanar-disjoint parallelism suffices for XI.9, but XI.10 must remember which rays correspond. The dedicated `HilbertSpaceDirectedParallelSegments` relation supplies exactly that missing information without adding coordinates or a global orientation of space.

The formal proof preserves Euclid's structure: two I.33 steps create a common parallel BE, XI.9 compares the opposite joining segments, a third I.33 creates the third side equality, and SSS closes the angle comparison. The additional plane-separation bridge explains rigorously how the directional information passes between the relevant plane slices.

The production theorem compiles, the full project build is clean, and the axiom audit is

```
[propext, Classical.choice,
 Geometry.bookZero_nullSegment1, Quot.sound]
```

with no new spatial or continuity axiom.

Chapter 11

Proposition XI.11

11.1 Introduction

Proposition XI.11 is the first Book XI construction in the present reconstruction whose target is the existence of a perpendicular from an arbitrary point outside a plane. Given a plane π and an elevated point $A \notin \pi$, the theorem constructs a line through A that is perpendicular to π.

The proposition is a natural test of the architecture developed in XI.4–XI.8. Its final statement is spatial, but the construction repeatedly moves into ordinary planar Hilbert geometry on explicitly chosen plane slices. In the nontrivial branch three planes are relevant:

$$\pi, \qquad \sigma = \text{plane through } A \text{ and } BC, \qquad \tau = \text{plane through } AD \text{ and } DE.$$

The proof also exposes an important distinction between the theorem proved and the proof route used. The current production proof follows Euclid's construction closely and therefore invokes XI.8. Since XI.8 uses the plane-local Euclidean parallel axiom, the public Lean theorem assumes `HilbertSpaceEuclidean`. This records the strength of the current proof, not a claim that the existence statement itself is axiomatically minimal.

11.2 Euclid's Statement

Euclid's Proposition XI.11 states:

> From a given elevated point to draw a straight line perpendicular to a given plane.

Let A be the elevated point and let π be the plane of reference. The required construction is a line AF through A such that

$$AF \perp \pi.$$

11.3 Classical Configuration

Choose a straight line BC at random in the plane of reference. Draw AD from A perpendicular to BC. If AD is already perpendicular to the reference plane, the construction is complete.

Otherwise draw DE in the reference plane through D, with

$$DE \perp BC.$$

Then draw AF from A perpendicular to DE, and through F draw GH parallel to BC.

Thus the reference plane contains

$$BC, \qquad DE, \qquad GH,$$

while the two lines

$$AD, \qquad AF$$

leave the reference plane and meet it at D and F, respectively.

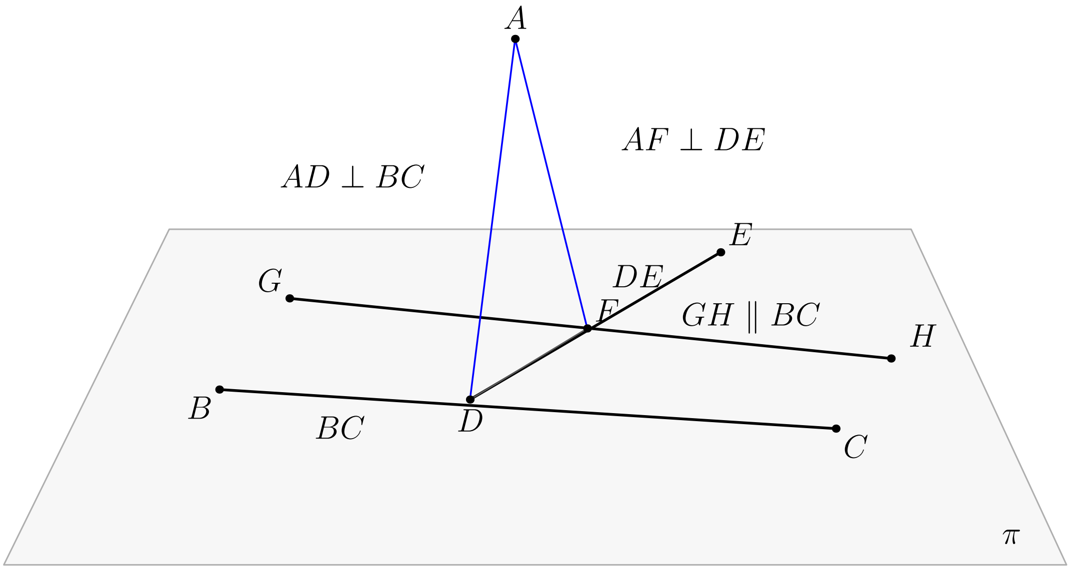


Figure 11.1: The classical configuration of Proposition XI.11. Black lines lie in the reference plane π; blue segments leave π. The construction starts with $AD \perp BC$, then uses $DE \perp BC$, $AF \perp DE$, and $GH \parallel BC$. The desired conclusion is $AF \perp \pi$.

11.4 Classical Proof

The classical proof is short but structurally dense.

First, choose $BC \subset \pi$ and draw

$$AD \perp BC$$

from the elevated point A by Proposition I.12. Euclid immediately splits into two cases.

If $AD \perp \pi$, the required perpendicular has already been drawn.

Assume therefore that AD is not perpendicular to π. In the reference plane draw

$$DE \perp BC$$

through D by Proposition I.11. Next draw

$$AF \perp DE$$

from A by a second application of I.12. Finally draw through F

$$GH \parallel BC$$

by I.31.

Now BC is perpendicular to both DA and DE. These two lines intersect at D, so Proposition XI.4 gives

$$BC \perp \text{plane}(ED, DA).$$

Since $GH \parallel BC$, Proposition XI.8 gives

$$GH \perp \text{plane}(ED, DA).$$

By XI.Def.3, a line perpendicular to a plane is perpendicular to every line of that plane through the foot. The line AF belongs to plane(ED, DA) and meets GH at F. Hence

$$GH \perp AF,$$

and therefore also

$$AF \perp GH.$$

But $AF \perp DE$ by construction. Thus AF is perpendicular at F to the two intersecting lines DE and GH of the reference plane. A second application of XI.4 yields

$$AF \perp \text{plane}(DE, GH).$$

The plane through DE and GH is the plane of reference, hence

$$AF \perp \pi.$$

The local classical dependency chain is therefore

$$\begin{gathered}
A \notin \pi, \quad BC \subset \pi \\
\Downarrow \\
\text{I.12} : AD \perp BC \\
\Downarrow \\
AD \perp \pi \quad \text{or} \quad AD \not\perp \pi \\
\Downarrow \\
\text{I.11} : DE \perp BC \\
\Downarrow \\
\text{I.12} : AF \perp DE \\
\Downarrow \\
\text{I.31} : GH \parallel BC \\
\Downarrow \\
\text{XI.4} : BC \perp \text{plane}(ED, DA) \\
\Downarrow \\
\text{XI.8} : GH \perp \text{plane}(ED, DA) \\
\Downarrow \\
\text{XI.Def.3} : GH \perp AF \\
\Downarrow \\
AF \perp DE, \quad AF \perp GH \\
\Downarrow \\
\text{XI.4} : AF \perp \pi.
\end{gathered}$$

No use of XI.9 or XI.10 occurs in this classical chain. The Book XI dependencies needed locally are XI.4 and XI.8.

11.5 What is genuinely three-dimensional here?

The proof is not simply a planar perpendicular construction repeated in space. The essential issue is selecting the correct plane for each planar construction and then transporting its conclusion back to ambient space.

The first auxiliary plane is

$$\sigma = \text{plane}(A, BC).$$

It exists because $A \notin BC$. Inside σ, the point A and the line BC are ordinary planar objects, so I.12 can be applied in the induced geometry PlaneGeo(σ).

After the foot D is obtained, the line DE is constructed inside the reference plane π.

The second auxiliary plane is

$$\tau = \text{plane}(AD, DE).$$

The lines AD and DE intersect at D and are distinct. Inside PlaneGeo(τ), the second I.12 construction produces the line $AF \perp DE$.

The three plane slices have a particularly transparent intersection pattern:

$$\pi \cap \sigma = BC, \qquad \sigma \cap \tau = AD, \qquad \pi \cap \tau = DE,$$

at the level of the carriers used in the construction.

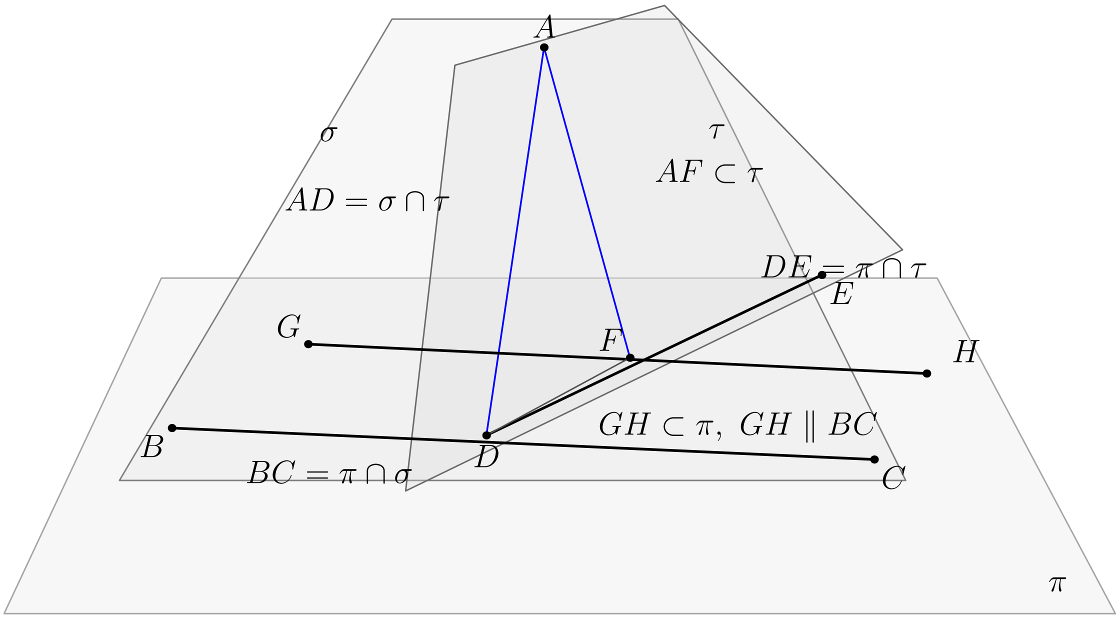


Figure 11.2: The three plane slices used by the reconstruction. The reference plane π contains BC, DE, GH. The plane σ carries the first planar perpendicular construction and shares BC with π. The plane τ carries AD, DE, AF; it shares AD with σ and DE with π. The final argument transfers perpendicularity from BC to GH through XI.8 and then applies XI.4 in π.

The figure should not be interpreted as a coordinate model. It is only a perspective rendering of incidence and plane membership. The Lean proof uses no coordinates, vectors, scalar products, normals, or numerical angle measure.

11.6 Formal Statement

The production theorem is:

```
theorem euclid_proposition_11_11
    [H : HilbertIncidence Geo]
    [HilbertPlaneIncidence Geo]
    [S : HilbertSpacePrimitive Geo]
    [HSI : HilbertSpaceIncidence Geo]
    [_HSO : HilbertSpaceOrder
      (Geo := Geo) (H := H) (S := S)]
    [HSC : HilbertSpaceCongruence
      (Geo := Geo) (H := H) (S := S)]
    [HSE : HilbertSpaceEuclidean Geo]
    (pi : S.Plane)
    (A : Geo.Point)
    (hApi : Not (S.OnPlane A pi)) :
    exists l F,
      H.OnLine A l /\
      HilbertLinePerpendicularPlaneAt Geo l pi F
```

The conclusion returns both the perpendicular carrier l and its foot F. It asserts that A lies on l and that l is perpendicular to π at F.

The predicate

```
HilbertLinePerpendicularPlaneAt Geo l pi F
```

is the spatial target relation established for XI.4. It expresses XI.Def.3 synthetically: F lies on the line and in the plane, and the line is perpendicular to every line of the plane through F.

11.7 Spatial / Planar Architecture Used

11.7.1 Ambient spatial layer

The ambient construction uses

```
HilbertSpacePrimitive
HilbertSpaceIncidence
HilbertSpaceOrder
HilbertSpaceCongruence
HilbertSpaceEuclidean
HilbertLineInPlane
HilbertSpaceLinesParallel
HilbertLinesPerpendicularAt
HilbertLinePerpendicularPlaneAt
```

There is no global ambient instance

```
HilbertOrder Geo
```

and no global ambient instance

```
HilbertCongruence Geo
```

installed merely to make planar theorems available in space.

11.7.2 Induced planar geometry

Planar constructions are performed in explicitly named plane slices using

```
PlanePoint
PlaneLine
PlaneGeo
```

The proof uses three instances of induced planar geometry:

$$\text{PlaneGeo}(\sigma), \qquad \text{PlaneGeo}(\pi), \qquad \text{PlaneGeo}(\tau).$$

The first I.12 construction is performed in PlaneGeo(σ). The construction of DE and the I.31

parallel construction are performed in PlaneGeo(π). The second I.12 construction is performed in PlaneGeo(τ).

11.7.3 Plane–space bridges

The principal bridges used by the proof include

```
planeGeo_rightAngle_iff_ambient
planeGeo_linesDisjoint_iff_ambient
HilbertSpaceIncidence.line_in_plane
HilbertLinePerpendicularPlaneAt.incidence
HilbertLinePerpendicularPlaneAt.perpendicular_to_line
```

The proof also uses the ambient symmetry normalizer

```
hilbert_space_linesPerpendicularAt_symm
```

when XI.4 requires the perpendicularity relation in the opposite carrier orientation.

11.8 Proof Architecture of the Reconstruction

The formal proof follows the classical route but expands the incidence and plane-selection steps that the printed diagram leaves implicit.

11.8.1 1. Choosing the reference line

The proof first obtains three noncollinear points of the reference plane:

```
hilbert_three_noncollinear_on_plane
```

Two of them, B, C, determine the line bc. The theorem

```
HilbertSpaceIncidence.line_in_plane
```

then proves

$$bc \subset \pi.$$

Since $A \notin \pi$, it follows immediately that

$$A \notin bc.$$

11.8.2 2. Constructing the first auxiliary plane

The helper

```
hilbert_plane_through_line_and_external_point
```

constructs the plane σ containing bc and A.

This step supplies exactly the missing incidence premise needed before a planar I.12 theorem can be applied. The classical phrase "draw AD from A perpendicular to BC" presupposes that A and BC have first been put in a common plane.

11.8.3 3. First I.12 construction in σ

Inside

$$\mathrm{PlaneGeo}(\sigma)$$

the theorem

```
hilbert_perpendicular_from_point_exists
```

constructs the foot D on bc and gives a right-angle witness. The ambient carrier ad through D, A is then constructed explicitly and proved to lie in σ.

The right-angle data are packaged as

```
HilbertLinesPerpendicularAt Geo bc ad D
```

using the bridge

```
planeGeo_rightAngle_iff_ambient
```

and explicit noncollinearity.

11.8.4 4. Euclid's case split

The formal proof uses the explicit split

```
by_cases hADpi :
    HilbertLinePerpendicularPlaneAt Geo ad pi D
```

If this proposition holds, the proof returns ad, D immediately.

This is not a Lean-specific mathematical invention. It is exactly Euclid's sentence: if AD is also perpendicular to the plane of reference, the construction is already complete.

11.8.5 5. Constructing DE in the reference plane

In the nontrivial branch the point D is known to lie in π. The formal proof constructs opposite points on bc in PlaneGeo(π) and then uses

```
hilbert_right_angle_exists_nondegenerate
```

to obtain a nondegenerate right-angle direction through D. The resulting carrier de satisfies

$$de \subset \pi, \qquad bc \perp de \text{ at } D.$$

Thus the formal proof realizes the role of Euclid I.11 without importing a separate I.11 proposition wrapper in the production file.

11.8.6 6. Constructing the second auxiliary plane

The proof establishes

$$ad \neq de.$$

Indeed, if $ad = de$, then A would lie on $de \subset \pi$, contrary to $A \notin \pi$.

The theorem

```
hilbert_plane_through_two_intersecting_lines
```

then constructs the plane τ containing ad and de.

11.8.7 7. Second I.12 construction in τ

Inside

$$\text{PlaneGeo}(\tau)$$

the proof applies

```
hilbert_perpendicular_from_point_exists
```

a second time, now from A to de. This yields the foot F and the carrier af, with

$$af \subset \tau, \qquad de \perp af \text{ at } F.$$

Because $de \subset \pi$, the point F also lies in the reference plane.

11.8.8 8. The hidden fact $F \notin BC$

Before I.31 can be applied through F, the formal proof must establish that F does not lie on bc.

Assume $F \in bc$. Since D, F lie on both bc and de, line uniqueness forces $F = D$. The two lines ad and af then contain the two distinct points D, A, so line uniqueness forces

$$ad = af.$$

The proof already knows

$$ad \perp bc, \qquad ad \perp de$$

at D. Since $bc, de \subset \pi$, XI.4 would imply

$$ad \perp \pi,$$

contradicting the nontrivial branch assumption.

Therefore

$$F \notin bc.$$

This is one of the main pieces of hidden formal data in XI.11.

11.8.9 9. I.31 in the reference plane

Now B, C, F are noncollinear in PlaneGeo(π). The proof applies

```
euclid_proposition_31
```

to obtain a point G and a line gh through F, G parallel to bc in the induced plane.

The result of I.31 is initially planar point-pair parallelism. The proof converts it to disjointness of the actual carriers using

```
hilbert_mem_pointLine_iff_onLine
```

and then transports planar disjointness back to ambient space by

```
planeGeo_linesDisjoint_iff_ambient
```

Finally it packages

```
HilbertSpaceLinesParallel Geo bc gh
```

with the reference plane π as the explicit common carrier plane.

11.8.10 10. First XI.4: $BC \perp \tau$

In the plane τ, the lines ad and de are distinct and meet at D. The proof has

$$bc \perp ad, \qquad bc \perp de.$$

Therefore

```
euclid_proposition_11_4
```

gives

$$bc \perp \tau.$$

This is the first use of XI.4 in the construction.

11.8.11 11. XI.8: transfer to the parallel GH

The proof now has

$$bc \parallel gh, \qquad bc \perp \tau.$$

It applies

```
euclid_proposition_11_8
```

and obtains a point K such that

$$gh \perp \tau \text{ at } K.$$

This is exactly the classical XI.8 step, but the formal theorem returns the foot existentially. Euclid's diagram already identifies the relevant intersection as F; Lean requires that identification to be proved.

11.8.12 12. Normalizing the XI.8 foot: $K = F$

The proof knows from the XI.8 conclusion that

$$K \in gh, \qquad K \in \tau,$$

and from the construction that

$$F \in gh, \qquad F \in \tau.$$

Suppose $K \neq F$. Since the two distinct points K, F lie on gh and in τ, spatial incidence gives

$$gh \subset \tau.$$

But $gh \perp \tau$ at K. Applying the universal line–plane-perpendicularity condition to the line gh itself would give

$$gh \perp gh.$$

The theorem

```
hilbert_linesPerpendicularAt_ne
```

then yields the absurd conclusion $gh \neq gh$.

Hence

$$K = F.$$

This argument is purely synthetic and uses the nondegeneracy already built into `HilbertLinesPerpendicularAt`.

11.8.13 13. XI.Def.3 gives $AF \perp GH$

After replacing K by F, the proof has

$$gh \perp \tau \text{ at } F.$$

Since

$$af \subset \tau, \qquad F \in af,$$

the universal line condition is unpacked by

```
HilbertLinePerpendicularPlaneAt.perpendicular_to_line
```

to obtain

$$gh \perp af.$$

The symmetry helper

```
hilbert_space_linesPerpendicularAt_symm
```

normalizes this to

$$af \perp gh.$$

The same helper normalizes the previously constructed $de \perp af$ to

$$af \perp de.$$

11.8.14 14. Proving $DE \neq GH$

The final XI.4 application requires two distinct lines of the reference plane. This distinctness is not left to the diagram.

Suppose

$$de = gh.$$

Since $D \in de$, it follows that $D \in gh$. But $D \in bc$, while the spatial parallelism package $bc \parallel gh$ contains disjointness of these two ambient carriers. Contradiction.

Thus

$$de \neq gh.$$

11.8.15 15. Final XI.4 in the reference plane

Both de and gh lie in π, meet at F, and are distinct. The proof has

$$af \perp de, \qquad af \perp gh.$$

A final call to

```
euclid_proposition_11_4
```

therefore gives

$$af \perp \pi \text{ at } F.$$

The returned witnesses are af and F.

11.9 Wyler-Style Flat Reconstruction

XI.11 begins with the canonical rank jump from a plane and an external point. For $A \notin \pi$, the flat layer records

$$\mathrm{Join}(\mathrm{carrier}(\pi), \{A\}) = E^3.$$

The converse characterization is also natural: if the point were already in the plane, adjoining it would not increase the generated flat. Thus externality is precisely the condition that the plane plus the point generate the whole three-space.

The perpendicular construction itself remains genuinely metric. The Wyler certificate organizes the ambient position but does not replace Euclid's construction:

```
A outside pi
     |
     v
Join(pi,{A}) = E^3
     |
choose a line bc in pi
     |
construct the required auxiliary plane slices
     |
I.12, I.31, XI.4, XI.8
     |
     v
AF perp pi
```

The value of the flat statement is therefore diagnostic: it proves at the outset that the construction uses the full ambient rank. The remaining steps explain how the perpendicular is actually obtained synthetically.

11.10 Lean Formalization

The production module imports

```
import CGJteamLab.Proposition11_8
import CGJteamLab.Proposition12
import CGJteamLab.Proposition31
```

The direct imports therefore expose XI.8, the Book I perpendicular-from- external-point construction used as I.12, and I.31.

XI.4 is invoked twice by the proof and is available transitively through the established Book XI import chain. The production theorem does not import or invoke XI.9 or XI.10.

The principal reusable declarations consumed by XI.11 are:

```
hilbert_three_noncollinear_on_plane
hilbert_plane_through_line_and_external_point
hilbert_perpendicular_from_point_exists
planeGeo_rightAngle_iff_ambient
planeGeo_opposite_points_on_line_congruent_to
hilbert_right_angle_exists_nondegenerate
hilbert_plane_through_two_intersecting_lines
hilbert_space_linesPerpendicularAt_symm
euclid_proposition_31
hilbert_mem_pointLine_iff_onLine
planeGeo_linesDisjoint_iff_ambient
euclid_proposition_11_4
euclid_proposition_11_8
HilbertLinePerpendicularPlaneAt.incidence
HilbertLinePerpendicularPlaneAt.perpendicular_to_line
```

```
hilbert_linesPerpendicularAt_ne
```

No XI.11-specific helper theorem is exported. The full construction is contained in the proof of `euclid_proposition_11_11`.

11.11 Hidden Formal Data

The classical diagram suppresses several obligations that are explicit in the Lean proof.

11.11.1 The first I.12 needs a plane

The statement $A \notin \pi$ does not by itself put A and the chosen line $BC \subset \pi$ in a common planar geometry. The plane σ through A and BC must be constructed before planar I.12 can be invoked.

11.11.2 Every constructed line needs a carrier proof

After constructing ambient lines ad, de, af, and gh, the proof records explicitly which plane contains each line. These are the data that permit later calls to XI.4, XI.8, and induced-plane theorems.

11.11.3 The two feet are not automatically the same

XI.8 returns a foot K of gh on τ. The earlier construction already supplied the point $F \in gh \cap \tau$. The equality $K = F$ is a real incidence theorem, not a definitional simplification.

11.11.4 Distinctness for the final XI.4

The statement $de \neq gh$ follows from the disjointness part of $bc \parallel gh$, but it must be proved explicitly before the two lines can be packaged as distinct `PlaneLine`s of π.

11.11.5 Orientation of perpendicularity

The project relation `HilbertLinesPerpendicularAt` carries ordered line arguments. Several classical sentences silently use symmetry of perpendicularity. The proof normalizes these orientations explicitly with `hilbert_space_linesPerpendicularAt_symm`.

11.12 Relationship with Earlier Book XI Propositions

The only earlier Book XI propositions invoked by the mathematical proof are XI.4 and XI.8.

XI.4 is used twice:

$$bc \perp ad, \quad bc \perp de \Longrightarrow bc \perp \tau,$$

and later

$$af \perp de, \quad af \perp gh \Longrightarrow af \perp \pi.$$

XI.8 is used once:

$$bc \parallel gh, \quad bc \perp \tau \Longrightarrow gh \perp \tau.$$

XI.9 and XI.10 are not local dependencies of XI.11. The production Lean source neither imports nor invokes them, so they are absent from the actual Lean DAG for XI.11 as well as from the classical local dependency chain.

XI.7 is also not a local dependency of XI.11. The spatial parallelism used here is created directly by I.31 inside the reference plane and then packaged as `HilbertSpaceLinesParallel`.

11.13 Relationship with Book I / Book Zero / Hilbert Infrastructure

The classical Book I dependencies are:

$$\text{I.12}, \quad \text{I.11}, \quad \text{I.12}, \quad \text{I.31}.$$

The formal proof calls the established I.12 theorem

```
hilbert_perpendicular_from_point_exists
```

twice, once in σ and once in τ.

The role of I.11 is reconstructed from the lower-level right-angle construction interface in PlaneGeo(π). The proof does not import a separate proposition-I.11 wrapper.

The I.31 step is represented directly by

```
euclid_proposition_31
```

inside PlaneGeo(π).

Book Zero enters transitively through the established planar Hilbert construction and right-angle infrastructure. XI.11 introduces no new Book Zero declaration.

The ambient three-space remains separated from planar order and congruence. All ordinary planar constructions are carried out in a named `PlaneGeo`; the spatial classes provide only the incidence, congruence transport, and plane-local Euclidean data required to connect those slices.

11.14 Relationship with the Foundations

The current Euclid-faithful proof of XI.11 uses Hilbert Group IV because it invokes Proposition XI.8. Thus the public theorem carries `HilbertSpaceEuclidean Geo` as a genuine geometric hypothesis of this proof route. No continuity principle is used.

This records the strength of the present reconstruction, not a minimality claim about the abstract target statement. The rest of the argument keeps planar order and congruence inside explicit plane slices rather than installing global planar instances on ambient three-space.

11.15 Dependency Summary

The classical local chain is

$$\begin{array}{c} A \notin \pi, \quad BC \subset \pi \\ \downarrow \\ \text{I.12} \\ \downarrow \\ AD \perp BC \\ \downarrow \\ \text{case } AD \perp \pi \text{ or not} \\ \downarrow \\ \text{I.11, I.12, I.31} \\ \downarrow \\ DE \perp BC, \quad AF \perp DE, \quad GH \parallel BC \\ \downarrow \\ \text{XI.4} \\ \downarrow \\ BC \perp \text{plane}(ED, DA) \\ \downarrow \\ \text{XI.8} \\ \downarrow \\ GH \perp \text{plane}(ED, DA) \\ \downarrow \\ \text{XI.Def.3} \\ \downarrow \\ AF \perp GH \\ \downarrow \\ \text{XI.4} \\ \downarrow \\ AF \perp \pi. \end{array}$$

The actual formal chain is

$$
\begin{array}{c}
\text{Hilbert spatial incidence/order/congruence} \\
\downarrow \\
\texttt{HilbertSpaceEuclidean}\ \text{(Group IV)} \\
\downarrow \\
bc \subset \pi, \quad A \notin bc \\
\downarrow \\
\sigma = \text{plane}(A, bc) \\
\downarrow \\
\text{I.12 in PlaneGeo}(\sigma) \\
\downarrow \\
bc \perp ad \\
\downarrow \\
\texttt{by\_cases}\ ad \perp \pi \\
\downarrow \\
de \subset \pi, \quad bc \perp de \\
\downarrow \\
\tau = \text{plane}(ad, de) \\
\downarrow \\
\text{I.12 in PlaneGeo}(\tau) \\
\downarrow \\
de \perp af, \quad F \in \pi \\
\downarrow \\
F \notin bc \\
\downarrow \\
\text{I.31 in PlaneGeo}(\pi) \\
\downarrow \\
\texttt{HilbertSpaceLinesParallel}\ bc\, gh \\
\downarrow \\
\text{XI.4 in } \tau \\
\downarrow \\
bc \perp \tau \\
\downarrow \\
\text{XI.8} \\
\downarrow \\
gh \perp \tau \text{ at } K \\
\downarrow \\
K = F \\
\downarrow \\
\text{XI.Def.3} : af \perp gh \\
\downarrow \\
de \neq gh \\
\downarrow \\
\text{XI.4 in } \pi \\
\downarrow \\
\texttt{euclid\_proposition\_11\_11}.
\end{array}
$$

The dependency classification is:

New proposition-specific global axiom	no
New spatial axiom class	no
Continuity	no
Group IV / Euclidean parallelism	yes, via XI.8
Direct production imports	XI.8, I.12 module, I.31
Classical explicit Book XI dependencies	XI.4, XI.8
XI.9 / XI.10	not used
New proposition-local helper theorem	no
New PlaneGeo bridge	no
Coordinates / vectors / scalar product	no
`sorry/admit`	no

11.16 Conclusion

Proposition XI.11 is a compact classical construction whose formal reconstruction exercises almost the entire plane–space interface developed so far. The key mathematical idea remains Euclid's: construct two successive planar perpendiculars, introduce a parallel in the reference plane, transfer a line–plane perpendicularity by XI.8, and close with XI.4.

Lean makes the spatial bookkeeping visible. The first I.12 needs the plane σ; the second needs the plane τ; the I.31 construction must be converted from planar parallelism to ambient spatial parallelism; the foot returned by XI.8 must be proved equal to F; and the final two reference-plane carriers must be proved distinct.

No analytic geometry enters. The production theorem is fully synthetic, compiles in the current project, and has the inherited axiom audit

$$[\texttt{propext}, \texttt{Classical.choice}, \texttt{Geometry.bookZero\_nullSegment1}, \texttt{Quot.sound}].$$

Its present Group IV hypothesis is inherited from the Euclid-faithful use of XI.8. Whether that hypothesis can be reduced is a separate strengthening question; the source-side comparison is kept in the appendices rather than repeated in this proposition chapter.

Chapter 12

Proposition XI.12

12.1 Introduction

Proposition XI.12 is the companion construction to XI.11. Proposition XI.11 constructs a perpendicular to a plane from a point outside the plane; XI.12 starts with a point A already lying in the reference plane π and constructs a line through A perpendicular to π.

The classical proof is exceptionally short. Euclid chooses an elevated point, uses XI.11 to obtain one perpendicular to the reference plane, draws through A a parallel to that perpendicular by I.31, and then uses XI.8 to transfer the line–plane perpendicularity to the new line.

The Lean reconstruction follows this classical route, but two incidence obligations that are invisible in the printed proof become explicit. First, the perpendicular obtained from XI.11 may already meet the reference plane at A. In that case it is itself the required line, and there is no new parallel to construct through A. Second, XI.8 returns the foot of the new perpendicular existentially; Lean must prove that this foot is the already specified point A.

These two obligations are handled by the proposition-local theorem `hilbert_XI12_perpendicular_foot_unique`. The other proposition-local helper, `hilbert_XI12_parallel_through_point_in_plane`, isolates the planar I.31 construction and packages its result as ambient spatial parallelism.

12.2 Euclid's Statement

Euclid's Proposition XI.12 states:

> To set up a straight line at right angles to a given plane from a given point in it.

Let A be the given point of the reference plane π. The required construction is a straight line through A satisfying

$$AD \perp \pi.$$

12.3 Classical Configuration

Let B be an arbitrary elevated point. By XI.11 draw BC perpendicular to the reference plane, with $C \in \pi$. Through the given point $A \in \pi$, draw AD parallel to BC.

The generic classical configuration is shown in Figure 12.1. The two blue lines leave the reference plane. Their hidden continuations below the plane are dashed, following the standing Book XI figure convention.

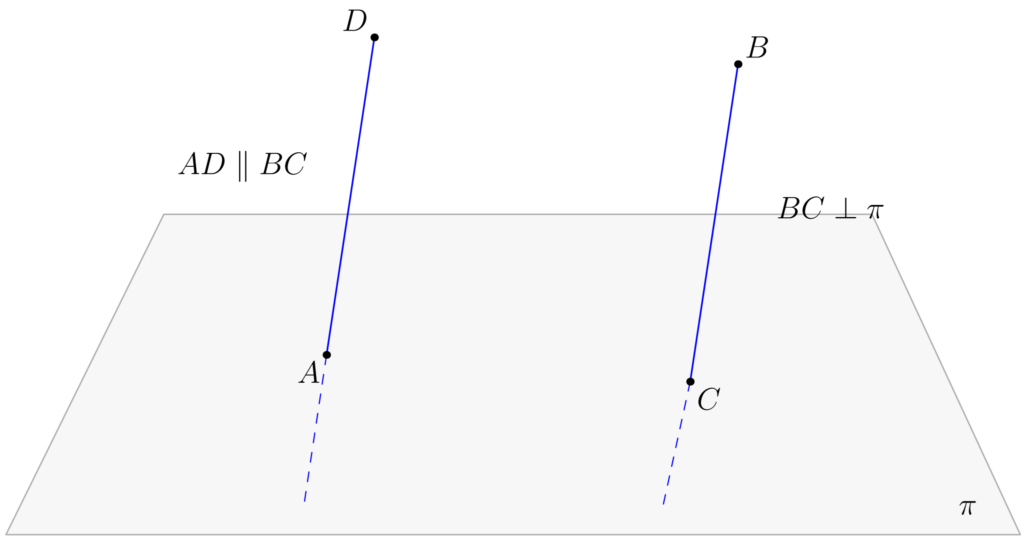


Figure 12.1: The classical configuration of Proposition XI.12. The points A, C lie in the reference plane π. XI.11 gives $BC \perp \pi$, and I.31 draws $AD \parallel BC$ through A. XI.8 then gives $AD \perp \pi$. The figure shows the generic case $A \neq C$; if $A = C$, the line BC already solves the construction.

12.4 Classical Proof

Let A be the given point in the reference plane. Choose any elevated point B. By Proposition XI.11 draw

$$BC \perp \pi.$$

Through A, draw

$$AD \parallel BC$$

by Proposition I.31. Since AD and BC are parallel and BC is perpendicular to the reference plane, Proposition XI.8 gives

$$AD \perp \pi.$$

Thus AD is the required straight line.

The local classical dependency chain is therefore

$$\begin{gathered}
A \in \pi \\
\Downarrow \\
\text{choose an elevated point } B \\
\Downarrow \\
\text{XI.11} \\
\Downarrow \\
BC \perp \pi \\
\Downarrow \\
\text{I.31} \\
\Downarrow \\
AD \parallel BC, \quad A \in AD \\
\Downarrow \\
\text{XI.8} \\
\Downarrow \\
AD \perp \pi.
\end{gathered}$$

Neither XI.9 nor XI.10 occurs in this local proof. The only earlier Book XI propositions used explicitly are XI.11 and XI.8.

12.5 What is genuinely three-dimensional here?

The I.31 step is planar, but the input line BC is not a line of the reference plane: it is perpendicular to that plane and meets it only at its foot C. To apply a planar parallel theorem through A, the formal reconstruction first has to place the spatial line and the point A in a common auxiliary plane.

In the nontrivial branch this plane is

$$\sigma = \text{plane}(BC, A).$$

Inside PlaneGeo(σ), I.31 becomes an ordinary planar construction. Its output must then be transported back to ambient three-space and packaged as

$$\text{HilbertSpaceLinesParallel}(BC, AD).$$

Only after this packaging can XI.8 be applied.

The other genuinely spatial ingredient is the meaning of a line perpendicular to a plane. The predicate

```
HilbertLinePerpendicularPlaneAt Geo l pi A
```

contains the incidence of the foot with both line and plane and the universal assertion that the line is perpendicular to every line of the plane through the foot. This universal structure is what makes the foot-uniqueness argument possible.

12.6 Formal Statement

The production theorem is:

```
theorem euclid_proposition_11_12
    [H : HilbertIncidence Geo]
    [HilbertPlaneIncidence Geo]
    [S : HilbertSpacePrimitive Geo]
    [HSI : HilbertSpaceIncidence Geo]
    [_HSO : HilbertSpaceOrder
      (Geo := Geo) (H := H) (S := S)]
    [HSC : HilbertSpaceCongruence
      (Geo := Geo) (H := H) (S := S)]
    [HSE : HilbertSpaceEuclidean Geo]
    (pi : S.Plane)
    (A : Geo.Point)
    (hApi : S.OnPlane A pi) :
    exists l : Geo.Line,
      HilbertLinePerpendicularPlaneAt Geo l pi A
```

The theorem therefore returns an ambient line l that is perpendicular to the given plane exactly at the prescribed point A.

12.7 Spatial / Planar Architecture Used

12.7.1 Ambient spatial layer

The ambient proof uses

```
HilbertSpacePrimitive
HilbertSpaceIncidence
HilbertSpaceOrder
HilbertSpaceCongruence
HilbertSpaceEuclidean
HilbertLineInPlane
HilbertSpaceLinesParallel
HilbertLinePerpendicularPlaneAt
```

As throughout the current Book XI architecture, there is no global ambient

```
HilbertOrder Geo
HilbertCongruence Geo
```

installed to make plane geometry available in space.

12.7.2 Induced planar geometry

The only new plane slice constructed specifically by XI.12 is σ, the plane through the XI.11 perpendicular and the point A, in the branch where A does not already lie on that line.

The I.31 construction is then carried out in

$$\mathrm{PlaneGeo}(\sigma).$$

The helper `hilbert_XI12_parallel_through_point_in_plane` selects two points of the ambient line, views them together with A as `PlanePoint`s of σ, applies the established planar parallel-existence theorem, and converts the result back to an ambient line.

12.7.3 Plane–space bridges

The principal bridge operations used locally are

```
planeGeo_primCollinear_to_ambient
hilbert_on_line_of_primCollinear_with_two_on_line
hilbert_mem_pointLine_iff_onLine
HilbertSpaceIncidence.line_in_plane
```

The planar I.31 output is not identified with ambient spatial parallelism by definition. The proof explicitly establishes ambient disjointness and supplies σ as the common carrier plane.

12.8 Proof Architecture of the Reconstruction

12.8.1 1. Choosing an elevated point and invoking XI.11

The proof first calls

```
hilbert_point_off_plane
```

to choose a point $B \notin \pi$. It then invokes

```
euclid_proposition_11_11
```

and obtains an ambient line l and a point F such that

$$B \in l, \qquad l \perp \pi \text{ at } F.$$

At this point the formal proof has exactly the spatial object that Euclid denotes by BC.

12.8.2 2. The exceptional incidence branch

The classical text immediately asks for a parallel through A. Formally, however, one must first decide whether A already lies on the XI.11 line l:

```
by_cases hAl : H.OnLine A l
```

If $A \in l$, then both A and the perpendicular foot F lie in the reference plane and on l. The helper

```
hilbert_XI12_perpendicular_foot_unique
```

proves

$$A = F.$$

Hence $l \perp \pi$ at A, and the theorem is complete in this branch. No I.31 construction is needed.

12.8.3 3. Why the perpendicular foot is unique

The helper has the normalized signature

```
theorem hilbert_XI12_perpendicular_foot_unique
    [H : HilbertIncidence Geo]
    [HilbertPlaneIncidence Geo]
    [S : HilbertSpacePrimitive Geo]
    [HSI : HilbertSpaceIncidence Geo]
    (pi : S.Plane)
    (l : Geo.Line)
    (C A : Geo.Point)
    (hPerp :
      HilbertLinePerpendicularPlaneAt Geo l pi C)
    (hAl : H.OnLine A l)
    (hApi : S.OnPlane A pi) :
    A = C
```

Suppose $A \neq C$. Since the two distinct points A, C lie both on l and in π, the spatial incidence axiom `line_in_plane` forces

$$l \subset \pi.$$

But $l \perp \pi$ at C. The universal defining clause of line–plane perpendicularity can therefore be applied to l itself, giving

$$l \perp l.$$

The established theorem `hilbert_linesPerpendicularAt_ne` then yields $l \neq l$, a contradiction. Thus the common point of a perpendicular line and its plane is unique.

12.8.4 4. Constructing the auxiliary carrier plane

In the second branch $A \notin l$. The theorem

```
hilbert_plane_through_line_and_external_point
```

constructs a plane σ satisfying

$$l \subset \sigma, \qquad A \in \sigma.$$

This is the precise formal replacement for the plane that the classical diagram leaves implicit when it draws a line through A parallel to the spatial line BC.

12.8.5 5. I.31 inside the induced plane

The helper

```
hilbert_XI12_parallel_through_point_in_plane
```

has the normalized signature

```
theorem hilbert_XI12_parallel_through_point_in_plane
    [H : HilbertIncidence Geo]
    [HilbertPlaneIncidence Geo]
    [S : HilbertSpacePrimitive Geo]
    [HSI : HilbertSpaceIncidence Geo]
    [_HSO : HilbertSpaceOrder
      (Geo := Geo) (H := H) (S := S)]
    [HSC : HilbertSpaceCongruence
      (Geo := Geo) (H := H) (S := S)]
    (sigma : S.Plane)
    (l : Geo.Line)
    (A : Geo.Point)
    (hlsigma : HilbertLineInPlane Geo l sigma)
    (hAsigma : S.OnPlane A sigma)
    (hAl : Not (H.OnLine A l)) :
    exists m : Geo.Line,
      H.OnLine A m /\
      HilbertSpaceLinesParallel Geo l m
```

The proof chooses two distinct points B, C on l, proves that B, C, A are noncollinear in $\mathrm{PlaneGeo}(\sigma)$, and then calls

```
hilbert_parallel_through_point_exists
```

inside the induced plane.

This is the formal I.31 step. The production module imports `Proposition31`, but the local helper calls the established lower-level parallel-existence theorem directly.

12.8.6 6. Packaging planar parallelism as spatial parallelism

The planar theorem initially gives disjoint `PointLine` carriers in $\mathrm{PlaneGeo}(\sigma)$. The proof constructs the actual ambient carrier m, proves that it lies in σ, and shows that an ambient intersection point of l and m would contradict the planar disjointness.

It can therefore package

$$l \parallel m$$

in the spatial sense

```
HilbertSpaceLinesParallel Geo l m
```

with σ as the explicit common carrier plane. This step excludes skew lines by construction.

12.8.7 7. XI.8 transfers perpendicularity

The proof now has

$$l \parallel m, \qquad l \perp \pi.$$

It applies

```
euclid_proposition_11_8
```

and obtains a point D such that

$$m \perp \pi \text{ at } D.$$

This is the exact classical XI.8 step. The only extra issue is that the formal theorem returns D existentially, whereas the target of XI.12 requires the foot to be the prescribed point A.

12.8.8 8. The second foot normalization

The constructed line m passes through A, and the original hypothesis gives $A \in \pi$. Applying `hilbert_XI12_perpendicular_foot_unique` to the XI.8 conclusion therefore gives

$$A = D.$$

After this substitution the conclusion is exactly

$$m \perp \pi \text{ at } A.$$

This closes the nontrivial branch and completes XI.12.

12.9 Wyler-Style Flat Reconstruction

The nontrivial branch of XI.12 has a particularly clean generated-flat normal form. XI.11 first supplies one normal l to π through an external point. If the prescribed point $A \in \pi$ does not already lie on l, the line and point generate an auxiliary plane

$$\mathrm{Join}(l, \{A\}) = \mathrm{carrier}(\sigma).$$

Inside σ, a line m through A is constructed parallel to l. The same carrier can then be written

$$\mathrm{Join}(l, m) = \mathrm{carrier}(\sigma).$$

Thus the middle of the proposition is normalized by the exact identity

$$\mathrm{Join}(l, \{A\}) = \mathrm{Join}(l, m) = \mathrm{carrier}(\sigma).$$

```
XI.11: l perp pi at F
        |
A in l? +-- yes --> foot uniqueness
        |
        no
        v
sigma = Join(l,{A})
        |
construct m through A, m || l
        |
Join(l,m) = sigma
        |
XI.8 transfers perpendicularity
        |
foot uniqueness
        |
        v
m perp pi at A
```

Here the flat calculus does more than shorten an incidence proof: it names the precise auxiliary slice that is preserved across the parallel construction.

12.10 Lean Formalization

The production module imports

```
import CGJteamLab.Proposition11_11
import CGJteamLab.Proposition11_8
import CGJteamLab.Proposition31
```

The two proposition-local declarations introduced by XI.12 are

```
hilbert_XI12_perpendicular_foot_unique
hilbert_XI12_parallel_through_point_in_plane
```

and the public theorem is

```
euclid_proposition_11_12
```

The first helper is a general uniqueness fact about a perpendicular line and its plane. The second is a general plane-slice packaging fact for parallel construction. Both are currently located in the proposition module; their signatures are sufficiently general that later promotion to `Hilbert3DInterface` may be considered if subsequent Book XI proofs reuse them.

12.11 Hidden Formal Data

12.11.1 The classical parallel construction has an exceptional case

Euclid chooses an arbitrary elevated point B, drops the XI.11 perpendicular, and then draws through A a parallel to it. The printed proof does not discuss the possibility that the foot of the XI.11 perpendicular is already A.

The formal proof cannot ignore this case. If $A \in l$, foot uniqueness shows that A is the foot and l is already the required perpendicular. Only when $A \notin l$ is the I.31 construction invoked.

12.11.2 I.31 needs an explicit plane slice

A planar parallel theorem cannot be applied directly to an arbitrary ambient spatial line and a point. The carrier plane σ through l and A is constructed first, and the entire I.31 step is performed inside PlaneGeo(σ).

12.11.3 Planar and spatial parallelism are different data

The I.31 result is initially a planar point-pair parallel relation. XI.8 expects `HilbertSpaceLinesParallel`, which includes an explicit common plane and ambient disjointness. The helper proves and packages these data rather than treating the two relations as definitionally identical.

12.11.4 XI.8 returns a new foot

XI.8 proves that the parallel line is perpendicular to π, but its formal conclusion contains an existentially returned foot D. The equality $D = A$ is a genuine incidence theorem and is discharged by the same foot-uniqueness helper used in the first branch.

12.12 Relationship with Earlier Book XI Propositions

The classical local Book XI dependencies are exactly XI.11 and XI.8.

XI.11 supplies a line perpendicular to the reference plane from an elevated point:

$$B \notin \pi \Longrightarrow \exists l, F, \quad B \in l, \quad l \perp \pi \text{ at } F.$$

XI.8 supplies the transfer along spatial parallel lines:

$$l \parallel m, \quad l \perp \pi \Longrightarrow m \perp \pi.$$

The production file imports both propositions directly. XI.9 and XI.10 are neither classical local dependencies nor Lean dependencies of XI.12.

XI.4 is not called directly by XI.12, but it remains in the transitive proof history through XI.11 and XI.8.

12.13 Relationship with Book I / Book Zero / Hilbert Infrastructure

The only explicit classical Book I proposition in XI.12 is I.31. In the formal proof its role is implemented by

```
hilbert_parallel_through_point_exists
```

inside PlaneGeo(σ).

The XI.12-specific parallel helper does not itself require `HilbertSpaceEuclidean`; its theorem signature stops at spatial order and congruence. The Euclidean Group IV assumption of the public XI.12 theorem is inherited from the calls to XI.11 and XI.8.

Book Zero contributes no new XI.12 declaration. The final public axiom profile contains the already accepted inherited dependency `bookZero_nullSegment1`, whereas both new proposition-local helpers audit without that dependency.

The ambient three-space continues to obey the Book XI instance discipline: ordinary order and congruence are used only inside explicit `PlaneGeo` slices, while the ambient layer uses the spatial classes.

12.14 Relationship with the Foundations

The current production proof of XI.12 uses Hilbert Group IV through the established XI.11–XI.8 chain. The public theorem therefore carries `HilbertSpaceEuclidean Geo`. No continuity principle is required.

Two proposition-local helpers isolate perpendicular-foot uniqueness and the construction of a parallel through a point in a fixed plane. Their role is mathematical and architectural: they make explicit two steps which the classical proof treats as immediate from the diagram and earlier Euclidean plane theory.

12.15 Dependency Summary

The classical local chain is

$$
\begin{array}{c}
A \in \pi \\
\downarrow \\
B \text{ elevated} \\
\downarrow \\
\text{XI.11} \\
\downarrow \\
BC \perp \pi \\
\downarrow \\
\text{I.31} \\
\downarrow \\
AD \parallel BC \\
\downarrow \\
\text{XI.8} \\
\downarrow \\
AD \perp \pi.
\end{array}
$$

The actual formal chain is

$$
\begin{array}{c}
A \in \pi \\
\downarrow \\
\texttt{hilbert\_point\_off\_plane} \\
\downarrow \\
B \notin \pi \\
\downarrow \\
\text{XI.11} : l \perp \pi \text{ at } F \\
\downarrow \\
\texttt{by\_cases} \;\; A \in l \\
\downarrow \\
\begin{array}{c|c}
\begin{array}{c}
A \in l \\
\downarrow \\
A = F \\
\downarrow \\
l \perp \pi \text{ at } A \\
\\
\\
\\
\\
\\
\\
\\
\end{array}
&
\begin{array}{c}
A \notin l \\
\downarrow \\
\sigma = \text{plane}(l, A) \\
\downarrow \\
\text{I.31 in } \text{PlaneGeo}(\sigma) \\
\downarrow \\
\texttt{HilbertSpaceLinesParallel} \;\; l\, m \\
\downarrow \\
\text{XI.8} : m \perp \pi \text{ at } D \\
\downarrow \\
D = A
\end{array}
\end{array} \\
\downarrow \\
\texttt{euclid\_proposition\_11\_12}.
\end{array}
$$

The dependency classification is:

New proposition-specific global axiom	no
New spatial axiom class	no
Continuity	no
Group IV / Euclidean parallelism	yes, via XI.11 and XI.8
Direct production imports	XI.11, XI.8, I.31 module
Classical explicit Book XI dependencies	XI.11, XI.8
XI.9 / XI.10	not used
New proposition-local helper theorem	yes, two
New PlaneGeo bridge	no
Coordinates / vectors / scalar product	no
`sorry/admit`	no

12.16 Conclusion

Proposition XI.12 has a minimal classical proof but a revealing formal shape. Euclid's three-step route remains visible:

$$\mathrm{XI.11} \longrightarrow \mathrm{I.31} \longrightarrow \mathrm{XI.8}.$$

Lean adds no alternative geometry. It makes explicit the carrier plane needed for the planar parallel construction, the exceptional case in which the XI.11 perpendicular already passes through the prescribed point, the conversion from planar to spatial parallelism, and the identification of the foot returned by XI.8.

The two new helpers are axiomatically clean relative to the inherited Book Zero dependency. The public theorem has the same global axiom profile as XI.11 and XI.8:

$$[\texttt{propext}, \texttt{Classical.choice}, \texttt{Geometry.bookZero\_nullSegment1}, \texttt{Quot.sound}].$$

The current result remains fully synthetic and uses no coordinates, vectors, scalar products, trigonometric functions, or numerical angle measure. Its Group IV hypothesis belongs to the chosen Euclid-faithful route through XI.11 and XI.8; it is not a claim that the abstract perpendicular-existence statement intrinsically requires Euclidean parallelism.

Chapter 13

Proposition XI.13

13.1 Introduction

Proposition XI.13 is the uniqueness theorem that follows the two perpendicular-construction propositions XI.11 and XI.12. Euclid now assumes that two straight lines have been erected from the same point at right angles to the same plane and proves that, on the same side of the plane, the configuration is impossible.

The classical proof is short but genuinely spatial. The two supposed perpendiculars determine an auxiliary plane. That plane intersects the reference plane in a straight line through the common foot. Each of the two perpendiculars is therefore perpendicular to the same intersection line. Euclid then reaches a planar impossibility inside the auxiliary plane.

The formal reconstruction keeps the same spatial core but packages the last planar uniqueness argument differently. Since `Geo.Line` is an unoriented extensional carrier, the phrase "on the same side" has no independent content at the level of line equality: opposite rays through the common point belong to the same ambient line. The formal target is therefore equality of the two line carriers.

Rather than reconstructing a separate proposition-local theorem for uniqueness of a perpendicular ray in the auxiliary plane, the Lean proof reuses XI.4. If the intersection line of the two planes were perpendicular to two distinct lines of the auxiliary plane, XI.4 would make that intersection line perpendicular to the whole auxiliary plane. But the intersection line itself lies in that plane, which is impossible.

13.2 Euclid's Statement

Euclid's Proposition XI.13 states:

> From the same point two straight lines cannot be set up at right angles to the same plane on the same side.

Assume, for contradiction, that AB and AC are two such straight lines erected from the common point A of the reference plane π.

13.3 Classical Configuration

Draw the plane through AB and AC. Denote it by σ. By XI.3 its section with the reference plane is a straight line through A; Euclid writes this line as DAE.

Thus the classical configuration contains

$$AB, AC, DAE \subset \sigma, \qquad DAE \subset \pi, \qquad A \in \pi \cap \sigma.$$

The configuration is shown in Figure 13.1. The reference plane π is the light gray plane. The auxiliary plane σ contains the two blue spatial lines and intersects π along the black line DAE.

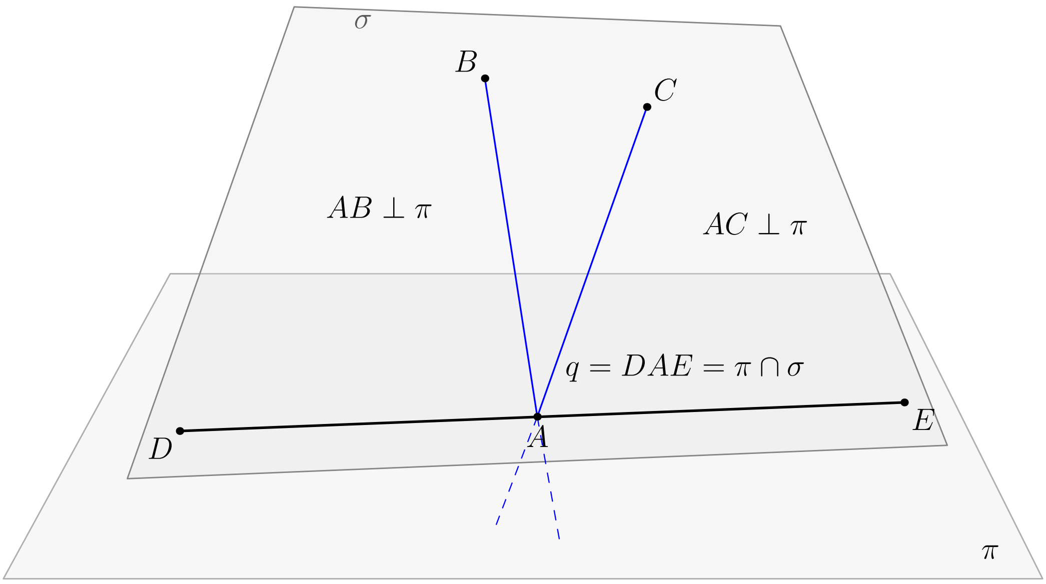


Figure 13.1: The classical configuration of Proposition XI.13. The two supposed perpendiculars AB and AC meet the reference plane π at the same point A and lie in the auxiliary plane σ. By XI.3 the planes π and σ intersect in the straight line DAE through A. Since both spatial lines are perpendicular to π, each is perpendicular to DAE.

13.4 Classical Proof

Suppose that AB and AC are distinct straight lines erected from A at right angles to the same plane π, on the same side.

Draw the plane σ through AB and AC. By XI.3 the intersection of σ with π is a straight line through A; write it as DAE.

Since $AC \perp \pi$, Definition XI.3 says that AC is perpendicular to every straight line in π meeting it at the foot. Hence

$$AC \perp DAE.$$

For the same reason,

$$AB \perp DAE.$$

Therefore both $\angle CAE$ and $\angle BAE$ are right angles and hence equal. The two angles lie in the same plane σ, share the same side AE, and the remaining sides AC and AB are assumed to lie on the same side. Euclid concludes that this is impossible.

The local classical dependency chain is therefore

$$\begin{gathered} AB \perp \pi, \quad AC \perp \pi, \quad A \in \pi \\ \Downarrow \\ \sigma = \mathrm{plane}(AB, AC) \\ \Downarrow \\ \text{XI.3} \\ \Downarrow \\ DAE = \pi \cap \sigma \\ \Downarrow \\ \text{XI.Def.3} \\ \Downarrow \\ AB \perp DAE, \quad AC \perp DAE \\ \Downarrow \\ \angle BAE, \angle CAE \text{ are equal right angles in } \sigma \\ \Downarrow \\ \text{planar uniqueness / contradiction} \\ \Downarrow \\ AB = AC \text{ as a line carrier.} \end{gathered}$$

Neither XI.9 nor XI.10 occurs in this proof. XI.11 and XI.12 are also not dependencies of XI.13: they establish existence, whereas XI.13 is a uniqueness statement.

13.5 What is genuinely three-dimensional here?

The proof has one essential spatial move and one spatial intersection move.

First, the two supposed distinct perpendicular lines determine the auxiliary plane

$$\sigma = \mathrm{plane}(l, m).$$

Second, this plane must be distinguished from the reference plane π, after which their common section is a line

$$q = \pi \cap \sigma$$

through the common point A.

Once q has been obtained, the classical argument is planar inside σ. The formal proof does not need to enter PlaneGeo(σ) explicitly, because the relevant planar conclusion has already been packaged synthetically by XI.4.

The point is therefore not that XI.13 is "really planar". The existence of the correct carrier plane, the distinctness of the two planes, and the common intersection line are genuine three-dimensional proof obligations.

13.6 Formal Statement

The production theorem is:

```
theorem euclid_proposition_11_13
    [H : HilbertIncidence Geo]
    [HilbertPlaneIncidence Geo]
    [S : HilbertSpacePrimitive Geo]
    [HSI : HilbertSpaceIncidence Geo]
    [_HSO : HilbertSpaceOrder
      (Geo := Geo) (H := H) (S := S)]
    [HSC : HilbertSpaceCongruence
      (Geo := Geo) (H := H) (S := S)]
    (pi : S.Plane)
    (l m : Geo.Line)
    (A : Geo.Point)
    (hLperp :
      HilbertLinePerpendicularPlaneAt Geo l pi A)
    (hMperp :
      HilbertLinePerpendicularPlaneAt Geo m pi A) :
    l = m
```

There is no `HilbertSpaceEuclidean` parameter. Thus the current formal XI.13 does not use Hilbert Group IV.

The difference between Euclid's phrase "on the same side" and the formal conclusion $l = m$ is representational. A `Geo.Line` is an unoriented carrier. Reversing the ray through A does not create a second line object, so the side qualification disappears when the target is equality of line carriers.

13.7 Spatial / Planar Architecture Used

13.7.1 Ambient spatial layer

The proof uses the ambient notions

```
HilbertSpacePrimitive
HilbertSpaceIncidence
HilbertSpaceOrder
HilbertSpaceCongruence
HilbertLineInPlane
HilbertLinesPerpendicularAt
HilbertLinePerpendicularPlaneAt
```

together with the general incidence constructors

```
hilbert_plane_through_two_intersecting_lines
hilbert_plane_intersection_line
```

and the universal elimination theorem

```
HilbertLinePerpendicularPlaneAt.perpendicular_to_line
```

As in the preceding Book XI development, no global ambient instances

```
HilbertOrder Geo
HilbertCongruence Geo
```

are installed.

13.7.2 Induced planar geometry

The final production proof makes no direct theorem call in PlaneGeo(σ). This is a genuine simplification of the formal DAG relative to the classical exposition.

The planar content has not disappeared. It is already contained in the earlier theorem

```
euclid_proposition_11_4
```

which states synthetically that a line perpendicular to two distinct intersecting lines of a plane is perpendicular to the entire plane.

13.7.3 Plane–space bridges

XI.13 introduces no new plane–space bridge theorem. The proof stays in ambient predicates and uses the already established spatial perpendicularity symmetry

```
hilbert_space_linesPerpendicularAt_symm
```

before applying XI.4.

13.8 Proof Architecture of the Reconstruction

13.8.1 1. Contradiction hypothesis

Lean begins with

```
by_contra hlm
```

so the working assumption is

$$l \neq m.$$

The incidence components of the two line–plane perpendicularities give

$$A \in l, \qquad A \in m, \qquad A \in \pi.$$

13.8.2 2. Constructing the auxiliary plane

Since l and m are distinct and meet at A, the theorem

```
hilbert_plane_through_two_intersecting_lines
```

produces a plane σ such that

$$l \subset \sigma, \qquad m \subset \sigma, \qquad A \in \sigma.$$

This is the formal version of Euclid's instruction to draw the plane through the two supposed perpendiculars.

13.8.3 3. Proving that the planes are distinct

The intersection theorem for two planes requires an explicit proof that

$$\sigma \neq \pi.$$

Suppose instead that $\sigma = \pi$. Since $l \subset \sigma$, this would imply $l \subset \pi$. But the already established theorem

```
hilbert_linePerpendicularPlaneAt_not_in_plane
```

says that a line perpendicular to a plane cannot itself lie in that plane. Hence $\sigma \neq \pi$.

This is one of the principal hidden spatial obligations of XI.13.

13.8.4 4. Constructing the intersection line

The theorem

```
hilbert_plane_intersection_line
```

is applied to π, σ at the common point A. It returns an ambient line q with

$$A \in q, \qquad q \subset \pi, \qquad q \subset \sigma.$$

This is the formal structural counterpart of Euclid's XI.3 step producing the line DAE.

13.8.5 5. Both candidate lines are perpendicular to the intersection

Since $q \subset \pi$ and $A \in q$, the universal clause of

```
HilbertLinePerpendicularPlaneAt
```

gives

$$l \perp q \text{ at } A, \qquad m \perp q \text{ at } A.$$

The proof then applies

```
hilbert_space_linesPerpendicularAt_symm
```

to reverse the two relations:

$$q \perp l \text{ at } A, \qquad q \perp m \text{ at } A.$$

13.8.6 6. XI.4 replaces the terminal planar uniqueness step

The lines l, m are now packaged as two distinct `PlaneLine`s of σ, and A as a `PlanePoint` of σ. Proposition XI.4 applies:

$$q \perp l, \quad q \perp m, \quad l \neq m, \quad l, m \subset \sigma \Longrightarrow q \perp \sigma.$$

This is the central difference between the classical and Lean DAGs. Euclid compares two right angles in the auxiliary plane and invokes their impossibility as distinct same-side angles. Lean reuses the already proved universal line–plane theorem XI.4.

The method remains synthetic: no metric formula, coordinate argument, or vector normal is introduced.

13.8.7 7. Final contradiction

The intersection construction preserved

$$q \subset \sigma.$$

But the XI.4 conclusion says

$$q \perp \sigma.$$

Applying

```
hilbert_linePerpendicularPlaneAt_not_in_plane
```

to q and σ gives the contradiction. Hence the assumption $l \neq m$ is false, and therefore

$$l = m.$$

13.9 Wyler-Style Flat Reconstruction

XI.13 gives the cleanest complete interaction of join, meet, and metric structure in the first fourteen propositions. Under the contradiction hypothesis $l \neq m$, the two candidate normals generate a plane σ. The reference plane π and σ then have an exact meet line q.

The entire proof can be read as the following graph:

```
assume l != m
    |
    v
sigma = Join(l,m)
    |
    v
q = Meet(pi,sigma)
    |
    v
l perp q, m perp q
    |
    v
q perp l, q perp m
    |
    v
XI.4 core: q perp sigma
    |
    v
q subset sigma
    |
    v
contradiction
```

The join identifies the canonical plane generated by the two supposed normals. The meet identifies the canonical section line of that plane with π. Only after these carriers have been fixed does the metric argument enter: because both normals are perpendicular to π, they are perpendicular to q; symmetry and XI.4 then force $q \perp \sigma$, which is impossible because the meet construction preserved $q \subseteq \sigma$.

This graph belongs at the theorem level. It is the most compact description of why the formal proof remains synthetic while exposing more structure than the diagram alone.

13.10 Lean Formalization

The production module has the single direct import

```
import CGJteamLab.Proposition11_12
```

and introduces no proposition-local helper theorem before the public result. The public declaration is

```
euclid_proposition_11_13
```

The proof consumes general infrastructure and earlier Book XI theorems available transitively through the current production chain. In particular, the substantive reused declarations are

```
hilbert_plane_through_two_intersecting_lines
hilbert_plane_intersection_line
hilbert_linePerpendicularPlaneAt_not_in_plane
HilbertLinePerpendicularPlaneAt.perpendicular_to_line
hilbert_space_linesPerpendicularAt_symm
euclid_proposition_11_4
```

XI.12 itself is not called in the proof. Its module is the immediate production import because XI.13 follows XI.12 in the current Book XI module sequence.

13.11 Hidden Formal Data

13.11.1 The auxiliary plane must be explicit

The classical diagram makes it visually immediate that the two supposed perpendiculars lie in a common plane. Lean constructs this plane from the two distinct intersecting line carriers and keeps both containment proofs.

13.11.2 The two planes must be proved distinct

Euclid immediately speaks of their section. The formal `hilbert_plane_intersection_line` theorem requires $\sigma \neq \pi$. This is proved by showing that equality of the planes would put a perpendicular line inside its reference plane.

13.11.3 The intersection carrier must survive

The proof must preserve all three facts returned with q:

$$A \in q, \qquad q \subset \pi, \qquad q \subset \sigma.$$

The first two are needed to extract the two line–line perpendicularities. The last is needed only at the final contradiction.

13.11.4 The "same side" phrase is not a missing formal hypothesis

The formal theorem does not introduce a relation saying that two lines are on the same side of a plane. This is not an omitted diagrammatic condition. Euclid's condition distinguishes erected rays. The formal objects l, m : `Geo.Line` are unoriented carriers, so opposite rays through the foot determine the same line.

13.12 Relationship with Earlier Book XI Propositions

The classical local dependency is XI.3 together with Definition XI.3. Propositions XI.9–XI.12 are absent from the classical proof.

The Lean proof reorganizes the dependency graph:

- the content of the XI.3 plane-section step is supplied by the general theorem `hilbert_plane_intersection_line`;
- XI.4 is used directly to package the final planar impossibility;
- the helper `hilbert_linePerpendicularPlaneAt_not_in_plane`, introduced in the XI.5 production module, proves both the distinctness of the planes and the final contradiction;
- the spatial symmetry theorem `hilbert_space_linesPerpendicularAt_symm`, introduced in the XI.6 production module, normalizes the two perpendicular relations before XI.4.

Thus XI.13 is formally dependent on earlier XI.4–XI.6 infrastructure, but not on the existence propositions XI.11 and XI.12 as theorem calls.

13.13 Relationship with Book I / Book Zero / Hilbert Infrastructure

No Book I proposition is called directly by XI.13. The planar right-angle machinery needed by XI.4 is inherited through the already proved Book XI infrastructure.

The current theorem uses spatial incidence, order, and congruence, but its signature has no

```
HilbertSpaceEuclidean Geo
```

parameter. Hence Group IV is absent from XI.13.

Book Zero contributes no new XI.13 declaration. The final axiom profile contains the already accepted inherited dependency `bookZero_nullSegment1`; XI.13 introduces no new global axiom constant.

13.14 Relationship with the Foundations

XI.13 is neutral in the current reconstruction. Its theorem signature does not contain `HilbertSpaceEuclidean Geo`; the proof uses spatial incidence, order, congruence, the plane-intersection infrastructure, and Proposition XI.4, but no Hilbert Group IV assumption and no continuity principle.

This is worth separating from the neighboring XI.11–XI.12 constructions: although all three concern perpendicularity in space, XI.13 closes by a neutral uniqueness argument inside an explicitly constructed auxiliary plane.

13.15 Dependency Summary

The classical local chain is

$$
\begin{array}{c}
l \perp \pi, \quad m \perp \pi, \quad A \in l \cap m \cap \pi \\
\downarrow \\
\sigma = \mathrm{plane}(l, m) \\
\downarrow \\
\text{XI.3} \\
\downarrow \\
q = \pi \cap \sigma \\
\downarrow \\
\text{XI.Def.3} \\
\downarrow \\
l \perp q, \quad m \perp q \\
\downarrow \\
\text{equal right angles in one plane} \\
\downarrow \\
\text{contradiction.}
\end{array}
$$

The actual formal chain is

$$
\begin{array}{c}
l \perp \pi, \quad m \perp \pi \text{ at } A \\
\downarrow \\
\texttt{by\_contra } l \neq m \\
\downarrow \\
\texttt{hilbert\_plane\_through\_two\_intersecting\_lines} \\
\downarrow \\
\sigma = \mathrm{plane}(l, m) \\
\downarrow \\
\texttt{hilbert\_linePerpendicularPlaneAt\_not\_in\_plane} \\
\downarrow \\
\sigma \neq \pi \\
\downarrow \\
\texttt{hilbert\_plane\_intersection\_line} \\
\downarrow \\
q \subset \pi \cap \sigma, \quad A \in q \\
\downarrow \\
\texttt{perpendicular\_to\_line} \\
\downarrow \\
l \perp q, \quad m \perp q \\
\downarrow \\
\texttt{hilbert\_space\_linesPerpendicularAt\_symm} \\
\downarrow \\
q \perp l, \quad q \perp m \\
\downarrow \\
\text{XI.4} \\
\downarrow \\
q \perp \sigma \\
\downarrow \\
q \subset \sigma \\
\downarrow \\
\texttt{hilbert\_linePerpendicularPlaneAt\_not\_in\_plane} \\
\downarrow \\
\bot \\
\downarrow \\
l = m.
\end{array}
$$

The dependency classification is:

New proposition-specific global axiom	no
New spatial axiom class	no
Continuity	no
Group IV / Euclidean parallelism	no
Direct production import	XI.12 module
Classical explicit Book XI dependency	XI.3
Classical definition dependency	XI.Def.3
XI.9 / XI.10	not used
XI.11 / XI.12 theorem calls	not used
Earlier formal Book XI theorem reused	XI.4
New proposition-local helper theorem	no
New PlaneGeo bridge	no
Coordinates / vectors / scalar product	no
`sorry`/`admit`	no

13.16 Conclusion

Proposition XI.13 is a compact uniqueness theorem with a particularly clean formal architecture. The classical proof uses an auxiliary plane, its intersection with the reference plane, and the impossibility of two distinct same-side perpendicular rays in one plane.

The Lean proof preserves the first two spatial steps exactly but reuses XI.4 for the terminal planar contradiction. The key hidden obligation is the proof that the auxiliary plane differs from the reference plane. Once the intersection line is retained with both carrier-plane proofs, the final contradiction is immediate from the fact that a line perpendicular to a plane cannot itself lie in that plane.

The theorem introduces no new helper, bridge, spatial axiom, Group IV assumption, or continuity principle. Its audited global axiom profile is

$$[\texttt{propext}, \texttt{Classical.choice}, \texttt{Geometry.bookZero\_nullSegment1}, \texttt{Quot.sound}].$$

The result is fully synthetic and uses no coordinates, vectors, scalar products, trigonometric functions, or numerical angle measure.

Chapter 14

Proposition XI.14

14.1 Introduction

Proposition XI.14 states that planes to which the same straight line is at right angles are parallel. The classical argument is short: if the two planes met, a point of their intersection together with the common perpendicular would produce a triangle with two right angles, contradicting Proposition I.17.

The formal reconstruction preserves this geometric core, but it exposes two points that the classical presentation leaves implicit.

First, Euclid's contradiction argument requires the two feet of the common perpendicular to be distinct. If the line meets both planes at the same point, the triangle used in the proof degenerates. The production proof therefore separates an equal-foot uniqueness theorem from the normalized case $A \neq B$.

Second, Euclid reaches a common point by first speaking about the line of intersection of the two planes. In the formal library, parallelism of two planes is defined directly as absence of a common point. Negating the target therefore gives the point K immediately; the proof does not need to construct the whole intersection line before beginning the contradiction.

These two differences make XI.14 a useful example of the project's general method. The classical geometry is retained, while diagrammatic assumptions and representational shortcuts are isolated as explicit proof obligations.

14.2 Classical Configuration

Let l be a straight line perpendicular to plane π at A and to plane ρ at B:

$$l \perp \pi \text{ at } A, \qquad l \perp \rho \text{ at } B.$$

The intended nontrivial configuration has

$$\pi \neq \rho, \qquad A \neq B.$$

Euclid's conclusion is that the two planes do not meet.

The formal public theorem also carries the hypothesis $\pi \neq \rho$. This matches the nontrivial reading of Euclid XI.Def.8 used in the project: parallel planes are distinct planes having no common point.

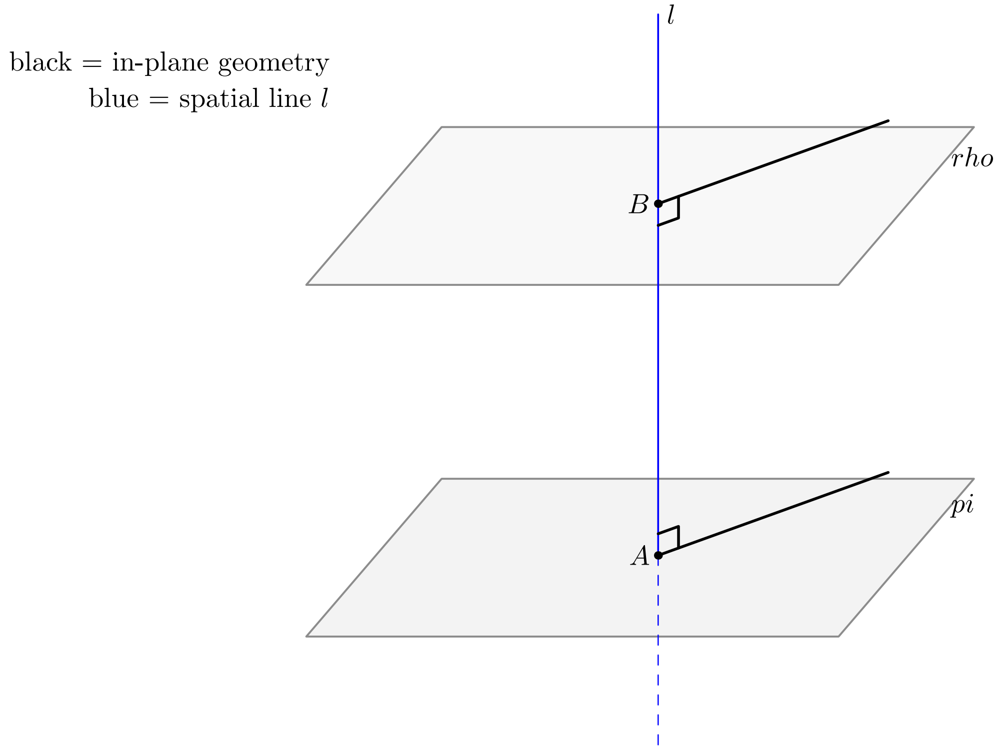


Figure 14.1: The basic spatial configuration of Proposition XI.14. The common perpendicular l meets the two reference planes at A and B. Following the Book XI drawing convention, geometry internal to a displayed plane is black, while genuinely spatial objects are blue.

Diagrammatic audit. Three stable geometric states are sufficient for the documentation of XI.14. Figure 14.1 records the input configuration. Figure 14.3 records the hypothetical common-point state and the reduction to a planar triangle. Figure 14.2 records the exceptional equal-foot uniqueness argument. No additional figure is needed for the internal SameRay/opposite-ray case split, because that split is a formal orientation normalization rather than a new stable Euclidean construction.

14.3 Statement

Theorem 2 (Euclid XI.14)**.** *Let π and ρ be distinct planes, and let l be a straight line. If l is perpendicular to π at A and perpendicular to ρ at B, then π and ρ are parallel.*

The current production declaration is:

```
theorem euclid_proposition_11_14
    [H : HilbertIncidence Geo]
    [HilbertPlaneIncidence Geo]
    [S : HilbertSpacePrimitive Geo]
    [HSI : HilbertSpaceIncidence Geo]
    [_HSO : HilbertSpaceOrder
      (Geo := Geo) (H := H) (S := S)]
    [_HSC : HilbertSpaceCongruence
      (Geo := Geo) (H := H) (S := S)]
    (pi rho : S.Plane)
    (l : Geo.Line)
```

```
    (A B : Geo.Point)
    (hPlanesNe : Ne pi rho)
    (hPerpPi :
      HilbertLinePerpendicularPlaneAt Geo l pi A)
    (hPerpRho :
      HilbertLinePerpendicularPlaneAt Geo l rho B) :
    HilbertSpacePlanesParallel Geo pi rho
```

The target predicate is proposition-local:

```
def HilbertSpacePlanesParallel
    [S : HilbertSpacePrimitive Geo]
    (pi rho : S.Plane) : Prop :=
  Not
    (exists X : Geo.Point,
      S.OnPlane X pi /\
      S.OnPlane X rho)
```

Thus the conclusion is represented directly as nonintersection of the two planes.

14.4 Classical Proof

Assume, for contradiction, that the two planes meet. By XI.3 their common section is a straight line; choose a point K on this section.

Join A to K and B to K. Since $K \in \pi$, the line AK lies in π. Because $l \perp \pi$ at A, Definition XI.3 gives

$$l \perp AK.$$

Similarly, since $K \in \rho$,

$$l \perp BK.$$

The lines l and the point K determine a plane σ. The points A and B lie on l, so the triangle ABK lies in σ. Inside this plane,

$$\angle BAK \text{ is right}, \qquad \angle ABK \text{ is right}.$$

But I.17 says that two angles of a triangle are together less than two right angles. A nondegenerate triangle therefore cannot have two right angles. This contradiction proves that π and ρ do not meet.

The local classical dependency pattern is

$$\begin{gathered} l \perp \pi, \quad l \perp \rho \\ \downarrow \\ \text{assume } \pi \cap \rho \neq \varnothing \\ \downarrow \\ \text{XI.3} \\ \downarrow \\ K \in \pi \cap \rho \\ \downarrow \\ \text{XI.Def.3} \\ \downarrow \\ \angle BAK, \angle ABK \text{ right} \\ \downarrow \\ \text{I.17} \\ \downarrow \\ \perp. \end{gathered}$$

The proof is neutral. No Euclidean parallel postulate is involved.

14.5 The Equal-Foot Issue

The classical triangle argument silently presupposes $A \neq B$. If $A = B$, the side AB collapses and the I.17 contradiction cannot even be formed.

The production proof handles this branch by proving a separate uniqueness theorem: for a fixed line l and a fixed point A, there is at most one plane perpendicular to l at A.

The construction is synthetic. Choose two distinct lines m, n through A in π. Since $l \perp \pi$, both are perpendicular to l. Now let p be any line through A in ρ. Since $l \perp \rho$, the line p is also perpendicular to l. Proposition XI.5 then places m, n, p in one plane. Because the two distinct intersecting lines m, n already determine π, the line p must lie in π.

Applying the same argument to two distinct lines through A in ρ shows that two distinct intersecting lines of ρ also lie in π. The uniqueness of the plane through two intersecting lines yields

$$\rho = \pi.$$

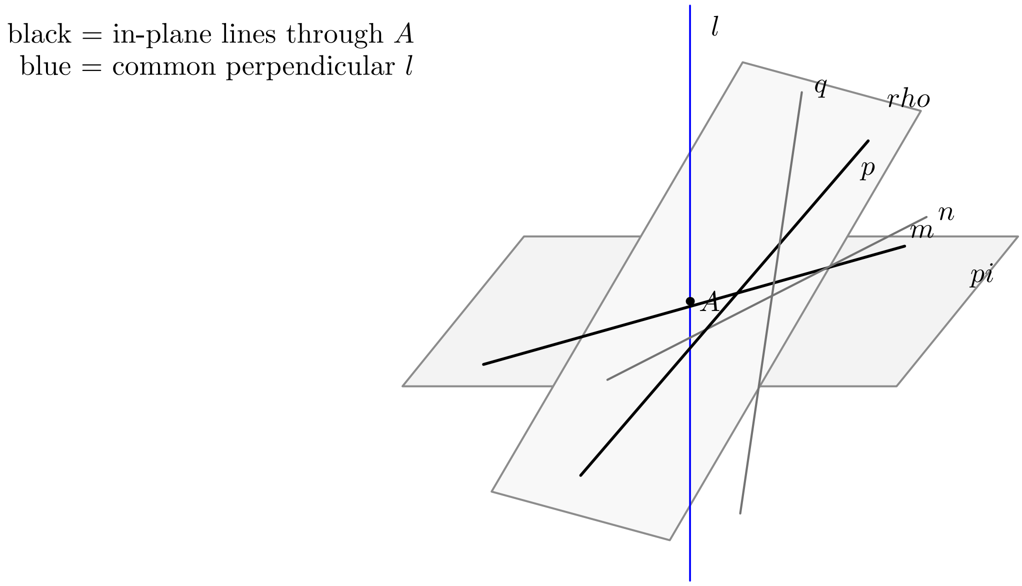
This is not an auxiliary side remark. It is exactly what allows the public theorem to derive $A \neq B$ from the assumed plane distinctness $\pi \neq \rho$.

14.6 What is genuinely three-dimensional here?

XI.14 has a particularly clear spatial-to-planar transition.

The genuinely three-dimensional data are:

- two ambient planes π, ρ;
- a single ambient line l perpendicular to each;



equal-foot uniqueness case

Figure 14.2: The exceptional case in which the same line l is perpendicular to both planes at the same point A. Two distinct in-plane lines through A, together with XI.5, force every line through A in the second plane into the first plane; hence the two planes coincide.

- a hypothetical point K common to the two planes;
- the auxiliary plane σ through l and the off-line point K;
- the two connector lines AK and BK, each simultaneously controlled by one reference plane and by σ.

Once σ has been constructed, the terminal contradiction is planar. The proof passes to

$$\mathrm{PlaneGeo}(\sigma)$$

and works with the induced Hilbert incidence, order, and congruence structure. The two spatial perpendicularities are converted into the exact right-angle statements for the triangle ABK, after which I.17 is purely two-dimensional.

Thus XI.14 is not merely a planar theorem drawn in space. The proof must first manufacture the correct plane slice and transport the ambient line–plane perpendicularity data into it.

14.7 Spatial / Planar Architecture Used

The ambient layer uses

```
HilbertSpacePrimitive
HilbertSpaceIncidence
HilbertSpaceOrder
HilbertSpaceCongruence
HilbertLineInPlane
HilbertLinesPerpendicularAt
HilbertLinePerpendicularPlaneAt
```

The key spatial incidence constructors are

```
hilbert_plane_through_line_and_external_point
hilbert_plane_through_two_intersecting_lines
```

For the normalized branch, the proof constructs σ from the common normal l and the hypothetical common point K. The points and lines are then re-packaged as

```
PlanePoint Geo sigma
PlaneLine Geo sigma
PlaneGeo Geo sigma
```

This passage is essential because Proposition I.17 is a planar theorem. The bridge layer transfers incidence and perpendicularity from the ambient spatial carriers to the induced plane geometry.

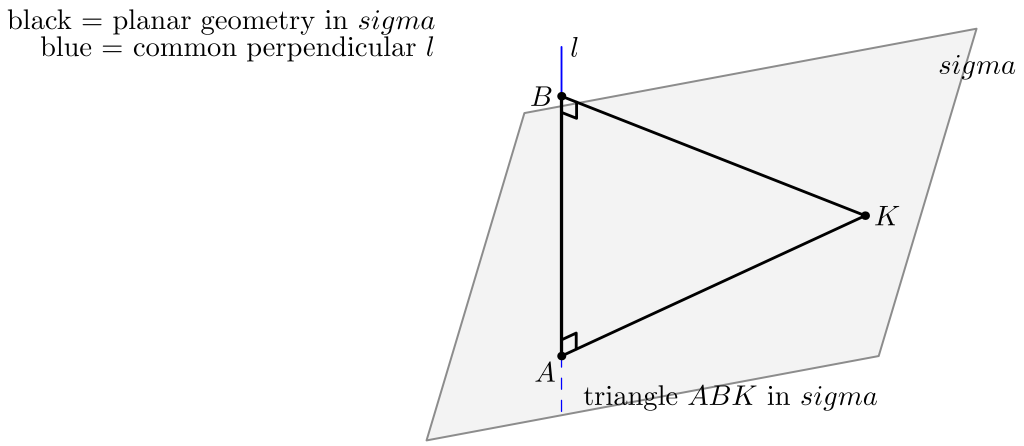


Figure 14.3: The hypothetical common point K and the common perpendicular l determine the auxiliary plane σ. In σ, the triangle ABK has right angles at both A and B. The spatial construction is therefore reduced to the planar contradiction supplied by I.17.

14.8 Proof Architecture of the Reconstruction

14.8.1 1. Excluding a common point on the normal

Assume $A \neq B$ and suppose K lies on both planes. If K also lay on l, the perpendicular-foot uniqueness theorem from the XI.12 infrastructure would give

$$K = A \qquad \text{and} \qquad K = B,$$

contradicting $A \neq B$.

This is formalized by `hilbert_XI14_common_point_off_normal`.

14.8.2 2. Constructing the auxiliary plane

Because $K \notin l$, the general spatial incidence theorem constructs a unique plane σ containing l and K. Since $A, B \in l$, all three vertices A, B, K lie in σ.

This stage is the formal replacement for what the classical diagram makes immediate.

14.8.3 3. Constructing the connector lines

The lines AK and BK are explicitly constructed. The first lies in both π and σ; the second lies in both ρ and σ. Definition XI.3, encoded by `HilbertLinePerpendicularPlaneAt.perpendicular_to_line`, gives

$$l \perp AK \text{ at } A, \qquad l \perp BK \text{ at } B.$$

14.8.4 4. Passing to `PlaneGeo`

The ambient configuration is transferred to the induced plane geometry of σ. The noncollinearity of A, B, K follows because $A, B \in l$ while $K \notin l$.

At this point the spatial part of the normalized contradiction is complete.

14.8.5 5. Recovering the exact triangle angles

The formal definition of line–line perpendicularity contains witnesses on the two carrier lines. Those witnesses need not initially be on the exact rays AB, AK and BA, BK used in the triangle.

The production proof therefore performs an orientation normalization. A collinear witness is either on the same ray from the vertex as the desired triangle point or on the opposite ray. SameRay transport and the two opposite-ray right-angle transport lemmas show that rightness is invariant under all four orientation combinations.

The result is the stable geometric statement

$$\angle BAK \text{ right}, \qquad \angle ABK \text{ right}.$$

This is packaged by `hilbert_XI14_triangle_has_two_right_angles`.

14.8.6 6. Proposition I.17 closes the planar contradiction

Apply the I.17 variant `euclid_proposition_17_BAC_ABC` to the triangle ABK. It extends the side BK through B to a point E and yields

$$\angle BAK < \angle ABE.$$

Since $K - B - E$ and $\angle ABK$ is right, the angle $\angle ABE$ is also right. All right angles are congruent, so

$$\angle BAK \cong \angle ABE.$$

A strict angle inequality cannot hold between congruent angles. This gives the contradiction.

The planar closure is packaged by `hilbert_XI14_two_right_angles_impossible`.

14.8.7 7. The normalized theorem

The theorem `euclid_proposition_11_14_normalized` now has a very short public shape: assume a common point K, construct the two-right-angle triangle, and invoke the planar impossibility theorem.

14.8.8 8. Removing the normalization

The public theorem must not assume $A \neq B$. It derives this from $\pi \neq \rho$.

If $A = B$, the equal-foot uniqueness theorem `hilbert_XI14_plane_perpendicular_to_line_at_unique` gives $\pi = \rho$, contradicting the hypothesis. Therefore $A \neq B$, and the normalized theorem applies.

14.9 Wyler-Style Flat Reconstruction

For distinct planes the flat layer first records the global rank fact

$$\mathrm{Join}(\pi, \rho) = E^3.$$

This is useful, but deliberately insufficient: two distinct planes may generate all of three-space whether or not they are parallel. XI.14 therefore requires a separate metric separation argument.

The complete structure is

```
pi != rho
   |
   +--> Join(pi,rho) = E^3        [flat certificate]
   |
A = B ?
   |
   +-- yes --> uniqueness of plane normal --> pi = rho --> contradiction
   |
   no
   v
assume K in pi cap rho
   |
sigma = plane(l,K)
   |
triangle ABK inside PlaneGeo(sigma)
   |
right angle at A and right angle at B
   |
I.17
   |
contradiction
   |
   v
pi cap rho = empty
```

Thus the meet of the two plane carriers is forced to be empty, which is the formal spatial parallelism conclusion. The proposition is a useful boundary case for the Wyler vocabulary: join detects global generation, but empty meet is proved by the metric geometry of a common normal and the planar I.17 contradiction.

14.10 Lean Formalization

The production module imports

```
import CGJteamLab.Proposition11_13
import CGJteamLab.Proposition11_5
import CGJteamLab.Proposition17
import CGJteamLab.HilbertRightAngle
```

The direct mathematical engines used by XI.14 are distributed across these imports:

- XI.5 supplies the equal-foot coplanarity argument;
- the XI.12 infrastructure supplies perpendicular-foot uniqueness;
- I.17 supplies the terminal planar angle inequality;
- the Hilbert right-angle library supplies transport and congruence of right angles.

The public proof itself is intentionally small. Almost all of the work is performed in reusable proposition-local lemmas that separate the spatial construction, the planar reduction, the angle-orientation normalization, and the equal-foot uniqueness theorem.

This organization mirrors the mathematical architecture more faithfully than a single monolithic tactic proof would.

14.11 Hidden Formal Data

14.11.1 Plane distinctness is part of the public statement

The project defines parallel planes by nonintersection, but XI.14 is intended for two distinct planes. The hypothesis `hPlanesNe` therefore appears explicitly.

It is also what rules out the equal-foot branch.

14.11.2 The common point must be off the normal

The classical picture places K away from AB automatically. Lean must prove this. The argument uses perpendicular-foot uniqueness in each reference plane.

14.11.3 The auxiliary plane must be carried explicitly

The sentence "draw the plane through AB and K" hides several facts: $K \notin l$, the existence of the plane, containment of l, and containment of K. All are retained because later bridge lemmas depend on them.

14.11.4 Perpendicularity witnesses have orientation

A statement such as $l \perp AK$ is not yet syntactically the statement that $\angle BAK$ is right. The witness points appearing in the definition of perpendicular lines may occupy either ray of the carrier. The production proof therefore has a genuine four-case SameRay/opposite-ray normalization.

This is formal data hidden completely by the Euclidean diagram.

14.11.5 I.17 returns an angle inequality, not the final contradiction

The formal I.17 theorem yields a strict angle comparison after extending one side of the triangle. XI.14 must still prove that the resulting exterior angle is right and then use congruence of all right angles to contradict the strict inequality.

14.12 Relationship with Earlier Book XI Propositions

The classical proof explicitly uses XI.3 and Definition XI.3.

The formal dependency graph is reorganized.

The content corresponding to XI.3 is supplied more economically by the definition of plane parallelism and by general spatial incidence constructors. Since the negation of `HilbertSpacePlanesParallel` already gives a common point K, the proof does not first need an explicit line representing the complete intersection $\pi \cap \rho$.

Definition XI.3 is represented directly by `HilbertLinePerpendicularPlaneAt`: every line of the reference plane through the foot is perpendicular to the given line.

XI.5 enters only in the equal-foot uniqueness branch. This is a genuine new formal dependency caused by making explicit the case $A = B$.

XI.12 contributes the previously proved uniqueness of the perpendicular foot of a fixed line in a fixed plane. XI.13 itself is not called by the proof; its module is an immediate production import in the Book XI sequence.

14.13 Relationship with Book I / Book Zero / Hilbert Infrastructure

The decisive Book I dependency is Proposition I.17. In the current library the relevant theorem is `euclid_proposition_17_BAC_ABC`. It supplies the strict comparison needed to rule out the two-right-angle triangle.

The right-angle library contributes two separate ingredients:

- transport of a right angle when one of its arms is replaced by the same or opposite ray;
- congruence of arbitrary right angles.

The spatial proof uses Hilbert incidence, order, and congruence only. There is no `HilbertSpaceEuclideanGeo` parameter in the theorem signature. Thus the reconstruction is neutral with respect to Hilbert Group IV.

No coordinate system, vector space, scalar product, normal vector, numerical angle measure, or trigonometric function occurs anywhere in the proof.

Book Zero contributes only inherited general infrastructure through the existing project layers; XI.14 introduces no new proposition-specific foundational principle.

14.14 Dependency Summary

The classical architecture is

$$\begin{array}{c} l \perp \pi, \quad l \perp \rho \\ \downarrow \\ \text{assume the planes meet} \\ \downarrow \\ \text{XI.3}: \ \text{intersection line} \\ \downarrow \\ K \in \pi \cap \rho \\ \downarrow \\ AK \subset \pi, \quad BK \subset \rho \\ \downarrow \\ \text{XI.Def.3} \\ \downarrow \\ \angle BAK, \angle ABK \text{ right} \\ \downarrow \\ \text{I.17} \\ \downarrow \\ \perp \\ \downarrow \\ \pi \parallel \rho. \end{array}$$

The formal architecture has two branches:

$$\begin{array}{c} \pi \neq \rho, \quad l \perp \pi \text{ at } A, \quad l \perp \rho \text{ at } B \\ \downarrow \\ A = B \ ? \\ \begin{array}{cc} \swarrow & \searrow \\ A = B & A \neq B \\ \downarrow & \downarrow \\ \text{XI.5} + \text{plane uniqueness} & \text{assume common point } K \\ \downarrow & \downarrow \\ \pi = \rho & K \notin l \\ \downarrow & \downarrow \\ \perp & \sigma = \text{plane}(l, K) \\ & \downarrow \\ & \text{PlaneGeo}(\sigma) \\ & \downarrow \\ & \angle BAK, \angle ABK \text{ right} \\ & \downarrow \\ & \text{I.17} \\ & \downarrow \\ & \perp \end{array} \\ \downarrow \\ \text{no common point of } \pi, \rho. \end{array}$$

The main conceptual difference is therefore clear. Euclid has one visible contradiction branch and suppresses the degenerate-foot possibility. The formal proof makes that possibility explicit and discharges it by a uniqueness theorem before entering the classical branch.

14.15 Audit Status

This documentation pass was checked against the current production `Proposition11_14.lean` and the Book XI source package through Proposition XI.14. The production theorem was reported to compile successfully, and the separate audit file was reported to compile without introducing a new proposition-specific geometric axiom.

The documentation deliberately does not reproduce the raw `#print axioms` output as a separate mathematical section. The relevant structural fact is recorded above: the theorem has no Group IV parameter and uses only the already accepted project infrastructure.

The Asymptote figures follow the Book XI convention exemplified by the arbitrary-transversal figure for XI.4: geometry internal to a displayed plane is black, genuinely spatial objects are blue, hidden spatial segments are blue dashed, and plane fills are kept light.

14.16 Conclusion

Proposition XI.14 has a simple classical idea but a rich formal anatomy.

The main Euclidean argument survives unchanged. A hypothetical common point of the two planes determines, together with the common perpendicular, an auxiliary plane. In that plane one obtains a nondegenerate triangle with two right angles, and Proposition I.17 gives the contradiction.

The formal reconstruction adds exactly what the classical diagram suppresses. It proves that the hypothetical common point is off the normal, constructs and retains the auxiliary plane explicitly, normalizes the orientation of the perpendicularity witnesses, and separates the degenerate equal-foot case. Proposition XI.5 then supplies a synthetic uniqueness theorem showing that two planes perpendicular to the same line at the same point must coincide.

The result is therefore both faithful to Euclid's spatial proof and sharper about its hidden assumptions. It also illustrates a recurring Book XI pattern: the genuinely spatial work consists in constructing and controlling the correct carrier plane; once that plane is available, the terminal argument returns to the planar Hilbert geometry already developed for Book I.

Chapter 15

Proposition XI.15

15.1 Introduction

Proposition XI.15 is a structural culmination of the parallelism and perpendicularity block developed in XI.8–XI.14. Euclid starts with two intersecting lines in one plane and two corresponding intersecting lines in another plane. If the corresponding pairs are parallel, then the two planes themselves are parallel.

The classical proof is remarkably economical. From the intersection point of the first pair, Euclid drops a perpendicular to the second plane by XI.11. Through its foot he draws two lines parallel to the two reference lines in that plane. Definition XI.3 makes the new normal perpendicular to both of these lines; XI.9 and I.29 transfer the two right angles back to the original pair. XI.4 then says that the same line is perpendicular to the first plane, and XI.14 finishes the proof.

The formal reconstruction preserves this outer architecture, but it exposes three hidden issues.

First, XI.11 requires the point B to lie outside the second plane ρ. Euclid's diagram makes this look automatic, but Lean proves it from the two spatial parallelisms and the distinctness of the two planes.

Second, Euclid invokes I.31 twice through the foot G. The formal parallel-through-a-point construction requires the point to be outside the given line. XI.15 does not guarantee that G lies outside either reference line. The production proof therefore treats the cases $G \in m_i$ and $G \notin m_i$ uniformly by constructing a line q_i which is either literally m_i or parallel to m_i.

Third, the classical I.29 step is not used as one opaque angle theorem. The formal proof factors the transport of perpendicularity through the mechanism already developed in XI.8: neutral I.27 gives disjointness of two perpendiculars to the same transversal, and Hilbert Group IV identifies the unique parallel through the required point.

The result is a proof whose top-level DAG is very close to Euclid's, while its local geometry is more explicit about plane carriers, exceptional incidence, and the exact place where Euclidean parallel uniqueness enters.

15.2 Classical Configuration

Let AB and BC be two straight lines meeting at B, and let DE and EF be two straight lines meeting at E. Suppose

$$AB \parallel DE, \qquad BC \parallel EF,$$

and suppose that the four lines are not all contained in one plane.

Write

$$\pi = \text{plane}(AB, BC), \qquad \rho = \text{plane}(DE, EF).$$

The conclusion is

$$\pi \parallel \rho.$$

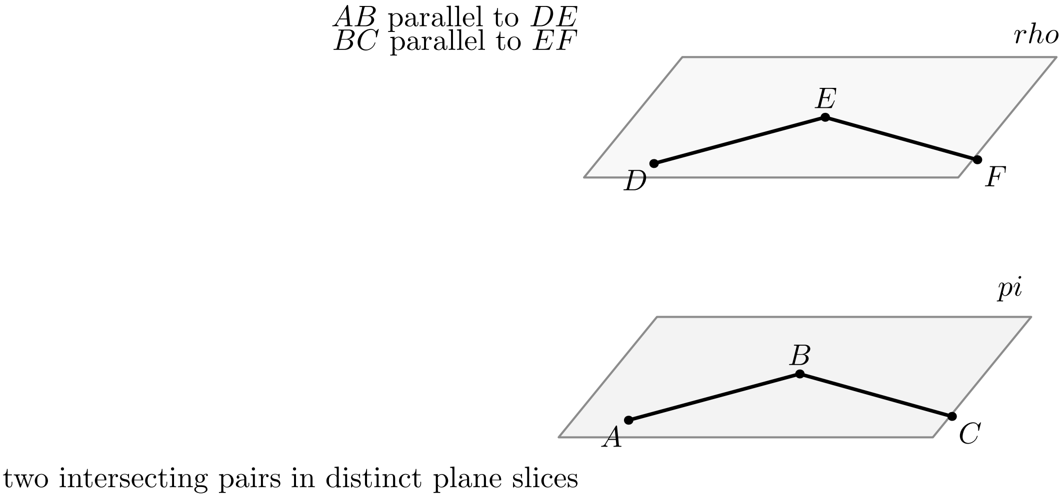


Figure 15.1: The classical spatial configuration of Proposition XI.15. The lines AB, BC meet at B in plane π, while DE, EF meet at E in plane ρ. Corresponding carriers are parallel: $AB \parallel DE$ and $BC \parallel EF$. Geometry internal to a displayed plane is black; the separation between the two plane slices is spatial rather than an additional geometric connector.

Diagrammatic audit. Three stable geometric states are sufficient for XI.15. Figure 15.1 records the classical input. Figure 15.2 records the XI.11 normal and the two formal replacements for Euclid's I.31 constructions, including the hidden exceptional branch $G \in m_i$. Figure 15.3 records the plane-slice mechanism which transports perpendicularity across a parallel pair.

No fourth figure is introduced for the final state

$$n \perp \pi, \qquad n \perp \rho,$$

because that state is exactly the input configuration of Proposition XI.14 and would duplicate the already documented XI.14 geometry.

15.3 Statement

Euclid's statement is:

> If two straight lines meeting one another be parallel to two straight lines meeting one another, not being in the same plane, the planes through them are parallel.

The source-style mathematical form is therefore:

Theorem 3 (Euclid XI.15)**.** *Let l_1, l_2 be distinct lines meeting in a plane π, and let m_1, m_2 be distinct lines meeting in a plane ρ. Suppose*

$$l_1 \parallel m_1, \qquad l_2 \parallel m_2,$$

and suppose the four lines do not lie in one common plane. Then

$$\pi \parallel \rho.$$

The production file exposes a slightly stronger public theorem. It requires the first pair l_1, l_2 to meet at B, but it does not require an explicit second intersection point E or the inequality $m_1 \neq m_2$. Those extra data are not needed by the proof once the second pair is known to lie in ρ and the two corresponding spatial parallelisms are given.

The current public declaration is:

```
theorem euclid_proposition_11_15
    [H : HilbertIncidence Geo]
    [HilbertPlaneIncidence Geo]
    [S : HilbertSpacePrimitive Geo]
    [HSI : HilbertSpaceIncidence Geo]
    [_HSO : HilbertSpaceOrder
      (Geo := Geo) (H := H) (S := S)]
    [HSC : HilbertSpaceCongruence
      (Geo := Geo) (H := H) (S := S)]
    [HSE : HilbertSpaceEuclidean Geo]
    (pi rho : S.Plane)
    (l1 l2 : PlaneLine Geo pi)
    (m1 m2 : PlaneLine Geo rho)
    (B : PlanePoint Geo pi)
    (hBl1 : H.OnLine B.1 l1.1)
    (hBl2 : H.OnLine B.1 l2.1)
    (hl12 : Ne l1 l2)
    (hParallel1 :
      HilbertSpaceLinesParallel Geo l1.1 m1.1)
    (hParallel2 :
      HilbertSpaceLinesParallel Geo l2.1 m2.1)
    (hNoCommonPlane :
      Not (exists omega : S.Plane,
        HilbertLineInPlane Geo l1.1 omega /\
        HilbertLineInPlane Geo l2.1 omega /\
        HilbertLineInPlane Geo m1.1 omega /\
        HilbertLineInPlane Geo m2.1 omega)) :
    HilbertSpacePlanesParallel Geo pi rho
```

For source comparison the file also provides

```
euclid_proposition_11_15_intersecting_pairs
```

which adds an explicit point $E \in m_1 \cap m_2$ and the hypothesis $m_1 \neq m_2$. Its proof is a direct projection to the stronger public theorem.

15.4 Classical Proof

Starting from B, Euclid applies XI.11 to draw

$$BG \perp \rho,$$

where $G \in \rho$.

Through G, Proposition I.31 gives two lines in ρ:

$$GH \parallel ED, \qquad GK \parallel EF.$$

Since $BG \perp \rho$, Definition XI.3 gives

$$BG \perp GH, \qquad BG \perp GK.$$

Now

$$BA \parallel ED \qquad \text{and} \qquad GH \parallel ED.$$

By XI.9,

$$BA \parallel GH.$$

Similarly,

$$BC \parallel GK.$$

Proposition I.29 transfers the two right angles:

$$BG \perp BA, \qquad BG \perp BC.$$

The two lines BA and BC meet at B, so XI.4 yields

$$BG \perp \pi.$$

But by construction

$$BG \perp \rho.$$

Therefore XI.14 applies to the same straight line BG and gives

$$\pi \parallel \rho.$$

The classical local dependency pattern is

$$\begin{gathered}
AB \parallel DE, \quad BC \parallel EF \\
\downarrow \\
\text{XI.11}: \ BG \perp \rho \\
\downarrow \\
\text{I.31}: \ GH \parallel ED, \ GK \parallel EF \\
\downarrow \\
\text{XI.Def.3}: \ BG \perp GH, \ BG \perp GK \\
\downarrow \\
\text{XI.9} + \text{I.29} \\
\downarrow \\
BG \perp BA, \quad BG \perp BC \\
\downarrow \\
\text{XI.4} \\
\downarrow \\
BG \perp \pi \\
\downarrow \\
\text{XI.14} \\
\downarrow \\
\pi \parallel \rho.
\end{gathered}$$

Thus XI.15 is classically a short synthesis theorem: XI.11 constructs the normal, XI.4 upgrades two line-line perpendicularities to a line-plane perpendicularity, and XI.14 converts a common normal into plane parallelism.

15.5 The Hidden I.31 Exceptional Case

The most important local correction in the formal reconstruction occurs before the classical I.31 step.

Euclid says simply: through G, draw a line parallel to ED and a line parallel to EF. In a formal Hilbert construction, however, a parallel through a point is constructed only when the point lies outside the given line. XI.15 does not imply

$$G \notin DE \qquad \text{or} \qquad G \notin EF.$$

The production theorem therefore uses the proposition-local helper

```
hilbert_XI15_parallel_or_equal_through_point_in_plane
```

with the following mathematical content. Given a line $m \subset \rho$ and a point $G \in \rho$, it returns a line $q \subset \rho$ through G such that

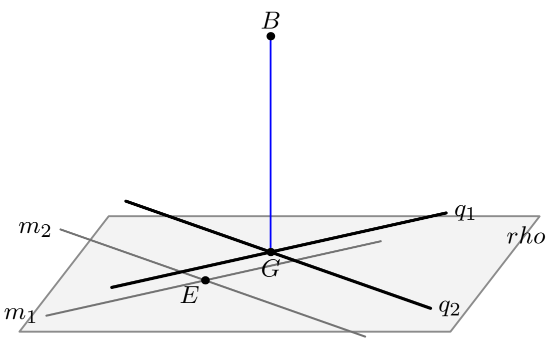
$$q = m \qquad \text{or} \qquad m \parallel q.$$

There are two branches:

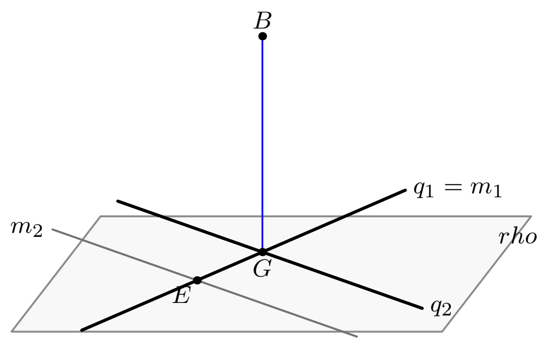
$$\begin{aligned} G \in m &\implies q := m, \\ G \notin m &\implies \text{use the XI.12 parallel-through-point construction.} \end{aligned}$$

In the second branch, the XI.12 helper returns spatial parallelism together with a carrier plane. XI.15 then proves that the constructed line actually lies in the intended plane ρ, using uniqueness of the plane through a line and an external point.

blue = XI.11 normal n



(a) generic case: G is off both reference lines



(b) exceptional case: G already lies on m1

Figure 15.2: The formal replacement for Euclid's two I.31 instructions. In the generic branch, lines $q_1, q_2 \subset \rho$ through G are parallel to the reference lines m_1, m_2. In the exceptional branch $G \in m_1$, the correct choice is simply $q_1 = m_1$; only the second reference line requires a new parallel. The blue line is the genuinely spatial normal n, while all black and gray lines lie in ρ.

This case split is not a Lean artifact. It is the precise formal content of the geometric fact that the instruction "draw a parallel through G" degenerates to "use the given line itself" when G already lies on the carrier.

15.6 What is genuinely three-dimensional here?

XI.15 alternates repeatedly between ambient space and induced plane geometry.

The genuinely spatial data are:

- the two reference planes π, ρ;
- the two pairs of corresponding spatially parallel lines;
- the condition that all four lines do not lie in one ambient plane;
- the point B, which must be proved to lie outside ρ;
- the XI.11 normal n from B to ρ;
- the final line-plane perpendicularities $n \perp \pi$ and $n \perp \rho$;
- the spatial target $\mathrm{HilbertSpacePlanesParallel}(\pi, \rho)$.

There are also auxiliary carrier planes which are genuinely spatial. Every assertion of

$$\mathrm{HilbertSpaceLinesParallel}(l, q)$$

contains a plane witnessing that l and q are coplanar. The perpendicularity-transport lemma uses precisely this carrier plane, denoted σ, to move from ambient line data to an induced $\mathrm{PlaneGeo}(\sigma)$.

The planar work occurs in two distinct kinds of slices:

1. the reference plane ρ, where the lines through the foot G are constructed;
2. a carrier plane σ for each parallel pair $l \parallel q$, where the I.27/Group IV perpendicularity transport is performed.

Thus XI.15 is not one planar proof placed in space. The proof changes planes for mathematically different reasons: first to construct reference directions inside ρ, then to compare a transversal with a particular parallel pair inside its own carrier plane.

15.7 Formal Statement and Normalized Core

The substantial theorem is first isolated under explicit plane distinctness:

```
theorem euclid_proposition_11_15_normalized
    ...
    (pi rho : S.Plane)
    (l1 l2 : PlaneLine Geo pi)
    (m1 m2 : PlaneLine Geo rho)
```

```
    (B : PlanePoint Geo pi)
    ...
    (hl12 : Ne l1 l2)
    (hParallel1 :
      HilbertSpaceLinesParallel Geo l1.1 m1.1)
    (hParallel2 :
      HilbertSpaceLinesParallel Geo l2.1 m2.1)
    (hPlanesNe : Ne pi rho) :
    HilbertSpacePlanesParallel Geo pi rho
```

The public theorem replaces the explicit hypothesis

$$\pi \neq \rho$$

by Euclid's source condition that the four lines are not contained in one common plane. If $\pi = \rho$, then all four displayed lines lie in that same plane, contradicting the public hypothesis. The public wrapper is therefore mathematically thin.

This separation is useful. The normalized theorem expresses the actual geometric engine. The public theorem expresses the source-facing configuration.

15.8 Spatial / Planar Architecture Used

The ambient layer is:

```
HilbertIncidence
HilbertPlaneIncidence
HilbertSpacePrimitive
HilbertSpaceIncidence
HilbertSpaceOrder
HilbertSpaceCongruence
HilbertSpaceEuclidean
```

The principal spatial relations are:

```
HilbertLineInPlane
HilbertSpaceLinesParallel
HilbertLinesDisjoint
HilbertLinesPerpendicularAt
HilbertLinePerpendicularPlaneAt
HilbertSpacePlanesParallel
```

The induced plane layer uses:

```
PlanePoint
PlaneLine
PlaneOnLine
PlaneGeo
```

and the standard ambient/plane bridges, in particular the bridge for line-line perpendicularity.

A permanent Book XI instance rule is respected throughout: the proof does not install global ambient instances

```
HilbertOrder Geo
HilbertCongruence Geo
```

for three-space. Order and planar congruence are supplied through explicit PlaneGeo slices; spatial congruence remains in `HilbertSpaceCongruence`.

15.9 Proof Architecture of the Reconstruction

15.9.1 Proving $B \notin \rho$

The normalized theorem cannot invoke XI.11 until it has proved that the intersection point B of l_1, l_2 lies outside the second plane.

The helper

```
hilbert_XI15_intersection_point_off_other_plane
```

is purely incidence-theoretic.

Assume $B \in \rho$. The parallelism $l_1 \parallel m_1$ provides a carrier plane σ_1 containing both lines. Since the two parallel lines are disjoint, $B \notin m_1$. Both σ_1 and ρ therefore contain the same line m_1 and the same external point B. Plane uniqueness gives

$$\sigma_1 = \rho,$$

hence

$$l_1 \subset \rho.$$

The same argument gives

$$l_2 \subset \rho.$$

But the distinct intersecting lines l_1, l_2 determine a unique plane. They already lie in π, and now they also lie in ρ. Therefore

$$\pi = \rho,$$

contradicting the normalized hypothesis.

The file also contains the symmetric package

```
hilbert_XI15_intersection_points_off_opposite_planes
```

which proves both $B \notin \rho$ and the analogous source-style statement $E \notin \pi$. The main normalized proof needs only the first half.

15.9.2 XI.11 constructs the normal

With $B \notin \rho$ established, the production proof invokes

```
euclid_proposition_11_11
```

and obtains a line n and a foot G satisfying

$$B \in n, \qquad n \perp \rho \text{ at } G.$$

This is packaged by

```
hilbert_XI15_normal_from_intersection_to_other_plane
```

and then combined with the two reference-line constructions in

```
hilbert_XI15_normal_and_reference_lines
```

to produce

$$q_1, q_2 \subset \rho, \qquad G \in q_1 \cap q_2,$$

with

$$q_i = m_i \quad \text{or} \quad m_i \parallel q_i.$$

15.9.3 Recovering $l_i \parallel q_i$

The original hypotheses give

$$l_i \parallel m_i.$$

If $q_i = m_i$, there is nothing to prove.

In the genuine parallel branch

$$m_i \parallel q_i,$$

the proof needs strict transitivity of spatial parallelism. This is slightly subtle because project parallelism is irreflexive: a line is not parallel to itself.

The helper

```
hilbert_XI15_spaceParallel_transitive_distinct
```

therefore assumes explicitly that the outer lines are distinct.

Its proof again separates two spatial configurations:

- if l, m, q lie in one plane, Hilbert Group IV gives uniqueness of the parallel to m through a point;
- if the three lines do not lie in one common plane, XI.9 supplies the required spatial transitivity.

In XI.15 the required distinctness $l_i \neq q_i$ follows from $B \notin \rho$. Indeed, if $l_i = q_i$, then $B \in q_i \subset \rho$.

Thus the proof obtains

$$l_1 \parallel q_1, \qquad l_2 \parallel q_2.$$

15.9.4 Definition XI.3 gives the two local perpendicularities

Since
$$n \perp \rho \text{ at } G$$
and
$$q_i \subset \rho, \qquad G \in q_i,$$
Definition XI.3 is represented directly by `HilbertLinePerpendicularPlaneAt.perpendicular_to_line`. It gives
$$n \perp q_1 \text{ at } G, \qquad n \perp q_2 \text{ at } G.$$

The remaining problem is to transport these perpendicularities from q_i to the parallel lines l_i.

15.9.5 The formal I.29 replacement

The central transport theorem is

```
hilbert_XI15_perpendicular_transfer_across_parallel
```

and its line-plane wrapper is

```
hilbert_XI15_plane_normal_perpendicular_to_parallel_line
```

Suppose
$$l \parallel q,$$
the transversal n meets q at G and l at B, and
$$q \perp n \text{ at } G.$$

The spatial parallelism $l \parallel q$ supplies a common carrier plane σ. Since $B, G \in n$ and $B \neq G$, the whole transversal n also lies in σ.

The proof now passes to
$$\mathrm{PlaneGeo}(\sigma).$$
Through B, it constructs a line p perpendicular to n. Hence
$$p \perp n \text{ at } B, \qquad q \perp n \text{ at } G.$$

The feet are distinct. The neutral XI.8 helper representing the I.27 mechanism therefore gives
$$p \cap q = \varnothing.$$

But l is also disjoint from q, and both l and p pass through B. Hilbert Group IV gives uniqueness of the parallel through B:
$$p = l.$$
Therefore
$$l \perp n \text{ at } B,$$
and by symmetry
$$n \perp l \text{ at } B.$$

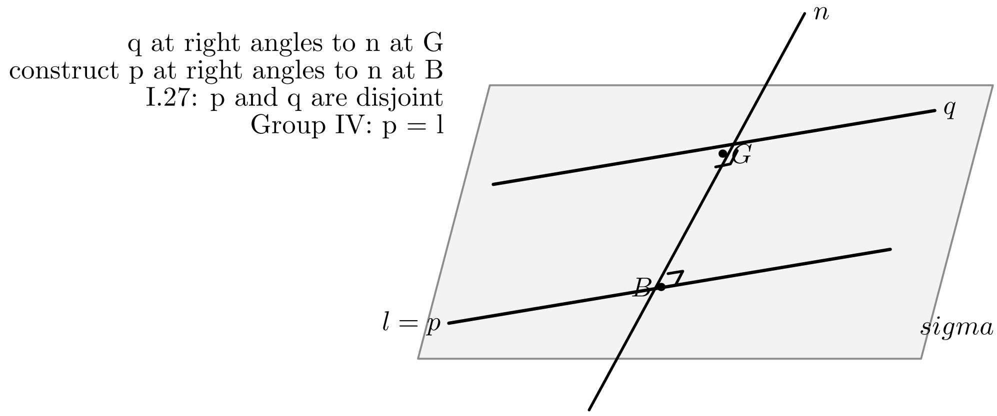


Figure 15.3: The formal replacement for Euclid's I.29 step, carried out in the common plane σ of $l \parallel q$. The line q is perpendicular to the transversal n at G. A perpendicular p to n is erected at B. Neutral I.27 gives $p \cap q = \varnothing$; Group IV then identifies p with the already given parallel l. The final lower carrier is therefore labelled $l = p$.

This is exactly the logical role of Euclid's I.29, but factored according to the current Hilbert architecture:

$$\text{I.27} \quad + \quad \text{Group IV uniqueness} \quad \Longrightarrow \quad \text{right-angle transport across parallels.}$$

15.9.6 XI.4 makes the normal perpendicular to π

Applying the preceding transport twice gives

$$n \perp l_1 \text{ at } B, \qquad n \perp l_2 \text{ at } B.$$

The lines l_1, l_2 are distinct and meet at B, so the production theorem

```
euclid_proposition_11_4
```

gives

$$n \perp \pi \text{ at } B.$$

This is one of the cleanest stages of the formal proof: the substantial work has already been normalized into exactly the XI.4 interface.

15.9.7 XI.14 closes the proof

From XI.11 we still have

$$n \perp \rho \text{ at } G.$$

From XI.4 we have

$$n \perp \pi \text{ at } B.$$

The normalized theorem assumes $\pi \neq \rho$, so XI.14 applies directly:

$$\pi \parallel \rho.$$

The final line of the normalized proof is therefore an actual invocation of

```
euclid_proposition_11_14
```

rather than a new plane-separation argument.

15.9.8 The source-facing wrapper

The public theorem assumes that the four displayed lines do not all lie in one plane. It first derives

$$\pi \neq \rho.$$

If the two planes were equal, then l_1, l_2, m_1, m_2 would all lie in π, contradicting the source hypothesis.

The public theorem then calls the normalized core.

The separate theorem

```
euclid_proposition_11_15_intersecting_pairs
```

adds the explicit second intersection point and line distinctness appearing in Euclid's wording. Since those data are not needed internally, this source-style theorem is intentionally only a wrapper.

15.10 Wyler-Style Flat Reconstruction

The Wyler reconstruction of XI.15 keeps the public Euclidean statement but changes the internal proof discipline. The main spatial states are recorded by exact generated-flat identities before the metric argument is used.

The source hypothesis that the four displayed lines are not all contained in one plane first gives

$$\pi \neq \rho.$$

The flat layer then records the global rank-three certificate

$$\mathrm{Join}(\pi, \rho) = E^3.$$

As in XI.14, this certificate is informative but not by itself sufficient: two distinct planes generate the whole ambient three-space whether or not they are parallel. The conclusion still requires the perpendicularity and plane-separation part of the argument.

The first substantial use of the Wyler vocabulary occurs in the proof that the intersection point B of l_1, l_2 lies outside ρ. Suppose instead that $B \in \rho$. From

$$l_1 \parallel m_1$$

let τ_1 be a carrier plane of the parallel pair. Since $B \in l_1$ and l_1, m_1 are disjoint, $B \notin m_1$. Therefore the same generators m_1 and B determine both τ_1 and ρ:

$$\mathrm{Join}(m_1, \{B\}) = \mathrm{carrier}(\tau_1),$$

$$\mathrm{Join}(m_1, \{B\}) = \mathrm{carrier}(\rho).$$

Hence the two plane carriers coincide and $l_1 \subset \rho$. Repeating the same argument for l_2, m_2 gives $l_2 \subset \rho$. Since the distinct intersecting lines l_1, l_2 generate π, this forces

$$\pi = \rho,$$

a contradiction.

Thus the Wyler proof does not treat "the plane through a line and an external point is unique" as a proposition-local geometric black box. It compares the exact joins generated by the same data.

After XI.11 constructs

$$B \in n, \qquad n \perp \rho \text{ at } G,$$

the same principle is used a second time. For each reference line m_i, there are two cases.

If $G \in m_i$, take

$$q_i = m_i.$$

Otherwise XI.12 constructs through G a line q_i parallel to m_i inside a generated slice σ_i, together with exact certificates

$$\mathrm{Join}(m_i, \{G\}) = \mathrm{carrier}(\sigma_i),$$

$$\mathrm{Join}(m_i, q_i) = \mathrm{carrier}(\sigma_i).$$

But $m_i \subset \rho$ and $G \in \rho$, so the same first join is also

$$\mathrm{Join}(m_i, \{G\}) = \mathrm{carrier}(\rho).$$

Consequently $q_i \subset \rho$. The containment of the constructed line in the intended plane is therefore recovered from equality of generated flats, not from a separate ad hoc plane-identification lemma.

The next step is the strict spatial parallelism

$$l_i \parallel q_i.$$

Again the proof separates the two possible flat configurations. If l_i, m_i, q_i lie in one plane, Hilbert Group IV gives the required uniqueness of the parallel through a point. If no common plane contains the three lines, Proposition XI.9 gives the spatial transitivity directly. The outer-line distinctness required by strict parallelism follows from $B \notin \rho$, because $B \in l_i$ whereas $q_i \subset \rho$.

Only after these incidence and flat identifications are complete does the proof enter the metric layer. Since

$$n \perp \rho \text{ at } G$$

and $q_i \subset \rho$ passes through G, Definition XI.3 gives

$$n \perp q_i \text{ at } G.$$

For each parallel pair $l_i \parallel q_i$, their common carrier plane contains both feet B, G of the transversal n, hence contains the whole line n. Inside the induced plane geometry one erects through B a line perpendicular to n. Neutral I.27 proves that this auxiliary line is disjoint from q_i, and Group IV identifies it with l_i. Therefore

$$n \perp l_i \text{ at } B.$$

The complete Wyler-style structure is therefore

```
four displayed lines not coplanar
   |
   v
pi != rho
   |
   +--> Join(pi,rho) = E^3             [rank-three certificate]
   |
   v
assume B in rho
   |
Join(m1,{B}) = carrier(tau1) = carrier(rho)
Join(m2,{B}) = carrier(tau2) = carrier(rho)
   |
l1,l2 subset rho
   |
pi = rho --> contradiction
   |
   v
B notin rho
   |
  XI.11
   v
n perp rho at G
   |
generated slices through G
   |
q1,q2 subset rho
m1 = q1 or m1 || q1
m2 = q2 or m2 || q2
   |
Group IV / XI.9
   |
l1 || q1, l2 || q2
   |
local PlaneGeo metric transport
   |
n perp l1, n perp l2
   |
  XI.4
   v
n perp pi
   |
  XI.14
   v
pi || rho
```

This decomposition illustrates the characteristic proof discipline of the current Wyler route. Generated-flat equalities identify the spatial configuration first; metric and Euclidean principles are invoked only after the relevant carriers have been fixed. The final theorem remains Euclid XI.15, but the proof exposes a separate incidence skeleton on which the perpendicularity argument acts.

15.11 Lean Formalization

The production file imports:

```
import CGJteamLab.Proposition11_14
import CGJteamLab.Proposition11_12
import CGJteamLab.Proposition11_9
import CGJteamLab.Proposition11_8
import CGJteamLab.Proposition11_4
```

Mathematically, the proof also calls XI.11 through the imported Book XI dependency chain.

The proposition-local helpers fall into four conceptual groups.

Spatial incidence normalization.

```
hilbert_XI15_spaceLinesParallel_symm
hilbert_XI15_intersection_point_off_other_plane
hilbert_XI15_intersection_points_off_opposite_planes
```

The XI.11/I.31 construction package.

```
hilbert_XI15_parallel_or_equal_through_point_in_plane
hilbert_XI15_normal_from_intersection_to_other_plane
hilbert_XI15_normal_and_reference_lines
```

Parallel and perpendicular transport.

```
hilbert_XI15_perpendicular_transfer_across_parallel
hilbert_XI15_plane_normal_perpendicular_to_parallel_line
hilbert_XI15_spaceParallel_transitive_distinct
```

Final theorem layer.

```
euclid_proposition_11_15_normalized
euclid_proposition_11_15
euclid_proposition_11_15_intersecting_pairs
```

This grouping reflects stable mathematical responsibilities, not the chronology of the test files used during development.

15.12 Hidden Formal Data

15.12.1 The XI.11 starting point must be outside ρ

Euclid begins XI.11 from B without a separate argument that $B \notin \rho$. The production proof derives this by a nontrivial spatial incidence argument. This is the first hidden obligation in XI.15.

15.12.2 The I.31 point may already lie on the reference line

The foot G may lie on m_1, on m_2, or on neither. The formal construction cannot assume the generic diagrammatic case. The equal-or-parallel helper packages all possibilities without changing the outer proof.

15.12.3 Spatial parallelism carries a plane witness

The predicate `HilbertSpaceLinesParallel` is not merely a disjointness relation. It also stores a common carrier plane. XI.15 uses that carrier plane to create the PlaneGeo slice in which perpendicularity transport is proved.

15.12.4 Spatial parallelism is strict

Because parallel lines are disjoint, the project relation is irreflexive. A transitivity lemma therefore needs the extra condition that the two outer lines are distinct. In XI.15 that condition is not assumed separately; it is derived from $B \notin \rho$.

15.12.5 The feet B and G must be distinct

The perpendicularity-transport theorem requires two distinct feet on the transversal n. If $B = G$, then B would lie on both parallel lines l and q, contradicting their disjointness.

15.12.6 The second source intersection is redundant

Euclid describes both pairs as intersecting. The public production theorem needs only the first explicit intersection $B \in l_1 \cap l_2$. The second intersection is mathematically compatible with the theorem but is not used by the proof.

This is a genuine strengthening, not an omission from the formal configuration. The source-style corollary records the literal Euclidean data separately.

15.13 Relationship with Earlier Book XI Propositions

XI.15 is unusually transparent as a consumer of the preceding Book XI infrastructure.

XI.4. This proposition is used exactly at the classical place. Once the formal proof has obtained

$$n \perp l_1, \qquad n \perp l_2$$

at their common point B, XI.4 upgrades the pair to

$$n \perp \pi.$$

XI.8. The final theorem XI.8 is not the target of the proof, but XI.15 reuses its neutral helper for two coplanar lines perpendicular to the same transversal at distinct points. This is the formal I.27 engine inside the replacement for I.29.

XI.9. XI.9 appears in two roles. It is part of the classical proof of $BA \parallel GH$ and $BC \parallel GK$. Formally, it is used inside the strict transitivity helper when three spatial parallel lines do not have one common carrier plane.

XI.11. XI.11 supplies the normal from B to the second plane. Before it can be called, XI.15 proves the hidden hypothesis $B \notin \rho$.

XI.12. The plane-local parallel-through-point helper developed in XI.12 is reused in the generic branch $G \notin m_i$. XI.15 is therefore the first later proposition where that helper becomes genuine reusable infrastructure rather than merely proposition-local convenience.

XI.14. XI.14 is the terminal theorem. Once the same line n is known to be perpendicular to both π and ρ, the conclusion of XI.15 is immediate.

XI.10 is conceptually related because it starts from the same kind of configuration—two pairs of corresponding parallel lines in distinct spatial plane slices. But XI.15 does not need XI.10's directed segment parallelism. Orientation of selected rays matters for equality of angles in XI.10; it is irrelevant to plane parallelism in XI.15.

15.14 Relationship with Book I / Book Zero / Hilbert Infrastructure

The classical Book I dependencies are I.31 and I.29.

The formal reconstruction preserves the role of I.31 but makes its precondition explicit. The generic branch uses the existing parallel-through-point construction in a plane slice; the exceptional branch uses the original line itself.

The classical I.29 step is represented more structurally. The current proof uses:

- a neutral I.27-type theorem to show that two lines perpendicular to the same transversal at distinct points are disjoint;
- Hilbert Group IV to identify the unique parallel through the required point.

Thus XI.15 genuinely uses Euclidean parallel geometry. This is visible already in the production signature:

```
[HSE : HilbertSpaceEuclidean Geo]
```

The use of Group IV is mathematically substantive, not merely inherited module noise.

No continuity principle is required. No coordinate system, vector, scalar product, numerical angle measure, or trigonometric function occurs in the reconstruction.

Book Zero contributes only the already established synthetic infrastructure used transitively by the right-angle and planar construction layers. XI.15 introduces no new proposition-specific geometric axiom.

15.15 Dependency Summary

The classical DAG is:

$$
\begin{array}{c}
l_1 \parallel m_1, \quad l_2 \parallel m_2 \\
\downarrow \\
\text{XI.11} \\
\downarrow \\
n \perp \rho \text{ at } G \\
\downarrow \\
\text{I.31 twice} \\
\downarrow \\
q_1 \parallel m_1, \quad q_2 \parallel m_2 \\
\downarrow \\
\text{XI.Def.3} \\
\downarrow \\
n \perp q_1, \quad n \perp q_2 \\
\downarrow \\
\text{XI.9} + \text{I.29} \\
\downarrow \\
n \perp l_1, \quad n \perp l_2 \\
\downarrow \\
\text{XI.4} \\
\downarrow \\
n \perp \pi \\
\downarrow \\
\text{XI.14} \\
\downarrow \\
\pi \parallel \rho.
\end{array}
$$

The formal DAG is:

$$
\begin{array}{c}
\pi \neq \rho, \quad l_1 \parallel m_1, \quad l_2 \parallel m_2, \quad l_1 \cap l_2 = \{B\} \\
\downarrow \\
B \notin \rho \\
\downarrow \\
\text{XI.11} \\
\downarrow \\
n \perp \rho \text{ at } G \\
\downarrow \\
G \in m_i \ ? \\
\swarrow \qquad \searrow \\
q_i := m_i \qquad \text{XI.12 parallel construction} \\
\downarrow \\
q_i \subset \rho, \quad q_i = m_i \text{ or } m_i \parallel q_i \\
\downarrow \\
l_i \parallel q_i \\
\downarrow \\
n \perp q_i \text{ at } G \\
\downarrow \\
\text{PlaneGeo}(\sigma_i) : \text{I.27} + \text{Group IV} \\
\downarrow \\
n \perp l_i \text{ at } B \quad (i = 1, 2) \\
\downarrow \\
\text{XI.4} \\
\downarrow \\
n \perp \pi \text{ at } B \\
\downarrow \\
\text{XI.14} \\
\downarrow \\
\pi \parallel \rho.
\end{array}
$$

The formal proof is longer around the two classical line constructions and the I.29 step, because the diagram suppresses exceptional incidence and carrier-plane information. It is shorter at the public boundary: the source-style second intersection point E is not required by the geometric core.

15.16 Audit Status

The documentation was checked against the current production `Proposition11_15.lean`. The user reported that the production file, the separate axiom-audit file, and the full project build all compile cleanly.

The separate audit reports the standard accepted project baseline and no new proposition-specific geometric axiom. The mathematically important foundation fact is the one visible in the theorem signature itself: XI.15 genuinely uses the Euclidean spatial layer `HilbertSpaceEuclidean`, while no continuity assumption enters.

The figures follow the established Book XI convention: displayed plane geometry is black or gray, genuinely spatial lines are blue, hidden spatial segments are blue dashed when needed, and plane fills remain light.

15.17 Conclusion

Proposition XI.15 is a particularly successful example of the Book XI architecture reaching maturity.

Euclid's top-level idea survives almost unchanged. Construct a normal to the second plane, transfer its right angles to the two lines of the first plane, use XI.4 to make it a normal to that plane as well, and conclude by XI.14 that the planes are parallel.

What formalization adds is not a different geometry but a precise account of the information carried by the classical diagram. It proves that the first intersection point is genuinely outside the second plane; it repairs the hidden exceptional case in the two I.31 constructions; it records the carrier planes implicit in spatial parallelism; and it factors the I.29 transport into neutral perpendicular geometry plus Group IV uniqueness.

The resulting proof also clarifies the relation between source statement and formal theorem. Euclid presents two intersecting pairs, but the second explicit intersection is not needed by the geometric core. The production theorem therefore proves a slightly stronger statement, while a thin source-style corollary retains the literal Euclidean configuration.

At the level of the Book XI DAG, XI.15 is an important closure theorem: XI.11 supplies the common normal, XI.12 supplies the reusable parallel-through-point construction, XI.8 and XI.9 supply the parallel transport machinery, XI.4 upgrades two right angles to a line-plane perpendicularity, and XI.14 converts the common normal into the final parallelism of planes. No coordinates or metric algebra are needed; the proof remains entirely synthetic.

Appendix A

Euclid in Heath's Edition: Source Audit

A.1 Scope

This appendix records the Book XI source audit against Euclid's *Elements* in Heath's edition through Proposition XI.13 [1]. The purpose is to control the classical statement, construction, and local proposition dependencies. Lean-specific incidence, subtype, and plane-transport obligations are documented in the proposition chapters and are not attributed to Euclid or Heath.

A.2 Proposition XI.4

Proposition XI.4 states that if a straight line is set up at right angles to each of two straight lines at their point of intersection, then it is also at right angles to the plane through those two lines.

The classical proof chooses equal segments on the two given lines and introduces an arbitrary line of the plane through the common point. Triangle-congruence arguments are then used to prove that the candidate spatial line makes a right angle with that arbitrary transversal.

For the formal reconstruction the significant classical features are:

- the target is universal over every line of the plane through the common point;
- the arbitrary transversal is part of the proposition's logical content, not merely a feature of the diagram;
- the proof is synthetic and based on incidence, right angles, equal segments, and triangle congruence.

A.3 Proposition XI.5

Proposition XI.5 says that if one straight line is perpendicular at a common point to three straight lines meeting there, then those three straight lines lie in one plane.

Euclid's contradiction argument uses the plane through two of the three candidate coplanar lines and another plane through the common perpendicular and the remaining line. The intersection line of the two planes produces, inside one plane, two distinct lines through the same point both perpendicular to the common perpendicular.

The local classical dependency is therefore governed by the preceding spatial propositions: the intersection-of-planes configuration and Proposition XI.4 supply the contradiction. The current Lean proof organizes this material differently, but preserves the same spatial core: perpendicularity to a plane forces the required intersection line to be perpendicular to the common normal.

A.4 Proposition XI.6

Heath gives the statement:

> If two straight lines be at right angles to the same plane, the straight lines will be parallel.

Let AB and CD be perpendicular to the reference plane at B and D. Euclid joins BD, draws DE in the reference plane perpendicular to BD, makes

$$DE = AB,$$

and joins BE, AE, and AD.

Since AB is perpendicular to the reference plane,

$$\angle ABD, \qquad \angle ABE$$

are right. Since CD is also perpendicular to the same plane, the corresponding lines of the plane through D are perpendicular to CD.

The first triangle comparison uses

$$AB = DE, \qquad BD = DB,$$

and equality of the included right angles. Proposition I.4 gives

$$AD = BE.$$

The second comparison uses

$$DE = BA, \qquad EA = AE, \qquad DA = BE.$$

Proposition I.8 therefore gives

$$\angle EDA = \angle ABE.$$

Hence $\angle EDA$ is right and

$$ED \perp DA.$$

At D, the line ED is now perpendicular to

$$DB, \qquad DA, \qquad DC.$$

Proposition XI.5 places these three lines in one plane. Proposition XI.2 supplies the corresponding triangle-plane incidence, bringing AB into that same plane. The two right angles made with the transversal BD then give the final parallelism by Proposition I.28.

The local classical DAG can be recorded as

$$\begin{gathered}
\text{construct } DE \perp BD, \quad DE = AB \\
\Downarrow \\
\text{I.4} \\
\Downarrow \\
AD = BE \\
\Downarrow \\
\text{I.8} \\
\Downarrow \\
ED \perp DA \\
\Downarrow \\
\text{XI.5} \\
\Downarrow \\
DB, DA, DC \text{ coplanar} \\
\Downarrow \\
\text{XI.2} \\
\Downarrow \\
AB, BD, DC \text{ coplanar} \\
\Downarrow \\
\text{I.28} \\
\Downarrow \\
AB \parallel CD.
\end{gathered}$$

A.5 Comparison with the Lean reconstruction at XI.6

The Lean proof preserves the central metric sequence of the classical argument:

$$\text{auxiliary } DE \longrightarrow \text{SAS} \longrightarrow \text{SSS} \longrightarrow \text{XI.5}.$$

The formal expansion is concentrated in the data suppressed by the classical diagram:

- each planar construction is performed in an explicit induced plane geometry;
- cross-plane segment and angle comparisons use the spatial congruence layer;
- noncollinearity and point-in-plane obligations are explicit;
- the common plane of the two final normals is produced as a witness;
- spatial parallelism records both coplanarity and disjointness, so skew lines are excluded;
- the last planar disjointness proof uses the established neutral alternate-angle infrastructure rather than a direct call to the public I.28 wrapper.

The last point is a dependency-graph difference, not a change in the geometric target. Euclid closes the configuration by I.28; the current Lean source imports the Proposition I.27 module and invokes the lower alternate-angle criterion in the induced common plane.

A.6 Checkpoint XI.6

Through XI.6 the reconstruction remains synthetic. No coordinate, vector, scalar-product, trigonometric, or numerical-angle representation is used in the proposition proofs.

The source comparison also isolates the new structural point of XI.6: in three-space the proof of parallelism has to establish coplanarity before planar nonintersection can be invoked. Euclid obtains this by XI.5 and XI.2. The Lean theorem records the same distinction directly in the target relation `HilbertSpaceLinesParallel`, which consists of an explicit common plane together with disjointness of the two ambient carriers.

A.7 Proposition XI.7

Proposition XI.7 concerns the plane determined by two parallel straight lines. In Book I the definition of parallel lines already requires the two lines to be in one plane. Thus the reference plane is part of the hypothesis, not an additional conclusion of XI.7.

Choose a point on each of the two parallel lines and join the chosen points. Euclid argues by contradiction. He supposes that the joining straight line is carried by another, more elevated plane. A plane through that alleged join is then intersected with the reference plane. Proposition XI.3 makes the common section a straight line. Since both chosen points lie in both planes, that section passes through both of them. The contradiction is expressed in the classical text by the impossibility of two straight lines enclosing an area. The join must therefore lie in the original plane of the parallels.

The local classical DAG is

$$\begin{gathered} AB \parallel CD \\ \Downarrow \\ AB, CD \subset \pi \quad \text{(Book I definition of parallels)} \\ \Downarrow \\ E \in AB, \quad F \in CD \\ \Downarrow \\ \text{suppose the join } EF \text{ is carried by another plane } \rho \\ \Downarrow \\ \text{XI.3} \\ \Downarrow \\ \pi \cap \rho \text{ is a straight line through } E, F \\ \Downarrow \\ \text{contradiction} \\ \Downarrow \\ EF \subset \pi. \end{gathered}$$

No Proposition XI.4, XI.5, or XI.6 occurs in this local proof. The explicit Book XI dependency is XI.3.

A.8 Comparison with the Lean reconstruction at XI.7

The Lean proof has a shorter incidence DAG. The formal relation `HilbertSpaceLinesParallel` already contains a witness plane σ for the two parallel carriers, together with their disjointness.

The selected points therefore lie in σ. Their distinctness follows from disjointness of the parallel carriers. After the joining carrier is constructed, `HilbertSpaceIncidence.line_in_plane` places the whole join in σ.

This final call represents the Hilbert incidence principle that two distinct points of a line lying in a plane force the entire line to lie in that plane. Consequently the formal proof does not reproduce Euclid's auxiliary second plane or his use of XI.3. This is a difference in foundational packaging, not in the geometric conclusion.

A.9 Checkpoint XI.7

Through XI.7 the reconstruction remains entirely synthetic. Proposition XI.7 uses only incidence: no order, congruence, angle theory, continuity, coordinate representation, vector formalism, or scalar product enters its proof.

The source comparison now exhibits two distinct ways to express the same closure principle. Euclid derives it by a plane-intersection contradiction; the Hilbert-style formalization uses an explicit common-plane witness for spatial parallelism and a general line-in-plane incidence axiom.

A.10 Proposition XI.8

Proposition XI.8 states that if two straight lines are parallel and one of them is perpendicular to a plane, then the other is perpendicular to the same plane.

Let $AB \parallel CD$, with AB perpendicular to the reference plane at B. Euclid lets CD meet the plane at D, joins BD, draws DE in the reference plane perpendicular to BD, makes $DE = AB$, and joins BE, AE, and AD.

The local classical dependencies are unusually informative because they cross the Euclidean parallel boundary.

- XI.7 places AB, CD, and BD in one plane.
- I.11 and I.3 supply the auxiliary perpendicular and equal segment.
- The definition of line–plane perpendicularity makes $\angle ABD$ and $\angle ABE$ right.
- I.29 converts the given parallelism into the right angle $\angle CDB$, hence $CD \perp BD$.
- I.4 and I.8 give $AD = BE$ and then show $\angle EDA$ right.
- XI.4 applied in the plane through BD, DA gives $ED \perp CD$.
- XI.4 applied again in the reference plane, to DB, DE, gives $CD \perp \pi$.

The incidence step placing CD in the plane through BD, DA uses the planarity of triangle ABD together with the common plane of AB, CD, BD. The printed diagram suppresses this plane identification.

The local classical DAG is

$$\begin{gathered}
AB \parallel CD, \quad AB \perp \pi \\
\Downarrow \\
\text{XI.7} \\
\Downarrow \\
DE \perp BD, \quad DE = AB \\
\Downarrow \\
\text{I.29} \\
\Downarrow \\
CD \perp BD \\
\Downarrow \\
\text{I.4}, \text{I.8} \\
\Downarrow \\
ED \perp AD \\
\Downarrow \\
\text{XI.4} \\
\Downarrow \\
ED \perp CD \\
\Downarrow \\
\text{XI.4} \\
\Downarrow \\
CD \perp \pi.
\end{gathered}$$

A.11 Comparison with the Lean reconstruction at XI.8

The Lean proof preserves the synthetic metric core but reorganizes two classical steps.

First, the point D where the second parallel meets the reference plane is not assumed. It is derived from the common carrier plane of the parallels, the intersection of that plane with the reference plane, and Group IV uniqueness.

Second, the proof does not call I.29 directly. It constructs through D a line perpendicular to BD, proves by the neutral I.27 alternate-angle criterion that this line is disjoint from the first parallel, and then uses Group IV uniqueness to identify it with the second parallel.

Thus the formal factorization is

$$\text{neutral I.27} + \text{Euclidean uniqueness of parallels} \Longrightarrow \text{the I.29-type conclusion needed in XI.8.}$$

The two applications of XI.4 remain visible in the production proof. The first is made in the common plane of the parallel lines; the second is made in the reference plane.

A.12 Checkpoint XI.8

Through XI.8 the reconstruction remains synthetic: no coordinates, vectors, scalar product, trigonometric functions, or numerical angle measure are introduced.

Unlike XI.4–XI.7, however, XI.8 is genuinely Euclidean in the axiomatic sense. The formal proof requires plane-local parallel uniqueness, represented by `HilbertSpaceEuclidean`. No new continuity assumption is used.

A.13 Proposition XI.9

Heath gives the proposition in the form:

> If two straight lines are parallel to the same straight line, then they are parallel to one another, if they do not lie in the same plane with it.

Euclid's printed proof is organized through the perpendicularity theory built in the immediately preceding propositions rather than through an affine plane-intersection argument. Its characteristic local dependency pattern is

$$\begin{gathered}\text{I.11 in suitable planes}\\ \Downarrow\\ \text{auxiliary perpendiculars}\\ \Downarrow\\ \text{XI.4}\\ \Downarrow\\ \text{XI.8}\\ \Downarrow\\ \text{XI.6}\\ \Downarrow\\ \text{XI.9.}\end{gathered}$$

The important source-level point is that XI.9 is not simply planar I.30 with "space" substituted for "plane". Euclid explicitly uses the preceding line–plane perpendicularity results to bridge between the distinct carrier planes of the given parallels.

A.14 Comparison with the Lean reconstruction at XI.9

The production Lean proof reaches the same conclusion by a different synthetic route. From

$$l \parallel n, \qquad m \parallel n,$$

it extracts the two carrier planes. It then chooses a point $B \in m$ not on l, constructs a plane ω through l and B, and intersects ω with the carrier plane τ of m, n.

The resulting line r passes through B. Inside τ, both r and m are parallels to n through the same point. Group IV uniqueness therefore gives $r = m$. Hence $l, m \subset \omega$, and a final disjointness argument yields

$$l \parallel m.$$

Thus the formal proof replaces Euclid's perpendicularity chain by

$$\text{spatial incidence} + \text{plane intersection} + \text{plane-local parallel uniqueness.}$$

It remains fully synthetic and uses no coordinates, vectors, or scalar product.

A.15 Checkpoint XI.9

The classical and Lean proofs establish the same spatial transitivity statement but expose different local DAGs. Euclid packages the theorem through XI.4, XI.8, and XI.6; the current formalization packages it through explicit carrier planes and one Group IV uniqueness step.

This is a difference in proof architecture, not in theorem content. No continuity principle is introduced by the Lean route.

A.16 Proposition XI.10

Heath gives the statement:

> If two straight lines meeting one another be parallel to two straight lines meeting one another not in the same plane, they will contain equal angles.

Let AB, BC meet at B, and let DE, EF meet at E, with the corresponding pairs parallel. Euclid first uses I.3 to cut equal segments so that

$$BA = BC = ED = EF.$$

He then joins

$$AD, \quad CF, \quad BE, \quad AC, \quad DF.$$

The first two I.33 applications give

$$AD \parallel BE, \qquad AD = BE,$$

and

$$CF \parallel BE, \qquad CF = BE.$$

Then XI.9 gives

$$AD \parallel CF,$$

while transitivity of equality through BE gives $AD = CF$. A third application of I.33 yields

$$AC \parallel DF, \qquad AC = DF.$$

Finally I.8 applied to the two triangles gives

$$\angle ABC = \angle DEF.$$

The exact local classical DAG is therefore

$$\begin{array}{c} \text{I.3} \\ \Downarrow \\ \text{I.33 twice} \\ \Downarrow \\ \text{XI.9} \\ \Downarrow \\ \text{I.33} \\ \Downarrow \\ \text{I.8} \\ \Downarrow \\ \text{XI.10.} \end{array}$$

A subtle point is carried by the classical diagram: the selected parallel segments must be taken in corresponding directions. Reversing exactly one ray changes the relevant angle to its supplement. Euclid does not isolate this orientation datum as a separate predicate.

A.17 Comparison with the Lean reconstruction at XI.10

The Lean proof preserves the same I.33–XI.9–I.33–SSS spine but makes the orientation information explicit through

```
HilbertSpaceDirectedParallelSegments
```

Ordinary spatial parallelism is sufficient for XI.9 but not for the third I.33 step.

The normalized formal core has the following actual DAG:

```
BA ~= ED, BC ~= EF
directed parallels
        |
   I.33 twice
        |
AD ~= BE, CF ~= BE
        |
       XI.9
        |
AD || CF
        |
Borsuk-Szmielew 52
        |
AD directed-parallel CF
        |
       I.33
        |
     AC ~= DF
        |
  spatial SSS
        |
angle ABC ~= angle DEF
```

The Borsuk–Szmielew 52 line in this DAG is not an additional Euclidean proposition. It records the formal section-line/plane-separation bridge used to recover the direction of AD relative to CF.

The public theorem also reconstructs Euclid's opening I.3 normalization by segment construction. It lays off points D_1, F_1 on the original rays ED, EF, applies the normalized theorem, and then transports the resulting angle back along SameRay in the explicit plane of D, E, F.

The final SameRay step is deliberately planar. The proof does not install a global ambient `HilbertOrder Geo`; it works inside the induced `PlaneGeo` of the target angle plane and maps the resulting angle data back to ambient space.

A.18 Checkpoint XI.10

Through XI.10 the source comparison shows two kinds of information which classical diagrams suppress but the formalization must expose:

- explicit carrier planes and noncoplanarity witnesses;

- orientation of selected segments on parallel carriers.

The Lean theorem remains synthetic. Its extra machinery is plane-side and ray-order bookkeeping, not coordinates or numerical angle measure. XI.9 is a genuine direct dependency of XI.10, whereas XI.11–XI.13 below have their own local DAGs and do not consume XI.9 or XI.10.

A.19 Proposition XI.11

Heath gives the statement:

> From a given elevated point to draw a straight line perpendicular to a given plane.

Let A be the elevated point and let π be the plane of reference. Euclid draws an arbitrary line $BC \subset \pi$, then draws AD from A perpendicular to BC by I.12. If $AD \perp \pi$, the problem is finished.

In the nontrivial branch he constructs

$$DE \perp BC \quad \text{(I.11)}, \qquad AF \perp DE \quad \text{(I.12)}, \qquad GH \parallel BC \quad \text{(I.31)}.$$

Since BC is perpendicular to both DA and DE, XI.4 gives

$$BC \perp \text{plane}(ED, DA).$$

Then XI.8 transfers this perpendicularity to the parallel line GH:

$$GH \perp \text{plane}(ED, DA).$$

By XI.Def.3, GH is perpendicular to AF, which meets it and lies in that plane. Hence AF is perpendicular to both GH and DE. A second application of XI.4 gives

$$AF \perp \text{plane}(ED, GH) = \pi.$$

Thus the exact local classical dependency chain is

$$\begin{array}{c} \text{I.12} \\ \Downarrow \\ \text{I.11, I.12, I.31} \\ \Downarrow \\ \text{XI.4} \\ \Downarrow \\ \text{XI.8} \\ \Downarrow \\ \text{XI.Def.3} \\ \Downarrow \\ \text{XI.4} \\ \Downarrow \\ \text{XI.11.} \end{array}$$

Neither XI.9 nor XI.10 is cited in Heath's proof, and neither is needed transitively for the local construction described there. This verifies that the current reconstruction may pass directly from XI.8 to XI.11 without silently skipping a classical dependency.

One incidence point is implicit in the classical prose. Before the first I.12 construction can be interpreted as a planar construction, the elevated point A and the chosen line BC must be placed in a common plane. The formal reconstruction supplies this plane explicitly.

A.20 Comparison with the Lean reconstruction at XI.11

The production proof follows the same construction and retains Euclid's initial case split. Its additional work is spatial bookkeeping rather than a change of method.

The principal formal expansions are:

- construction of the plane σ through A and BC before the first planar I.12;
- construction of the plane τ through AD and DE before the second planar I.12;
- proof that the second foot F does not lie on BC, needed before I.31 is applied in the reference plane;
- conversion of planar I.31 parallelism into the ambient `HilbertSpaceLinesParallel` relation;
- proof that the foot K returned existentially by XI.8 equals the already constructed point F;
- proof that $DE \neq GH$ before the final XI.4 application.

The two XI.4 applications and the single XI.8 application remain exactly visible in the formal proof architecture.

A.21 Checkpoint XI.11

Through XI.11 the reconstruction remains synthetic: no coordinates, vectors, scalar products, trigonometric functions, or numerical angle measure are introduced.

The classical proof of XI.11 explicitly uses XI.8 and therefore inherits the Euclidean parallel content of that proposition. No continuity principle is cited. XI.9 and XI.10 are absent from the local source DAG.

A.22 Proposition XI.12

Heath gives the statement:

> To set up a straight line at right angles to a given plane from a given point in it.

Let A be the given point in the plane of reference. Euclid lets any elevated point B be conceived and applies XI.11 to draw BC perpendicular to the reference plane. Through A he then draws

$$AD \parallel BC$$

by I.31. Proposition XI.8 immediately transfers the line–plane perpendicularity:

$$BC \perp \pi, \qquad AD \parallel BC \Longrightarrow AD \perp \pi.$$

The complete local classical dependency chain is therefore

$$\begin{gathered} A \in \pi \\ \Downarrow \\ B \text{ elevated} \\ \Downarrow \\ \text{XI.11} \\ \Downarrow \\ BC \perp \pi \\ \Downarrow \\ \text{I.31} \\ \Downarrow \\ AD \parallel BC \\ \Downarrow \\ \text{XI.8} \\ \Downarrow \\ AD \perp \pi. \end{gathered}$$

Neither XI.9 nor XI.10 is cited or required in this local proof.

There is one exceptional incidence configuration that the classical prose does not discuss. If the foot C of the XI.11 perpendicular happens to equal the prescribed point A, then the line BC itself already solves XI.12. In that configuration I.31 should not be read as producing a distinct parallel through a point lying on BC. The formal proof therefore separates this completed case before carrying out the generic parallel construction.

A.23 Comparison with the Lean reconstruction at XI.12

The production proof preserves the source-level chain

$$\text{XI.11} \longrightarrow \text{I.31} \longrightarrow \text{XI.8},$$

but expands three pieces of hidden incidence data.

First, it explicitly tests whether the prescribed point A already lies on the perpendicular returned by XI.11. A proposition-local foot-uniqueness theorem then identifies A with the XI.11 foot and closes that branch.

Second, in the nontrivial branch the spatial line and the point A must be placed in a common plane σ before I.31 can be used. The parallel construction is performed inside PlaneGeo(σ), after which its planar disjointness data are packaged as ambient `HilbertSpaceLinesParallel`.

Third, XI.8 returns the new perpendicular foot existentially. The same foot-uniqueness theorem proves that this returned point is exactly the prescribed point A.

These are formal expansions of incidence and carrier-plane information; they do not replace Euclid's geometric method.

A.24 Checkpoint XI.12

Through XI.12 the reconstruction remains synthetic: no coordinates, vectors, scalar products, trigonometric functions, or numerical angle measure enter the construction.

The classical proof explicitly uses XI.11, I.31, and XI.8. Hence the present Euclid-faithful route inherits the Euclidean parallel content already present in XI.8 and XI.11. No continuity principle is cited, and XI.9 and XI.10 remain absent from the local source DAG.

A.25 Proposition XI.13

Heath gives the statement:

> From the same point two straight lines cannot be set up at right angles to the same plane on the same side.

Assume that AB and AC are two such straight lines erected from the common point A. Euclid draws the plane through BA and AC. By XI.3 its section with the reference plane is a straight line through A, written DAE.

Definition XI.3 then applies to each supposed perpendicular. Since DAE lies in the reference plane and meets both lines at A,

$$AC \perp DAE, \qquad AB \perp DAE.$$

Hence $\angle CAE$ and $\angle BAE$ are both right and therefore equal. They lie in the same plane, share the side AE, and the other two sides are assumed to be erected on the same side. Euclid concludes that this is impossible.

The complete local source DAG is therefore

$$\begin{gathered}
AB \perp \pi, \quad AC \perp \pi \\
\Downarrow \\
\text{plane through } AB, AC \\
\Downarrow \\
\text{XI.3} \\
\Downarrow \\
DAE = \pi \cap \text{plane}(AB, AC) \\
\Downarrow \\
\text{XI.Def.3} \\
\Downarrow \\
AB \perp DAE, \quad AC \perp DAE \\
\Downarrow \\
\text{two equal right angles in one plane on the same side} \\
\Downarrow \\
\text{contradiction.}
\end{gathered}$$

Neither XI.9 nor XI.10 occurs in the proof. XI.11 and XI.12 are also not source dependencies: they are existence constructions, while XI.13 proves uniqueness.

A.26 Comparison with the Lean reconstruction at XI.13

The formal proof preserves Euclid's two genuinely spatial steps:

1. construct the plane σ through the two supposed distinct perpendicular carriers;
2. construct the intersection line $q = \pi \cap \sigma$ through the common foot.

The production proof makes one hidden condition explicit before the second step: it proves $\sigma \neq \pi$. If the planes were equal, one of the perpendicular lines would lie in the reference plane, which is impossible.

The terminal planar contradiction is then packaged differently. From Definition XI.3, represented by the universal clause of `HilbertLinePerpendicularPlaneAt`, the proof gets

$$l \perp q, \qquad m \perp q.$$

After symmetry, XI.4 is applied in the auxiliary plane σ. If $l \neq m$, it yields $q \perp \sigma$. But the plane-intersection construction already gives $q \subset \sigma$, producing a contradiction.

Thus the Lean DAG replaces Euclid's final same-side angle-uniqueness sentence by an earlier Book XI theorem, but it does not replace the synthetic method by coordinates or metric formulas.

A.27 Checkpoint XI.13

Through XI.13 the source DAG remains independent of XI.9 and XI.10. The classical dependencies specific to the proposition are XI.3 and Definition XI.3.

The proposition is also independent of XI.11 and XI.12 as source dependencies: those propositions establish the two existence cases, whereas XI.13 supplies the corresponding uniqueness statement at a point of the plane.

The current reconstruction remains synthetic and introduces no coordinate, vector, scalar-product, trigonometric, or numerical-angle argument.

Appendix B

Hilbert: Spatial Axiomatic Architecture

B.1 Purpose and scope

This appendix records the Hilbert-side source audit for Euclid Book XI at the current checkpoint, Proposition XI.13. The purpose is architectural: identify which layers of incidence, order, congruence, and Euclidean parallelism enter at each theorem, and distinguish them from continuity assumptions.

B.2 Spatial architecture

The formal project separates the spatial hierarchy into

```
HilbertSpacePrimitive
HilbertSpaceIncidence
HilbertSpaceOrder
HilbertSpaceCongruence
HilbertSpaceEuclidean
```

and derives ordinary planar Hilbert geometry on each fixed plane via

```
PlanePoint
PlaneLine
PlaneGeo
```

with explicit bridges for incidence, betweenness, congruence, same rays, angles, and right angles.

This separation is essential at XI.4. There is deliberately no global `HilbertCongruence Geo` or `HilbertOrder Geo` instance installed on the ambient three-space. Planar order arguments are carried out inside a specified `PlaneGeo`; spatial congruence transports only the data needed between such plane slices.

B.3 Neutral status of XI.4 in the current hierarchy

The public theorem assumes the spatial incidence/order/congruence hierarchy but no `HilbertEuclideanPlane` instance. In particular, Hilbert Group IV – the Euclidean parallel axiom – does not occur in the proof of XI.4.

The final axiom audit reported by Lean is

```
'Geometry.euclid_proposition_11_4' depends on axioms:
[propext,
 Classical.choice,
 Geometry.bookZero_nullSegment1,
 Quot.sound]
```

The project deliberately retains the inherited `bookZero_nullSegment1` dependency at this checkpoint. XI.4 introduces no proposition-specific geometric axiom.

B.4 What XI.4 adds to the interface

The theorem forced one useful correction to the general 3D API: `HilbertLinesPerpendicularAt` now carries an explicit noncollinearity witness. Consequently distinctness of perpendicular carriers is derived by `hilbert_linesPerpendicularAt_ne` rather than supplied as a separate hypothesis.

The universal target is represented by

```
HilbertLinePerpendicularPlaneAt
```

which quantifies over every incidence line of the plane through the foot. This is a direct synthetic representation and introduces no metric structure.

B.5 Checkpoint XI.4

At XI.4 Hilbert's role is therefore not a one-to-one proposition-number match. It is the axiomatic architecture in which Euclid's spatial proof can be made explicit. The most significant result of the audit is the boundary: XI.4 is completed in the current incidence/order/congruence 3D hierarchy without Group IV and without a new continuity assumption.

B.6 Incidence axiom I.6 and Proposition XI.7

Proposition XI.7 isolates a much smaller part of the spatial hierarchy than XI.4–XI.6. The decisive Hilbert incidence principle is represented in the current formal interface by

```
HilbertSpaceIncidence.line_in_plane
```

Its geometric content is: if two distinct points of a line lie in a plane, then the whole line lies in that plane. In the numbering adopted by the current project this is Hilbert incidence axiom I.6.

The public XI.7 theorem assumes only

```
[HilbertIncidence Geo]
[HilbertPlaneIncidence Geo]
[HilbertSpacePrimitive Geo]
[HilbertSpaceIncidence Geo]
```

It does not assume `HilbertSpaceOrder`, `HilbertSpaceCongruence`, or an induced `PlaneGeo` instance. The Euclidean parallel axiom, Group IV, and continuity are therefore absent from the proof architecture.

B.7 Spatial parallelism as reusable incidence data

The relation

```
HilbertSpaceLinesParallel
```

is now part of `Hilbert3DInterface`. It packages two assertions: a common carrier plane and disjointness of the two ambient lines. The common plane is essential because ambient disjointness alone would also admit skew lines.

This notion was introduced during the XI.6 development and promoted to the general 3D interface before XI.7. Proposition XI.7 is the first subsequent Book XI theorem that consumes the common-plane witness directly.

B.8 Axiomatic status of XI.7

The final audit command is

```
import CGJteamLab.Proposition11_7

#print axioms Geometry.euclid_proposition_11_7
```

and Lean reports

```
'Geometry.euclid_proposition_11_7' does not depend on any axioms
```

This result concerns global Lean axiom constants. It does not remove the explicit Hilbert incidence structures from the theorem signature: those structures remain hypotheses of the theorem. What the audit establishes is that no additional globally declared axiom constant occurs in the proof term.

B.9 Checkpoint XI.7

At XI.7 the spatial reconstruction reaches a pure incidence theorem. The formal proof needs neither Group IV nor continuity, and it introduces no new proposition-specific geometric or spatial axiom. The decisive spatial content is exactly the combination of a common-plane witness for parallel lines and Hilbert's line-in-plane incidence principle.

B.10 Group IV becomes active at Proposition XI.8

Proposition XI.8 is the first theorem in the present Book XI sequence whose production signature assumes

```
[HSE : HilbertSpaceEuclidean Geo]
```

The class is the spatial representation of Hilbert Group IV. Its parallel-uniqueness field is explicitly restricted to one ambient plane: through a point of that plane outside a given line, two lines in the same plane that are disjoint from the given line must coincide.

This formulation is essential in three-space. The project does not install a global

```
HilbertEuclideanPlane Geo
```

instance on the ambient spatial geometry. Instead, every fixed plane inherits the planar Euclidean interface through

```
planeGeo_linesDisjoint_iff_ambient
planeGeoHilbertEuclidean
```

The first theorem identifies planar disjointness in a fixed `PlaneGeo` with ambient disjointness of the underlying spatial lines. The second packages the plane-local Group IV field as the existing planar `HilbertEuclideanPlane` instance.

B.11 Neutral I.27 and Euclidean uniqueness in XI.8

The formal replacement for Euclid's I.29 step exposes the logical boundary between neutral and Euclidean geometry.

Inside the common plane σ of the parallel lines, a line p is constructed through D perpendicular to the transversal $d = BD$. The established I.27 alternate-angle theorem shows that p is disjoint from the first parallel l. This part uses order and congruence but not Group IV.

The given second parallel m is also disjoint from l. Both p and m pass through D and lie in σ. Only now is Group IV used: uniqueness forces $p = m$. Therefore $m \perp d$.

The formal proof consequently separates

$$\text{existence/nonintersection} \quad \text{from} \quad \text{Euclidean uniqueness.}$$

B.12 Axiomatic status of XI.8

The final audit command is

```
import CGJteamLab.Proposition11_8

#print axioms Geometry.euclid_proposition_11_8
```

and Lean reports

```
'Geometry.euclid_proposition_11_8' depends on axioms: [propext,
 Classical.choice,
 Geometry.bookZero_nullSegment1,
 Quot.sound]
```

These are inherited global Lean/project dependencies. No new proposition-specific global axiom constant is introduced.

This audit output must be distinguished from the theorem's explicit geometric hypotheses. The typeclass `HilbertSpaceEuclidean Geo` is a genuine mathematical assumption of XI.8: it represents Group IV. It is passed as a theorem parameter, rather than appearing as an additional global Lean axiom constant.

No continuity assumption is used.

B.13 Checkpoint XI.8

At XI.8 the spatial hierarchy first crosses from the neutral incidence/order/congruence layers into the Euclidean parallel layer. The transition is plane-local: Group IV is applied only inside a specified ambient plane and is inherited by its induced `PlaneGeo` geometry.

The proof remains synthetic. Its Euclidean content is exactly the uniqueness step required to turn neutral nonintersection into the I.29-type perpendicularity conclusion.

B.14 Proposition XI.9 in the current Hilbert architecture

The production XI.9 theorem assumes

```
[HilbertIncidence Geo]
[HilbertPlaneIncidence Geo]
[HilbertSpacePrimitive Geo]
[HilbertSpaceIncidence Geo]
[HilbertSpaceOrder (Geo := Geo) (H := H) (S := S)]
[HilbertSpaceCongruence (Geo := Geo) (H := H) (S := S)]
[HilbertSpaceEuclidean Geo]
```

although the mathematical proof uses the order/congruence layers only through inherited spatial infrastructure. The decisive Euclidean call is

```
HilbertSpaceEuclidean.parallel_unique_in_plane
```

in one explicitly chosen carrier plane.

The formal proof does not reconstruct Euclid's perpendicularity chain. It uses two proposition-local incidence helpers,

```
hilbert_XI9_point_on_second_line_off_first
hilbert_XI9_planes_eq_of_common_line_and_external_point
```

together with the general spatial theorems

```
hilbert_plane_through_line_and_external_point
hilbert_plane_intersection_line
```

to create a common carrier plane for the two target lines.

No global ambient `HilbertOrder Geo` or `HilbertCongruence Geo` instance is installed. In fact, the public proof does not need to enter an induced `PlaneGeo`; Group IV is already available in its ambient plane-indexed form.

B.15 Axiomatic status of XI.9

The audit reports

```
'Geometry.euclid_proposition_11_9' depends on axioms:
[propext, Classical.choice, Quot.sound]
```

The theorem therefore introduces no proposition-specific global axiom and does not inherit `Geometry.bookZero_nullSegment1`.

The explicit parameter

```
[HSE : HilbertSpaceEuclidean Geo]
```

remains a genuine geometric hypothesis. As elsewhere, an explicit theorem parameter is not listed as a hidden global constant by `#print axioms`. No continuity assumption is used.

B.16 Checkpoint XI.9

At XI.9 the current hierarchy isolates a particularly clean decomposition:

$$\begin{gathered}\text{spatial incidence}\\ +\text{ one plane-local Group IV uniqueness step}\\ \Downarrow\\ \text{spatial transitivity of parallelism}\end{gathered}$$

The theorem adds no new reusable axiom class and no analytic representation.

B.17 Proposition XI.10 in the current Hilbert architecture

The production XI.10 theorem assumes the full spatial hierarchy

```
[HilbertIncidence Geo]
[HilbertPlaneIncidence Geo]
[HilbertSpacePrimitive Geo]
[HilbertSpaceIncidence Geo]
[HilbertSpaceOrder (Geo := Geo) (H := H) (S := S)]
[HilbertSpaceCongruence (Geo := Geo) (H := H) (S := S)]
[HilbertSpaceEuclidean Geo]
```

and concludes an ambient angle-congruence statement.

XI.10 introduces a new proposition-local orientation relation:

```
HilbertSpaceDirectedParallelSegments
```

This relation refines ordinary

```
HilbertSpaceLinesParallel
```

by adding a connector line through the terminal endpoints and a same-side condition for the initial endpoints in the common carrier plane. It is not a new axiom class; it is a definition built from incidence, disjointness, and plane-local order.

The proof also defines proposition-local plane-separation language

```
HilbertSegmentMeetsPlane
HilbertSameSideOfPlane
```

and proves the section bridge

```
hilbert_XI10_borsuk_szmielew_52
```

from the existing spatial incidence/order infrastructure. Thus no primitive half-space axiom is added to the Hilbert hierarchy.

The normalized proof spine is

```
BA ~= ED, BC ~= EF
directed parallels
        |
   I.33 twice
        |
AD ~= BE, CF ~= BE
```

```
        |
       XI.9
        |
AD || CF
        |
Borsuk-Szmielew 52
        |
AD directed-parallel CF
        |
       I.33
        |
     AC ~= DF
        |
  spatial SSS
        |
angle ABC ~= angle DEF
```

The I.33 steps and XI.9 consume the Euclidean parallel layer. The final SSS step is synthetic congruence. The Borsuk–Szmielew 52 bridge consumes spatial order but no continuity.

The public theorem removes the normalized equal-length assumptions by spatial segment construction. The constructed points D_1, F_1 lie on the original rays ED, EF. After the normalized angle theorem is applied, the final SameRay angle transport is performed inside the explicit induced geometry PlaneGeo(plane(D, E, F)).

This last point is architecturally important. XI.10 does not install ambient

```
[HilbertOrder Geo]
```

merely to reuse planar ray lemmas. The order-theoretic angle transport stays inside a genuine plane slice and is transferred back by the existing PlaneGeo/ambient bridge.

B.18 Axiomatic status of XI.10

The validated audit is

```
'Geometry.euclid_proposition_11_10' depends on axioms:
[propext,
 Classical.choice,
 Geometry.bookZero_nullSegment1,
 Quot.sound]
```

These are inherited global Lean/project dependencies. XI.10 introduces no proposition-specific global axiom constant, no new spatial axiom class, and no continuity principle.

The appearance of `Geometry.bookZero_nullSegment1` distinguishes the public XI.10 wrapper from XI.9 at the level of global audit output. It is inherited through existing construction/congruence infrastructure rather than introduced by the new directed-parallel or plane-side definitions.

B.19 Checkpoint XI.10

At XI.10 the Hilbert architecture acquires no new primitive layer, but the formal proof exposes a new interface problem: orientation data must move between several plane slices and ambient three-space.

The solution is entirely synthetic:

$$\text{plane-local same side} \longleftrightarrow \text{ambient same side of a plane} \longrightarrow \text{directed spatial parallelism.}$$

This is the principal new structural lesson of XI.10. Group IV is reused, spatial order becomes genuinely active, and no continuity or analytic geometry is introduced.

B.20 Proposition XI.11 in the current Hilbert architecture

The production theorem XI.11 assumes the same full spatial hierarchy as XI.8:

```
[HilbertIncidence Geo]
[HilbertPlaneIncidence Geo]
[HilbertSpacePrimitive Geo]
[HilbertSpaceIncidence Geo]
[HilbertSpaceOrder (Geo := Geo) (H := H) (S := S)]
[HilbertSpaceCongruence (Geo := Geo) (H := H) (S := S)]
[HilbertSpaceEuclidean Geo]
```

The proof does not install ambient `HilbertOrder Geo` or ambient `HilbertCongruence Geo`. Instead, Book I perpendicular and parallel constructions are applied inside explicit induced planes PlaneGeo(σ), PlaneGeo(π), and PlaneGeo(τ).

The Group IV hypothesis enters through the call to

```
euclid_proposition_11_8
```

which transfers perpendicularity from BC to the constructed parallel GH. No independent Group IV argument is added by XI.11 itself.

The two calls to

```
euclid_proposition_11_4
```

remain in the incidence/order/congruence spatial layer. The first proves $BC \perp \tau$; the second proves the final $AF \perp \pi$.

B.21 Axiomatic status of XI.11

The final audit command is

```
import CGJteamLab.Proposition11_11
```

```
#print axioms Geometry.euclid_proposition_11_11
```

and Lean reports

```
'Geometry.euclid_proposition_11_11' depends on axioms: [propext,
 Classical.choice,
 Geometry.bookZero_nullSegment1,
 Quot.sound]
```

These are inherited global Lean/project dependencies. XI.11 introduces no proposition-specific global axiom constant, no new spatial axiom declaration, and no continuity assumption.

The explicit parameter

```
[HSE : HilbertSpaceEuclidean Geo]
```

is nevertheless a genuine hypothesis of the current theorem because XI.8 is used in the proof. As at XI.8, this theorem parameter does not appear as an extra global constant in the output of `#print axioms`.

B.22 Checkpoint XI.11

At XI.11 the existing plane-local Group IV layer is consumed but not extended. The proposition introduces no new general 3D interface theorem. Its main architectural role is to coordinate three induced plane slices and to return a spatial line–plane perpendicularity statement.

The current formal proof is therefore Euclid-faithful with respect to its parallel-theoretic step: it inherits the Euclidean strength of XI.8. The audit does not claim that this hypothesis is minimal for the abstract existence statement.

B.23 Proposition XI.12 in the current Hilbert architecture

The public XI.12 theorem assumes the same full spatial hierarchy as XI.11 and XI.8:

```
[HilbertIncidence Geo]
[HilbertPlaneIncidence Geo]
[HilbertSpacePrimitive Geo]
[HilbertSpaceIncidence Geo]
[HilbertSpaceOrder (Geo := Geo) (H := H) (S := S)]
[HilbertSpaceCongruence (Geo := Geo) (H := H) (S := S)]
[HilbertSpaceEuclidean Geo]
```

The proof again avoids global ambient `HilbertOrder Geo` and `HilbertCongruence Geo`. The only new planar construction is the parallel through the prescribed point. It is carried out in an explicit auxiliary plane σ via PlaneGeo(σ).

XI.12 introduces two proposition-local theorems.

The first,

```
hilbert_XI12_perpendicular_foot_unique
```

is an incidence consequence of the universal definition of line–plane perpendicularity. If two distinct points of the perpendicular line lay in the plane, Hilbert spatial incidence would force the whole line into the plane; the universal perpendicularity clause would then make the line perpendicular to itself.

The second,

```
hilbert_XI12_parallel_through_point_in_plane
```

performs the I.31 construction inside a fixed induced plane and packages the resulting planar disjointness as `HilbertSpaceLinesParallel`. Its signature does not assume `HilbertSpaceEuclidean`; the helper itself remains below Group IV.

The public theorem nevertheless assumes Group IV because it consumes XI.11 and XI.8. No additional Euclidean uniqueness argument is introduced by the two new XI.12 helpers.

B.24 Axiomatic status of XI.12

The public audit command is

```
import CGJteamLab.Proposition11_12

#print axioms Geometry.euclid_proposition_11_12
```

and Lean reports

```
'Geometry.euclid_proposition_11_12' depends on axioms: [propext,
 Classical.choice,
 Geometry.bookZero_nullSegment1,
 Quot.sound]
```

The two local helpers were audited separately:

```
'Geometry.hilbert_XI12_perpendicular_foot_unique'
depends on axioms: [propext, Classical.choice, Quot.sound]

'Geometry.hilbert_XI12_parallel_through_point_in_plane'
depends on axioms: [propext, Classical.choice, Quot.sound]
```

Thus the XI.12-specific incidence and plane-slice packaging steps add no new global geometric axiom constant and do not themselves depend on `bookZero_nullSegment1`. The public theorem inherits that existing project dependency through the established earlier construction chain.

The explicit parameter

```
[HSE : HilbertSpaceEuclidean Geo]
```

remains a genuine geometric hypothesis of the current theorem. It enters through XI.11 and XI.8, not through a new XI.12-specific Group IV step.

No new continuity assumption, spatial axiom class, or `sorry`/`admit` is introduced.

B.25 Checkpoint XI.12

XI.12 adds no new primitive 3D structure and no new plane–space bridge declaration. Its two local helpers expose reusable consequences of the existing architecture: uniqueness of the perpendicular foot and the packaging of a planar parallel construction as spatial parallelism.

The current public proof is Euclid-faithful and therefore retains the Group IV strength inherited from XI.11 and XI.8. The axiom audit distinguishes that explicit theorem hypothesis from the global Lean/project axiom constants appearing in the proof term.

B.26 Proposition XI.13 in the current Hilbert architecture

The production XI.13 theorem assumes

```
[HilbertIncidence Geo]
[HilbertPlaneIncidence Geo]
[HilbertSpacePrimitive Geo]
[HilbertSpaceIncidence Geo]
[HilbertSpaceOrder (Geo := Geo) (H := H) (S := S)]
[HilbertSpaceCongruence (Geo := Geo) (H := H) (S := S)]
```

and, unlike XI.8, XI.11, and XI.12, does not assume

```
HilbertSpaceEuclidean Geo
```

Thus XI.13 returns to the neutral spatial incidence/order/congruence layer of the current hierarchy.

The proof needs no new induced-plane construction theorem and no new plane–space bridge. Its principal spatial ingredients are:

- the plane through two distinct intersecting lines;
- the line of intersection of two distinct planes;
- the universal clause of line–plane perpendicularity;
- symmetry of line–line perpendicularity;
- XI.4;
- the fact that a line perpendicular to a plane cannot itself lie in that plane.

The explicit proof that the auxiliary plane differs from the reference plane is an incidence consequence of the existing line–plane perpendicularity interface. No parallel uniqueness principle is needed.

B.27 Axiomatic status of XI.13

The audit command is

```
import CGJteamLab.Proposition11_13

#print axioms Geometry.euclid_proposition_11_13
```

and Lean reports

```
'Geometry.euclid_proposition_11_13' depends on axioms: [propext,
 Classical.choice,
 Geometry.bookZero_nullSegment1,
 Quot.sound]
```

These are the accepted inherited global Lean/project dependencies. XI.13 introduces no proposition-specific global axiom constant.

The absence of `HilbertSpaceEuclidean` from the theorem signature is a genuine geometric distinction from XI.8, XI.11, and XI.12. Group IV is not used in the current XI.13 proof. No continuity assumption, new spatial axiom declaration, or `sorry`/`admit` is introduced.

B.28 Checkpoint XI.13

At XI.13 the current reconstruction isolates uniqueness of the normal at a point without crossing the Euclidean parallel boundary. The proof uses the existing spatial incidence/order/congruence hierarchy and reuses XI.4 as the synthetic closure theorem.

No global ambient `HilbertOrder Geo` or `HilbertCongruence Geo` instance is installed, and no new `Hilbert3DInterface` declaration is required.

Appendix C

Borsuk–Szmielew: Axiomatic Layers and Spatial Geometry

C.1 Scope

This appendix records the Book XI audit against Karol Borsuk and Wanda Szmielew, *Foundations of Geometry*, through Proposition XI.13. Their primitive language includes points, lines, planes, betweenness, and equidistance. Their axioms are organized into incidence, order, congruence, continuity, and the Euclidean axiom.

C.2 Congruence results relevant to XI.4

The Book I audit already identifies the triangle-congruence results used in the XI.4 argument within the pre-Euclidean congruence layer. In particular:

- the SSS theorem corresponding to Euclid I.8 occurs as Borsuk–Szmielew Theorem 21;
- right-angle and perpendicular constructions belong to their synthetic congruence development and are not formulated by numerical angle measure.

The formal XI.4 proof likewise treats SAS/SSS and right-angle transport as congruence infrastructure rather than as consequences of the Euclidean parallel axiom.

C.3 Spatial geometry

Borsuk–Szmielew develop three-dimensional geometry with points, lines, planes, perpendicular lines, perpendicular projection, and later an absolute coordinate system in space. In that spatial development a line perpendicular to a plane is part of the geometric language before coordinates are introduced as a later representation.

At the current checkpoint no theorem number has been identified in the audited source whose statement is exactly Euclid XI.4. Accordingly the Lean theorem is not attributed here to a specific Borsuk–Szmielew theorem.

C.4 Axiomatic comparison

For XI.4 the comparison is:

$$\begin{gathered}\text{incidence and order in planes}\\ + \\ \text{synthetic segment and angle congruence}\\ + \\ \text{spatial incidence/congruence transport}\\ \Downarrow \\ \text{line–plane perpendicularity.}\end{gathered}$$

The formal proof does not use the Euclidean parallel axiom. This agrees with the placement of the triangle-congruence ingredients below the Euclidean layer in the Borsuk–Szmielew organization.

C.5 Checkpoint XI.4

The current audit records no exact Borsuk–Szmielew counterpart for XI.4 itself. Exact theorem-number correspondence should be added only if it is located in the source.

C.6 Proposition XI.7 and incidence Axiom I7

For XI.7 the Borsuk–Szmielew source contains an exact structural counterpart of the decisive Hilbert incidence clause.

Their Axiom I7 states that for a line L and a plane P, if there exist two distinct points a, b belonging both to L and to P, then

$$L \subset P.$$

This is exactly the line-in-plane principle used in the formal proof of XI.7. Borsuk–Szmielew also classify Axioms I1–I4 as plane axioms and Axioms I5–I9 as space axioms. Hence I7 belongs to the spatial incidence layer.

Their general organization is especially transparent here: the first group consists of incidence axioms concerning points, lines, and planes; order, congruence, continuity, and the Euclidean axiom are introduced only in later groups. Proposition XI.7 therefore lies strictly below all of those stronger layers.

The correspondence with the Lean proof is

$$\begin{gathered}E, F \in \sigma, \qquad E \neq F\\ \Downarrow \\ \text{line } n \text{ through } E, F\\ \Downarrow \\ \text{Borsuk–Szmielew I7 / Hilbert I.6}\\ \Downarrow \\ n \subset \sigma.\end{gathered}$$

No Borsuk–Szmielew theorem number is needed here: the relevant source item is the incidence axiom itself.

C.7 Checkpoint XI.7

At XI.7 the comparison becomes sharper than at XI.4. The source contains an explicit spatial-incidence axiom whose content matches the decisive formal step exactly.

Thus XI.7 requires only incidence structure in the Borsuk–Szmielew hierarchy. It does not require their axioms of order, congruence, continuity, or the Euclidean axiom. This agrees with the Lean theorem, which assumes only the ambient incidence classes and whose axiom audit reports no additional global axiom constants.

C.8 Proposition XI.8 and the Euclidean parallel boundary

No exact Borsuk–Szmielew theorem number matching the full statement of Euclid XI.8 has been identified in the audited source. The source does, however, give a precise axiomatic placement for the parallel-theoretic step on which XI.8 depends.

The Book I source audit places the I.27-type implication

$$\text{appropriate angle equality} \Longrightarrow \text{nonintersection}$$

in the congruence development before continuity and before the Euclidean axiom.

After the Euclidean axiom E is introduced, the source obtains uniqueness of the parallel and develops the converse Euclidean parallel-angle theory. In particular, the material corresponding to the I.29 direction lies on the Euclidean side of this boundary.

This is exactly the split exposed by the Lean proof of XI.8. The proposition-local helper first uses the neutral alternate-angle theorem to construct a nonintersecting line. The spatial Group IV class then supplies uniqueness in the fixed plane and identifies that line with the given parallel.

Thus the source-layer comparison is

$$\begin{gathered}
\text{incidence} + \text{order} + \text{congruence} \\
\Downarrow \\
\text{I.27-type nonintersection} \\
\Downarrow \\
+ \text{ Euclidean axiom } E \text{ / Group IV} \\
\Downarrow \\
\text{unique parallel and I.29-type converse} \\
\Downarrow \\
\text{XI.8 perpendicularity transport.}
\end{gathered}$$

The metric SAS/SSS and right-angle ingredients used around this step remain in the pre-Euclidean congruence layer. The new axiomatic input is parallel uniqueness, not a new metric or continuity principle.

C.9 Checkpoint XI.8

At XI.8 the Borsuk–Szmielew comparison records the first Book XI checkpoint in this reconstruction that requires the Euclidean axiom layer. No new continuity assumption is needed.

The source does not justify assigning a one-to-one Borsuk–Szmielew theorem number to XI.8 itself. The durable source observation is instead the same boundary made explicit by the formal API: neutral nonintersection precedes E, while the converse parallel-angle information requires Euclidean uniqueness.

C.10 Proposition XI.9 and the spatial parallel-transitivity boundary

No exact Borsuk–Szmielew theorem number matching Euclid XI.9 has been identified in the audited source. The axiomatic placement of the theorem is, however, clear enough to compare with the current Lean proof.

The formal XI.9 reconstruction combines two ingredients:

$$\text{spatial incidence} \quad + \quad \text{uniqueness of a parallel in a fixed plane.}$$

The first ingredient belongs to the spatial incidence layer: planes through a line and an exterior point, intersections of distinct planes, and uniqueness of a plane from a line plus an exterior point. The second ingredient belongs to the Euclidean parallel layer.

This agrees with the Borsuk–Szmielew organization. Their incidence axioms supply the line–plane and plane–plane closure principles, while uniqueness of a parallel is obtained only after the Euclidean axiom is active. Thus the source comparison does not suggest a new metric or continuity assumption for XI.9.

The current Lean proof is more affine than Euclid's printed perpendicularity argument. It constructs a plane through one target line and a point of the second target line, intersects this plane with the carrier plane of the second parallel pair, and uses plane-local parallel uniqueness to identify the intersection line. The resulting theorem is therefore naturally classified as

$$\text{3D incidence} + \text{Group IV} \Longrightarrow \text{spatial transitivity of parallelism under Euclid's proviso.}$$

C.11 Checkpoint XI.9

At XI.9 the Borsuk–Szmielew comparison confirms the same Euclidean boundary already visible at XI.8. The theorem requires no continuity principle and no new proposition-specific spatial axiom. The decisive non-incidence content is plane-local uniqueness of a parallel, not a new theorem of congruence.

No numbered Borsuk–Szmielew theorem is assigned to XI.9 at this checkpoint. That attribution should be added only if an exact source counterpart is located.

C.12 Proposition XI.10 and Borsuk–Szmielew Theorem 52

Proposition XI.10 adds a new issue which is not visible in XI.9: orientation of parallel segments. Ordinary carrier parallelism records coplanarity and disjointness, but it does not determine which of the two possible directions on each carrier corresponds to the intended angle.

The present Lean reconstruction makes this information explicit through the relation

```
HilbertSpaceDirectedParallelSegments
```

whose geometric content is expressed by a same-side condition with respect to a connector line inside the common carrier plane.

The key source comparison is exact. In Chapter I, Section 19, Borsuk–Szmielew prove Theorem 52: if a plane Q intersects a plane P along a line K, then for points $p, q \in Q$, the plane P separates p, q if and only if the line K separates them inside Q. Equivalently, under the corresponding off-boundary hypotheses, same-side in a plane slice with respect to the section line is the same as same-half-space in ambient space with respect to the cutting plane.

This is exactly the structural bridge used by

```
hilbert_XI10_borsuk_szmielew_52
```

and by the orientation theorem

```
hilbert_XI10_directed_AD_CF_from_common_BE
```

The formal proof first lifts two plane-local same-side statements to same-side-of-plane information, composes them in ambient space, and then cuts the result back down to the plane containing the final pair of parallel segments.

The comparison is important for axiomatic reasons. The Lean development does *not* postulate a new global plane-separation axiom. The required bridge is reconstructed from the existing spatial incidence and order structure, in the same synthetic spirit as Borsuk–Szmielew Theorem 52.

The remaining ingredients of XI.10 lie in already identified layers:

- I.33-type equal-parallel-segment transport uses the Euclidean parallel layer together with congruence;
- XI.9 supplies the unoriented spatial parallel conclusion;
- SSS belongs to the synthetic congruence layer;
- no numerical angle measure or coordinate representation is used.

Thus the source-level architecture of XI.10 can be summarized as

$$\begin{gathered}\text{spatial incidence + order}\\ \Downarrow \\ \text{Borsuk–Szmielew 52 type section bridge}\\ \Downarrow \\ \text{directed parallel transport}\\ \Downarrow \\ \text{+ Euclidean parallel uniqueness and congruence}\\ \Downarrow \\ \angle ABC \cong \angle DEF.\end{gathered}$$

C.13 Checkpoint XI.10

At XI.10 the Borsuk–Szmielew audit identifies an exact numbered theorem for the new formal ingredient: Theorem 52 is the plane-section separation bridge needed to transport orientation between plane slices and ambient space.

This does not introduce a continuity dependency. Nor does it justify a new primitive half-space axiom in the Lean hierarchy. The production theorem continues to use the existing Hilbert spatial incidence/order/congruence architecture plus the Euclidean parallel layer already active at XI.8–XI.9.

The source comparison therefore sharpens the interpretation of the formal proof: the extra XI.10 machinery is not analytic geometry and not topology added from outside the Hilbert system; it is synthetic order geometry made explicit at the plane–space interface.

C.14 Proposition XI.11 and Borsuk–Szmielew Theorem 89

At XI.11 the source comparison becomes stronger than a general architectural analogy. In Chapter II, Section 31, Borsuk–Szmielew prove Theorem 89: through a given point there exists exactly one line perpendicular to a given plane.

Their theorem covers both cases:

- the given point lies in the plane;
- the given point does not lie in the plane.

The second case contains the existence statement corresponding directly to Euclid XI.11, while the full theorem also includes uniqueness and the on-plane case later treated by Euclid XI.12.

The construction in their off-plane case is recognizably related to Euclid's synthetic strategy. Starting from a line in the plane, they use perpendicular projections and a second in-plane perpendicular, then invoke their two-line criterion for perpendicularity to a plane (Theorem 87). Their proof is organized inside the congruence chapter, before the later continuity and Euclidean-axiom layers.

This yields an important comparison with the current Lean theorem. The production proof of XI.11 deliberately follows Euclid's route through XI.8, and therefore assumes `HilbertSpaceEuclidean`. Borsuk–Szmielew Theorem 89 shows that the existence-and-uniqueness result itself admits a synthetic treatment below the Euclidean parallel layer in their axiomatic organization.

Consequently the present Group IV hypothesis should be read as a property of the chosen Euclid-faithful Lean proof, not as evidence that line–plane perpendicular construction is intrinsically Euclidean.

C.15 Comparison of the two proof architectures at XI.11

The comparison can be summarized as

Euclid / current Lean route	Borsuk–Szmielew route
two successive line perpendiculars	successive perpendicular projections
parallel $GH \parallel BC$	no Euclidean parallel step required
XI.8 / Group IV	Theorem 87 in congruence layer
XI.4	Theorem 87
existence	existence + uniqueness

This does not invalidate the current formalization. It identifies a possible later strengthening: a neutral theorem proving the existence (and potentially uniqueness) of the perpendicular through a point could be added independently, after its proof is reconstructed in the project's current Hilbert spatial API.

C.16 Checkpoint XI.11

Borsuk–Szmielew Theorem 89 is the first exact numbered source result in this appendix whose statement directly subsumes the target of Euclid XI.11. It also sharpens the axiomatic audit: the current Lean proof uses the Euclidean layer because it reuses XI.8, whereas the target theorem has a pre-Euclidean synthetic proof in the Borsuk–Szmielew development.

No continuity assumption is suggested by this comparison, and no new axiom should be added to the Lean project on its basis.

C.17 Proposition XI.12 and Theorem 89: the on-plane case

Theorem 89 already identified at the XI.11 checkpoint is stronger than the separate Euclidean construction problems XI.11 and XI.12. It states that through a given point there exists exactly one line perpendicular to a given plane, and its proof treats both positions of the point.

For XI.12 the relevant branch is precisely

$$A \in \pi.$$

Thus the on-plane existence statement proved by Euclid XI.12 is contained directly in Borsuk–Szmielew Theorem 89, while the source also supplies uniqueness. The theorem remains in Chapter II, Section 31, in the synthetic congruence development before the later continuity and Euclidean axiom layers.

This sharpens the comparison with the current Lean proof. The production XI.12 deliberately follows Euclid:

$$\text{XI.11} \longrightarrow \text{I.31} \longrightarrow \text{XI.8}.$$

Consequently it assumes `HilbertSpaceEuclidean`, because both XI.11 and XI.8 in the present library use the plane-local Group IV layer. Borsuk–Szmielew Theorem 89 shows that this

Euclidean strength belongs to the chosen proof route rather than to the abstract existence-and-uniqueness result.

The source also clarifies the status of the new Lean foot-uniqueness helper. Euclid XI.12 asks only for existence, while Theorem 89 packages existence and uniqueness together. The formal helper is therefore not an ad hoc strengthening foreign to the synthetic theory; it isolates one incidence consequence needed to make the Euclid-faithful construction total over all configurations.

C.18 Checkpoint XI.12

At XI.12 the Borsuk–Szmielew comparison is now exact for both Euclidean perpendicular-construction problems: the off-plane case of Theorem 89 subsumes XI.11, and the on-plane case subsumes XI.12.

The source therefore provides a precise future strengthening target for the formal library: a neutral existence-and-uniqueness theorem for a perpendicular through an arbitrary point. No new axiom should be added to obtain such a theorem; its reconstruction would belong to the existing synthetic incidence/order/congruence architecture.

No continuity assumption is suggested by this source comparison.

C.19 Proposition XI.13 and the uniqueness part of Theorem 89

Theorem 89, already identified at the XI.11–XI.12 checkpoints, states that through a given point there exists exactly one line perpendicular to a given plane. Proposition XI.13 corresponds directly to the uniqueness part of this result when the given point lies in the plane.

Thus the relationship among the three Euclidean propositions and Borsuk–Szmielew Theorem 89 is

Euclid	Theorem 89
XI.11	existence, point outside the plane
XI.12	existence, point in the plane
XI.13	uniqueness of the perpendicular carrier

Borsuk–Szmielew formulate the theorem for an arbitrary given point, so their uniqueness statement is not restricted to the on-plane position of XI.13.

Theorem 89 belongs to Chapter II, Section 31, within the synthetic congruence development before the later continuity and Euclidean-axiom layers. This source placement agrees with the current Lean XI.13 signature, which uses spatial incidence/order/congruence but no `HilbertSpaceEuclidean` parameter.

The two proof architectures are not identical. The current Lean XI.13 follows Euclid's auxiliary-plane geometry and closes the contradiction through XI.4. Borsuk–Szmielew package existence and uniqueness as one structural perpendicular-to-plane theorem. The source comparison therefore identifies an exact theorem-level counterpart without forcing a line-by-line proof correspondence.

C.20 Checkpoint XI.13

At XI.13 the comparison with Borsuk–Szmielew completes the three-part Euclidean sequence XI.11–XI.13. Theorem 89 subsumes both existence cases and the uniqueness conclusion.

The source also confirms the axiomatic placement exposed by the final Lean theorem: uniqueness of the perpendicular carrier is available below continuity and below the Euclidean parallel axiom. No new axiom should be introduced for XI.13.

Appendix D

Wyler: Flats, Closure, and the 3D Incidence Calculus

D.1 Purpose and scope

This appendix records the structural comparison between the Wyler-inspired three-dimensional flat calculus developed in CGJteamLab and the classical incidence-geometric line of work represented by Oswald Wyler, the *Handbook of Incidence Geometry*, and the modern theory of locally projective geometries.

The comparison has a deliberately limited purpose. The current Lean library is not a formalization of Wyler's 1953 paper *Incidence geometry* and does not claim to reproduce a historical axiom system declaration by declaration. Instead, Book XI has been used as a laboratory for a smaller question:

> Can the spatial incidence structure already present in the Hilbert layer be organized by flats, generated flats, joins, meets, and a short rank ladder in a way that exposes the geometry of Euclid's arguments?

The answer through Proposition XI.14 is affirmative. The resulting calculus is recognizably close to the closure-and-flat language used in the classical incidence literature, but it remains intentionally three-dimensional and does not yet contain a general exchange theory.

D.2 A source-critical note on the 1953 papers

Wyler published two papers in 1953 that are relevant here. The Duke paper *Incidence geometry* is the standard reference for his incidence-geometric program [5]. The companion paper *Order in projective and in descriptive geometry* is particularly useful for the present audit because its full axiomatic text makes the flat language explicit [6].

The latter paper states that its theory of order is based on Hilbert incidence axioms modified for geometry of arbitrary dimension, together with seven order axioms. It also emphasizes that no metric concepts are involved in this order theory. This separation is important for the present project: incidence, order, congruence, and Euclidean parallelism should not be fused merely because they coexist in ordinary Euclidean space.

A caution is necessary concerning modern summaries of Wyler. Some secondary accounts describe the 1953 incidence system by labels such as "IG1–IG5" or speak globally of an "exchange lattice". Those labels are not used in this appendix as primary-source terminology. The source material checked here gives something more precise for our purposes: explicit incidence axioms, an explicit definition of a flat, an explicit generated-flat construction, and, in the later Handbook treatment, a closure-space formulation with strong exchange. These are sufficient for the architectural comparison without retroactively rewriting Wyler in modern matroid terminology.

D.3 Wyler's incidence axioms and the three-dimensional core

Section 1 of *Order in projective and in descriptive geometry* begins with points, lines, and planes and lists eight incidence axioms [6]. Their roles can be summarized as follows:

1. a line contains at least two points;
2. two distinct points lie on a line;
3. two distinct lines have at most one common point;
4. a plane contains three noncollinear points;
5. three noncollinear points lie in a plane;
6. if two distinct points of a line lie in a plane, the whole line lies in that plane;
7. planes generated from two lines of one plane together with an external point have a line as their intersection;
8. there are noncollinear points and there are four noncoplanar points.

This list should not be read as an assertion that the current Hilbert classes are definitionally identical to Wyler's system. It does, however, explain why the comparison with the present 3D API is so direct. The same structural moves occur repeatedly:

Wyler-side geometric role	Current Lean realization or role
line through two distinct points	`HilbertPlaneIncidence.line_through`
uniqueness of a line from two points	`HilbertPlaneIncidence.line_unique`
plane through three noncollinear points	`HilbertSpaceIncidence.plane_through`
line determined to lie in a plane	`HilbertSpaceIncidence.line_in_plane`
nontrivial spatial dimension	points outside a plane and explicit noncoplanarity witnesses
plane–plane section line	`hilbertPlaneCarrier3D_inter_eq_lineCarrier`

The table records mathematical roles, not a declaration-by-declaration translation. The significance is that our flat layer is built over exactly the kind of incidence information for which the classical flat language was designed.

D.4 The decisive definition: flats and generated flats

Wyler's 1953 order paper defines a set of points to be a *flat* by two closure clauses [6]:

- if two distinct points of a line belong to the set, the whole line is contained in the set;
- if three noncollinear points belong to the set, a plane through them is contained in the set.

It then observes that lines and planes are flats, that an intersection of a family of flats is again a flat, and that the smallest flat containing a set of generators is the intersection of all flats containing those generators.

This is the closest historical counterpart of the definitions introduced in

```
CGJteamLab/Wyler/Hilbert3DFlats.lean
```

The current definitions are

```
def HilbertFlat3D ... (F : Set Geo.Point) : Prop :=
  -- closure under the line through two distinct points
  ... /\
  -- closure under the plane through three noncollinear points
  ...

def HilbertSpan3D ... (X : Set Geo.Point) : Set Geo.Point :=
  fun P =>
    forall F : Set Geo.Point,
      HilbertFlat3D Geo F ->
      Set.Subset X F ->
      F P
```

Thus `HilbertSpan3D` is not merely analogous to a generated flat in an informal sense. It is defined by the same closure principle:

$$\mathrm{Span}(X) = \bigcap\{F : \; F \text{ is a flat and } X \subseteq F\}.$$

This observation is the main historical justification for the name "Wyler route" used in the project. It is stronger than a superficial resemblance based on the words line, plane, or dimension: the closure object itself is organized in the same way.

D.5 From Wyler to dimensional linear spaces

The 1995 *Handbook of Incidence Geometry* gives a mature closure-theoretic formulation in Anne Delandtsheer's chapter *Dimensional Linear Spaces* [7, 8]. In the descriptive finite-dimensional presentation, points and higher varieties are organized by dimension, and the intersection of any family of varieties is again a possibly improper variety. Consequently the varieties form a closure system.

The general definition is sharper. A dimensional linear space on a point set P is a simple closure space $(P, \mathcal{C})$, whose closed sets are called varieties, satisfying the *strong exchange axiom*. In the finitary case the Handbook states that this is equivalent to the classical Mac Lane–Steinitz exchange axiom.

The associated lattice

$$\mathrm{Lat}(P, \mathcal{C})$$

is the lattice of closed sets ordered by inclusion. The Handbook records that this lattice is complete, atomic, and upper semimodular, and that in finite dimension it is called a geometric lattice. It then develops dimension, independence, bases, hyperplanes, and several equivalent forms of exchange [8].

The bibliography of that chapter explicitly lists Wyler's *Incidence geometry* [5]. Historically, therefore, the route

$$\text{incidence} \longrightarrow \text{flats/varieties} \longrightarrow \text{closure} \longrightarrow \text{exchange and lattice structure}$$

is not an artificial reinterpretation invented for the Lean development. It belongs to the established incidence-geometric tradition.

D.6 Modern matroid language: what may safely be said

The closure-and-exchange formulation is also the natural meeting point with modern matroid theory. In finite rank, geometric lattices are the standard lattice-theoretic objects associated with simple matroid geometries. For the present project, however, this modern language should be used as an interpretive bridge rather than as a historical attribution.

In particular, the current Lean library does *not* yet define:

- a matroid on `Geo.Point`;
- a general rank function;
- independent sets or bases;
- the Mac Lane–Steinitz exchange axiom;
- upper semimodularity as a lattice theorem;
- a dimension-free lattice of all flats.

Accordingly it would be premature to say that `HilbertFlat3D` has already been proved to form a geometric lattice. What has been proved is the concrete rank-three fragment needed to operate with line carriers, plane carriers, generated spans, joins, and meets.

D.7 The current 3D rank ladder

The main structural theorems in

```
Hilbert3DFlats.lean
Hilbert3DWyler.lean
```

produce a short rank ladder without introducing a numerical rank function:

$$\begin{aligned} \{A, B\},\ A \neq B \quad &\Longrightarrow \quad \mathrm{Span}\{A, B\} = \mathrm{carrier}(AB), \\ \{A, B, C\},\ A, B, C \text{ noncollinear} \quad &\Longrightarrow \quad \mathrm{Span}\{A, B, C\} = \mathrm{carrier}(\pi), \\ \{A, B, C, D\} \text{ noncoplanar} \quad &\Longrightarrow \quad \mathrm{Span}\{A, B, C, D\} = E^3. \end{aligned}$$

The corresponding Lean theorems are

```
hilbertSpan3D_pair_eq_lineCarrier
hilbertSpan3D_triple_eq_planeCarrier
hilbertSpan3D_four_noncoplanar_eq_univ
```

Further generator rules give

```
hilbertJoin3D_line_external_point_eq_planeCarrier
hilbertJoin3D_plane_external_point_eq_univ
hilbertJoin3D_two_distinct_planes_eq_univ
```

Hence the informal rank pattern is already visible:

$$\text{point pair}o\text{line}, \qquad \text{noncollinear triple}o\text{plane}, \qquad \text{noncoplanar extension}oE^3.$$

This is deliberately weaker than a general dimension theory. The gain is that it is proved directly over the project's existing Hilbert 3D incidence API and therefore can be tested immediately against Euclid Book XI.

D.8 Join and meet

For flats F, G, the current join is the span generated by their union:

```
def HilbertJoin3D ... (F G : Set Geo.Point) : Set Geo.Point :=
  HilbertSpan3D Geo (F union G)
```

The basic API then proves extensivity and leastness. At the level of geometric carriers this gives exact identities such as

$$\mathrm{Join}(l, m) = \pi$$

for two distinct coplanar intersecting lines, and

$$\mathrm{Join}(\pi, \rho) = E^3$$

for two distinct planes.

Meet is represented concretely by set-theoretic intersection. The crucial rank-three result is

```
hilbertPlaneCarrier3D_inter_eq_lineCarrier
```

for distinct planes with a common point, together with the normal form

```
hilbertPlaneCarrier3D_inter_eq_empty_or_lineCarrier
```

for arbitrary distinct planes.

This use of intersection is again exactly compatible with the classical closure-space viewpoint: closed sets are stable under arbitrary intersection, so meet is the set-theoretic intersection of flats.

D.9 Why Book XI is a good laboratory

Book XI repeatedly moves between three kinds of information:

1. *generation*: which line or plane is determined by given data;
2. *intersection*: what is the common flat of two carriers;
3. *metric structure*: perpendicularity, congruence, order, and later Euclidean parallel uniqueness.

The flat calculus makes the first two kinds explicit and prevents them from being buried inside proposition-local uniqueness arguments. The metric layer is then added only where Euclid actually needs it.

This separation is particularly close in spirit to Wyler's 1953 order paper, which explicitly distinguishes incidence and order from metric concepts [6]. The Lean architecture is not derived from that paper, but the structural agreement is useful:

```
Hilbert3DWyler
      |
      v
HilbertWylerMetric
      |
      v
HilbertWylerParallel
      |
      v
HilbertWylerEuclidean
      |
      v
HilbertWylerPlanes
      |
      v
HilbertWylerInterface
```

The flat layer therefore does not pretend that all of Euclidean spatial geometry is incidence geometry. It isolates the incidence skeleton on which the stronger layers act.

D.10 Book XI as a test bed

The second pass through Propositions XI.1–XI.14 was designed to test whether the new vocabulary was mathematically useful at theorem level. Each public Wyler proposition keeps the production theorem signature; flat identities occur inside the proof rather than replacing Euclid's theorem by a weaker structural surrogate.

The proposition-specific normal forms, dependency graphs, and join–meet calculations are documented in the corresponding proposition chapters. This appendix deliberately does not repeat them. Its role is to explain the common incidence-geometric language—flats, generated flats, closure, exchange, and rank—against which those local reconstructions should be read.

Across Book XI the recurring structural ladder is

$$\text{two points} \longrightarrow \text{line}, \qquad \text{three noncollinear points} \longrightarrow \text{plane},$$

$$\text{line} + \text{external point} \longrightarrow \text{plane}, \qquad \text{plane} + \text{external point} \longrightarrow E^3.$$

The theorem chapters show where this incidence skeleton is sufficient and where order, congruence, perpendicularity, or Euclidean parallel uniqueness must enter as genuinely stronger layers.

D.11 Locally projective geometries and the special role of dimension three

A modern continuation of Wyler's line is the theory of locally projective geometries. Sancho de Salas defines a geometry X to be locally projective when every quotient or residue $X//x$ at a point is a projective space [9]. He records the classical embedding boundary: under general conditions Wyler obtained projective embedding in dimension at least four, whereas dimension three requires an additional bundle condition and belongs to the theorem of Kahn.

The same paper gives a point–line–plane incidence formulation of local projectivity. In dimension three one may use axioms expressing unique joining lines, unique planes through noncollinear triples, line closure inside a plane, intersection of distinct meeting planes in a line, and the existence of four noncoplanar points [9].

This dimension-three exception is conceptually relevant to the present work. It warns us not to infer an arbitrary-dimensional projective or exchange theory merely from the success of the 3D calculus. Book XI shows that the flat language is productive in E^3; it does not by itself settle the higher-rank axioms required for a genuine dimension-free theory.

D.12 What has been formalized, and what has not

The current checkpoint can be summarized precisely.

Formalized in the Wyler 3D route

- flats closed under generated lines and planes;
- span as the least flat containing a set;
- line and plane carriers as flats;

- monotonicity and leastness of span;
- exact pair, triple, and noncoplanar generation theorems;
- join as generated union;
- plane meet as set intersection with line/empty normal forms;
- the rank-three generator ladder;
- interaction of this incidence calculus with perpendicularity, spatial parallelism, and plane separation through XI.14.

Not yet formalized

- arbitrary-dimensional flats;
- a rank function and dimension theorem;
- independence and bases;
- strong exchange or Mac Lane–Steinitz exchange;
- geometric-lattice or matroid equivalence;
- residues and local projectivity;
- projective completion or ideal elements;
- Wyler's higher-dimensional embedding theorem;
- any orthocomplementation theory for metric flats.

This boundary is intentional. The present 3D theory was introduced to discover the right objects and relations in a domain already rich enough to test them, not to declare a general lattice theory before the geometry had been understood.

D.13 Assessment

The Book XI experiment suggests that the flat language is not merely a change of notation. It identifies several recurring geometric states that otherwise appear as repeated local constructions:

generated line, generated plane, join of carriers, meet of planes, rank-three extension.

These states organize proofs across incidence, metric, and Euclidean layers and make dependency graphs such as XI.9 and XI.13 visible at the theorem level.

At the same time, the experiment also identifies the correct limit of the current development. Exchange, bases, rank, and the geometric-lattice viewpoint are natural next questions because the Handbook places them directly above closure by generated flats [8]. They are not yet necessary assumptions of the 3D Book XI reconstruction and should therefore not be introduced merely to make the theory look more general.

The durable conclusion at XI.14 is consequently modest but substantial:

$$\begin{gathered}\text{Hilbert 3D incidence}\\ \Downarrow\\ \text{flat/span/join/meet calculus}\\ \Downarrow\\ \text{readable structural decomposition of Book XI}\end{gathered}$$

This is the point at which the Wyler route has earned further investigation. The next step, if pursued, should be to test whether an exchange principle can be derived or naturally axiomatized from the existing synthetic geometry, rather than to assume in advance that the full matroid or geometric-lattice formalism is the desired endpoint.

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